Brine Dispersion Modelling Case Study

Universal Brine Dispersion Solver (UBDS): Near-, Medium- and Far-Field Assessment of the Bahia Azul Seawater Desalination & Brine Outfall

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Solver: UBDS v1.0.0 | Document: case.docx

Executive Summary

This study presents a holistic near-, medium- and far-field assessment of the brine discharge from the proposed Bahia Azul seawater desalination and brine outfall, performed entirely with the Universal Brine Dispersion Solver (UBDS) - a novel, open, physics-based solver developed by the author. UBDS unifies, in a single dynamically-coupled and mass-conserving framework, the near-field integral jet/plume modelling and the far-field hydrodynamic transport modelling that conventionally require two separate, proprietary codes. Its sign-agnostic buoyancy treatment and dense-water gravity-current closure make it universal across negatively buoyant (brine), positively buoyant (thermal) and neutral discharges.

For the proposed inclined 4-port diffuser, the near-field initial dilution of the reverse-osmosis (RO) concentrate at first seabed contact ranges between x58 and x186 across the flow and tidal range, corresponding to seabed salinities of 36.7-37.0 psu against a background of 36.5 psu.

The medium- and far-field assessment confirms that, at the maximum discharge of 1200 m3/hr, no 0.5 psu influence area is resolved on the regional grid, whose seabed increment peaks at 0.39 psu. The reported 0.000 km2 is therefore a statement about the 60 m regional cell, not a claim that the increment is nowhere above 0.5 psu: at the near-field impact footprint at slack water it reaches 0.54 psu, over a scale the regional grid cannot resolve, with the 38.5 psu threshold not exceeded beyond 0 m from the diffuser - well within the 100 m regulatory mixing zone. All trace impurities remain below their Environmental Quality Standards after dilution. Every near-field run lies inside the Froude range over which the entrainment closure is validated (see section 4); the saturated-cavern-brine worst case governs the seabed-salinity assessment.

The solver was tested against canonical self-similar jet and plume dilution laws, published inclined dense-jet experimental data, and reference seawater/brine densities. The equation of state, the dense-jet return distance and the dense-jet return-point dilution fall inside their reference ranges; the round-jet dilution slope, the pure-plume exponent and the dense-jet terminal rise fall a few per cent outside theirs. Each deviation is quantified in the Validation section rather than asserted away. Data sources are recorded there also. The hydrodynamic skill scores in section 3 are not part of this evidence: the ADCP series they score against is synthetic, derived from the model solution itself.

1 Introduction and the UBDS Solver

Brine discharges from desalination plants and solution-mined storage caverns are negatively buoyant and sink towards the seabed, where elevated salinity can affect benthic ecology. Their assessment requires resolving both the small-scale turbulent jet/plume mixing in the immediate vicinity of the diffuser (the near field) and the tidally-driven transport and dilution over the wider water body (the medium and far field). Conventionally these two regimes are handled by separate, proprietary software, with manual transfer of results between them.

1.1 Novelty and gaps bridged

- Unified near-to-far field coupling in a single open framework, removing the manual hand-off and the associated mass-conservation errors between separate codes.
- Sign-agnostic buoyancy: the same governing equations describe dense (brine), buoyant (thermal/freshwater) and neutral discharges - a genuinely universal solver.
- A dense-water gravity-current closure in the sigma-layer transport engine that reproduces the strong bottom-trapping of brine that 2-D depth-averaged models cannot capture and that 3-D models obtain only at large computational cost.
- Built-in regulatory-metric engine: mixing-zone radius, salinity-increment areas, diffusion distances and EQS compliance computed automatically from the solution.
- Fully open and reproducible: physically-standard, documented entrainment and turbulence closures, with all inputs and outputs exposed as data.

1.2 Governing equations

Near field - steady Eulerian flux-integral equations (top-hat) integrated along the jet arclength s, with combined shear + cross-flow entrainment and a sign-agnostic buoyancy body force:

d(rho_e Q)/ds = rho_a E ;   d(rho_e Q V)/ds = rho_a E Va + (0,0,(rho_a-rho_e) g pi b^2)

d(rho_e Q S)/ds = rho_a E Sa ;   E = 2 pi b alpha_s |V-Va| + 2 b alpha_f |Va_perp|

Far field - depth-averaged shallow-water hydrodynamics (mass + momentum with Manning friction, Coriolis and turbulent mixing) driving a sigma-layer advection-dispersion equation augmented with the buoyancy-scaled inter-layer gravity-current flux:

d(eta)/dt + d(Hu)/dx + d(Hv)/dy = 0

d(hs)/dt + d(uhs)/dx + d(vhs)/dy = d/dx(hDx ds/dx) + d/dy(hDy ds/dy) + sources

2 Case Study Site and Input Data

Site: Bahia Azul Seawater Desalination & Brine Outfall (Bahia Azul (fictional), open Mediterranean-type coast). A mid-size seawater reverse-osmosis desalination plant (product capacity 60,000 m3/d) discharging hypersaline concentrate through an offshore inclined multiport diffuser, with provision for later receipt of saturated leaching brine from an adjacent salt-cavern energy-storage scheme. Latitude 36.5 N.

2.1 Discharge and ambient inputs

Product (freshwater) capacity	60,000 m3/d
Recovery ratio	0.45
Brine (concentrate) flow - full	1200 m3/hr (28,800 m3/d)
RO brine salinity / excess temp	68.0 psu / +0.3 C
Cavern brine salinity / excess temp	250.0 psu / +2.0 C
Background salinity / temperature	36.5 psu / 18.0 C
Outfall depth / offshore distance	18.0 m / 850.0 m
Diffuser	4 x 6" ports @ 20.0 m, 60.0 deg, 0.8 m above bed
Mixing-zone radius / threshold	100.0 m / 38.5 psu

2.2 Flow scenarios

Table. Hydrodynamic and transport model parameters used in the simulations.

Parameter	Value
Minimum time step	0.4 s
CFL number	0.45
Manning coefficient n	0.025 (0.022-0.04)
Horizontal eddy viscosity	2.0 m2/s
Smagorinsky constant	0.2
Horizontal dispersion	1.0 m2/s
Vertical diffusivity	0.0005 m2/s
Gravity-current coefficient	12.0
Background salinity	36.5 psu
Background temperature	18.0 C
RO brine salinity	68.0 psu
Cavern brine salinity	250.0 psu
Outfall depth	18.0 m
Shear entrainment coefficient (jet)	0.057
Shear entrainment coefficient (plume)	0.083
Forced entrainment coefficient	0.9
Ports installed	4
Port diameter	0.1524 m (6 inch)
Port discharge angle	60.0 deg
Port spacing	20.0 m
Port height above bed	0.8 m

3 Model Geometry, Mesh and Setup

Two nested structured-grid domains were configured: a high-resolution medium-field domain (2500 x 2500 m at 25 m, 9 sigma layers) and a regional far-field domain (7200 x 7200 m at 60 m, 4 layers). The bathymetry deepens offshore and includes a southern headland whose lee generates a tidal eddy.

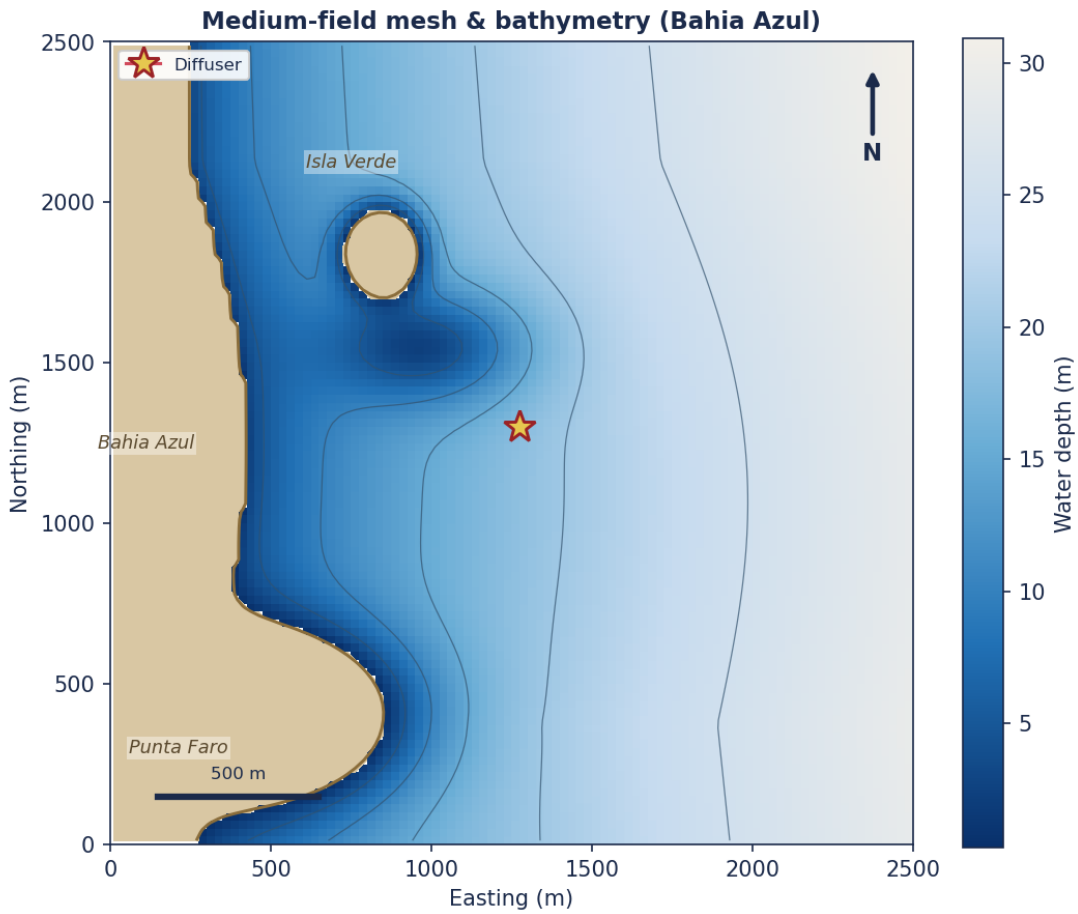


Figure. Medium-field mesh & bathymetry (Bahia Azul)

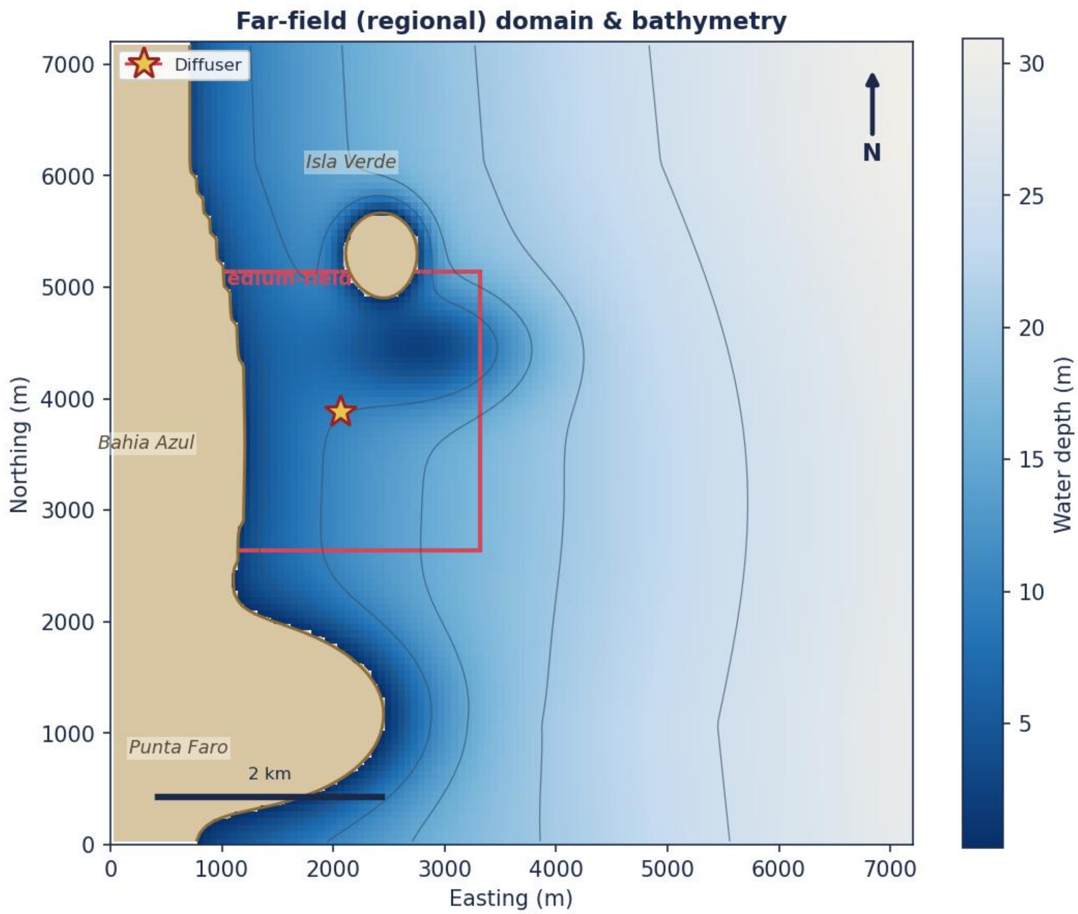


Figure. Far-field (regional) domain & bathymetry

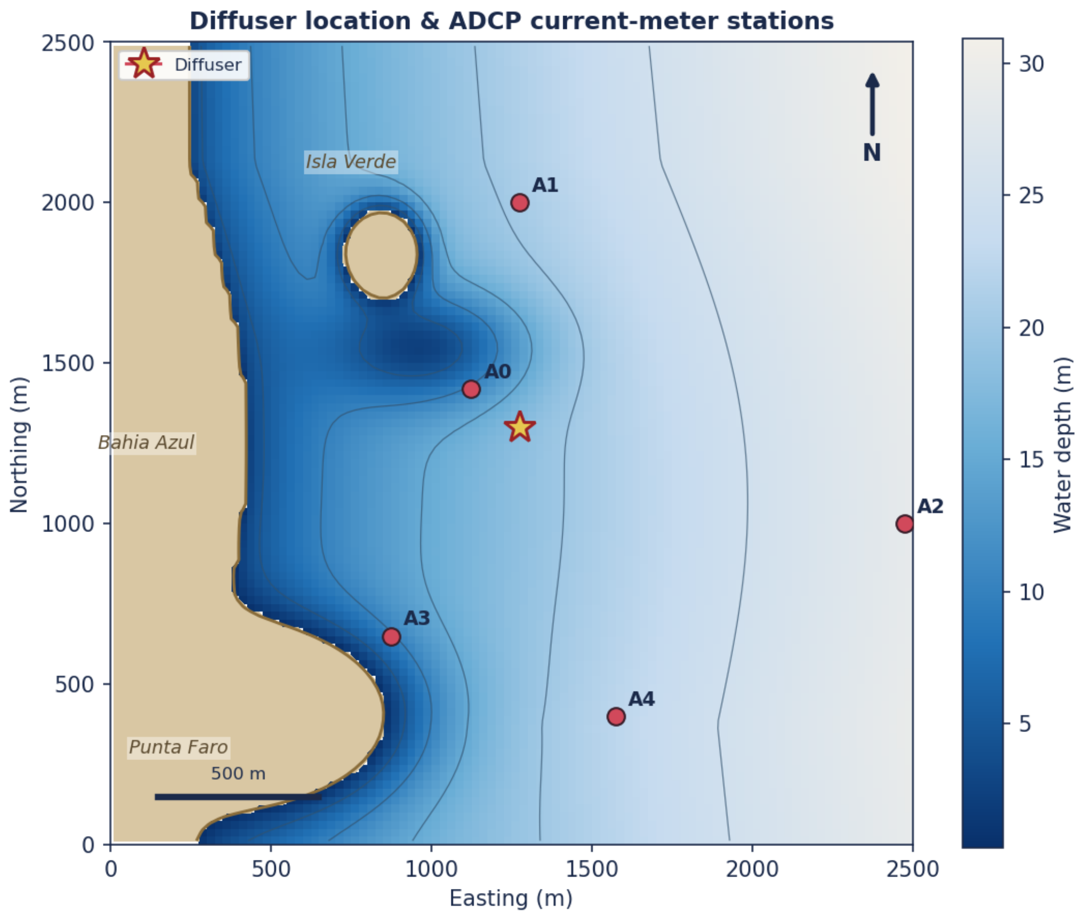


Figure. Diffuser location & ADCP current-meter stations

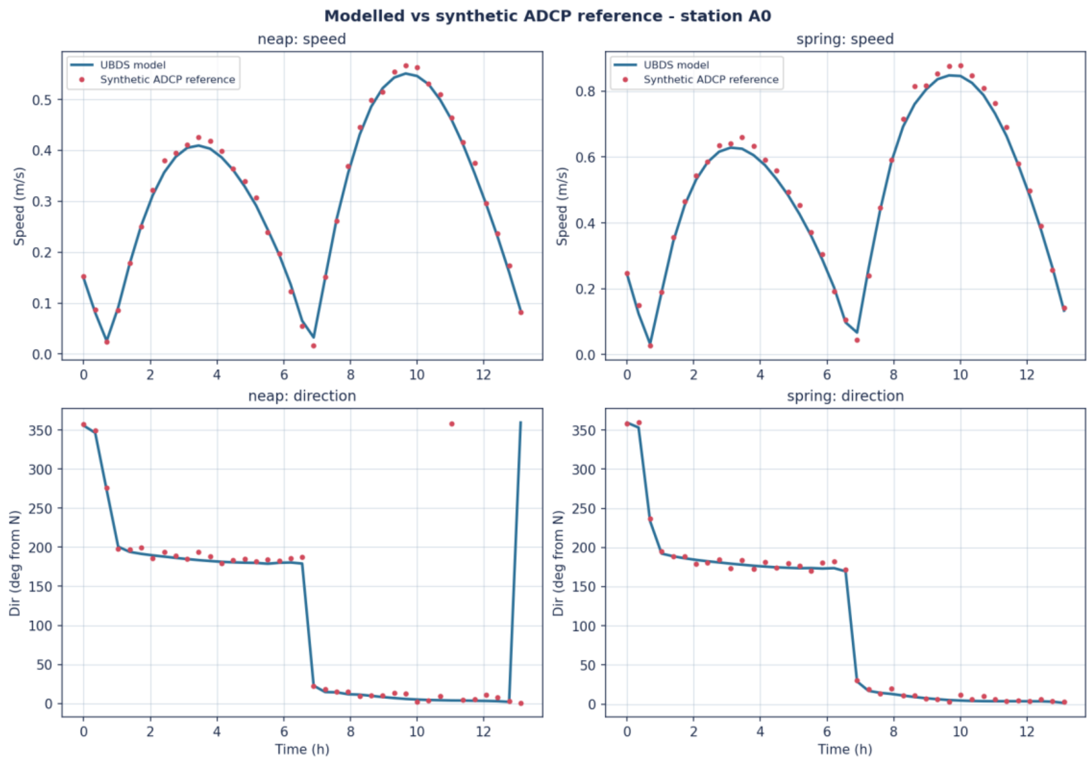


Figure. Modelled current speed & direction at station A0 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.

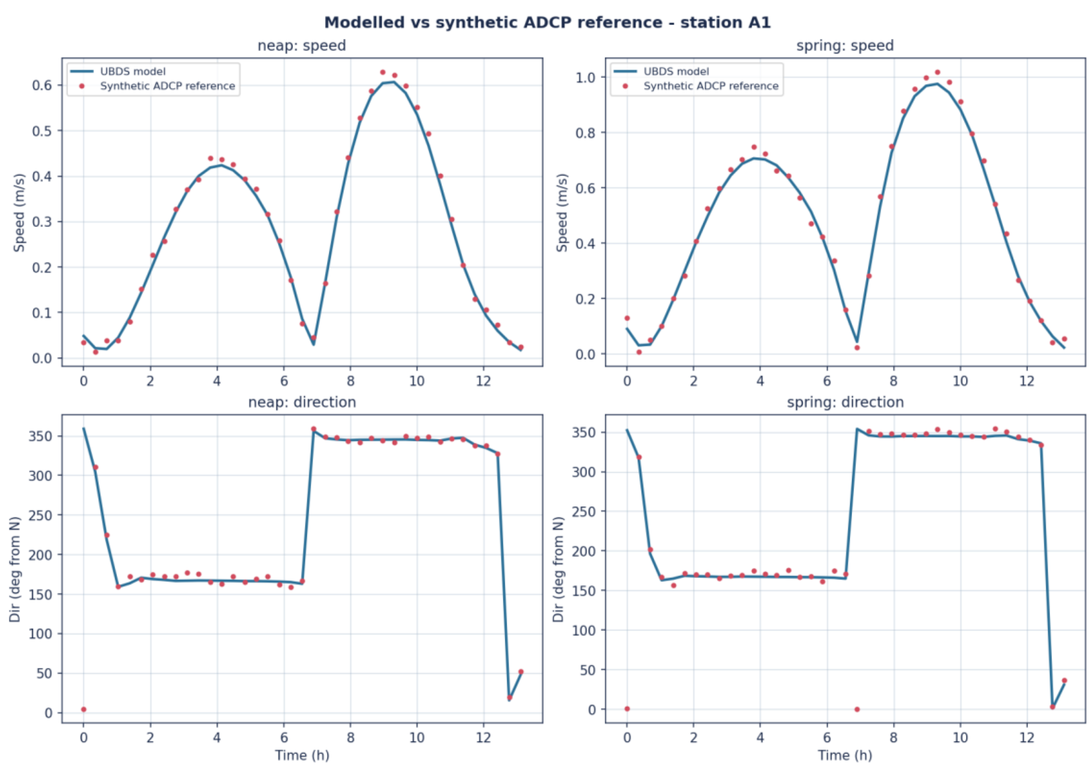


Figure. Modelled current speed & direction at station A1 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.

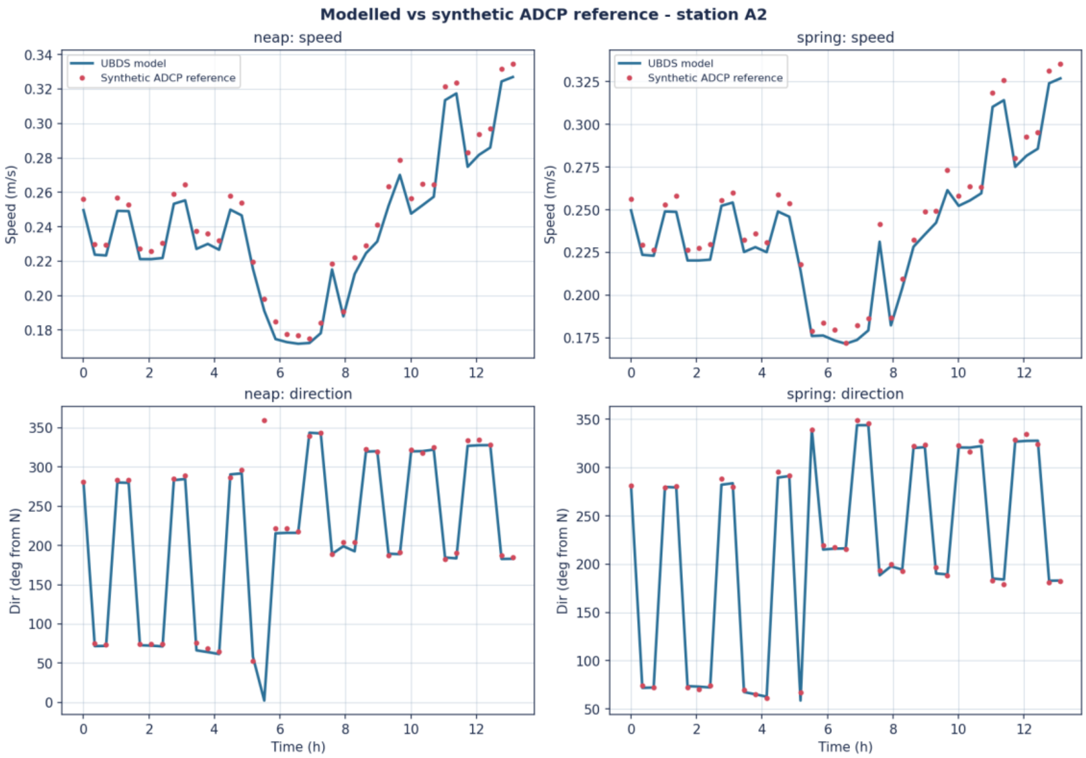


Figure. Modelled current speed & direction at station A2 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.

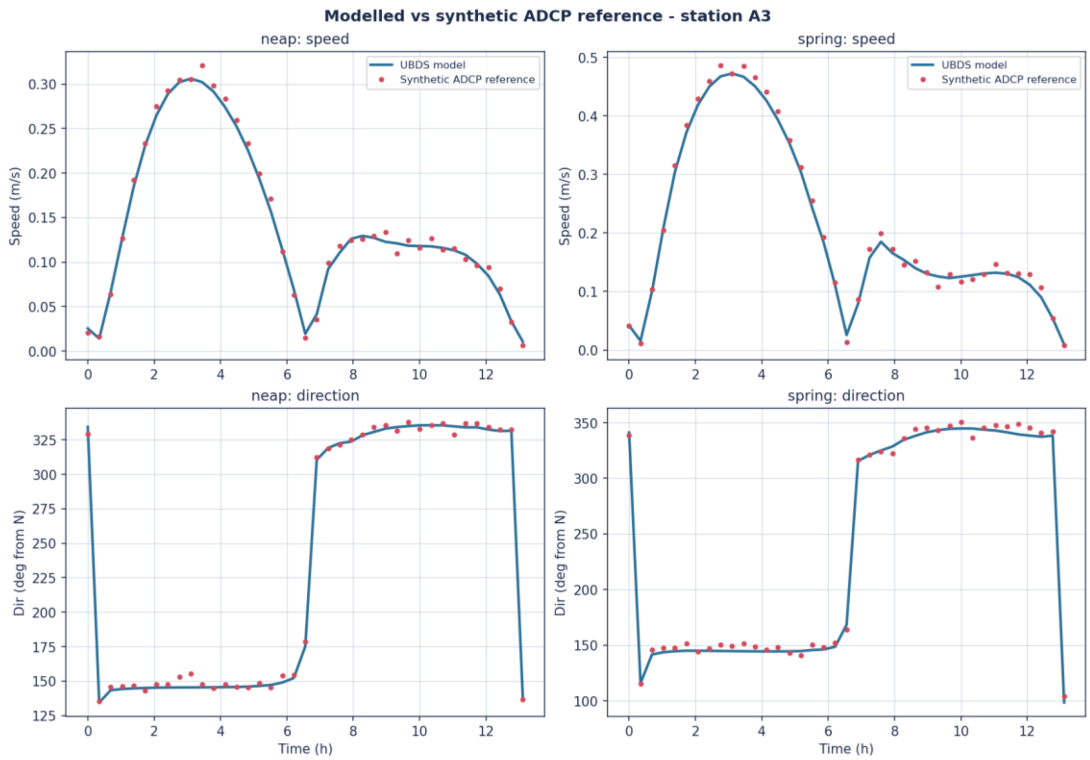


Figure. Modelled current speed & direction at station A3 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.

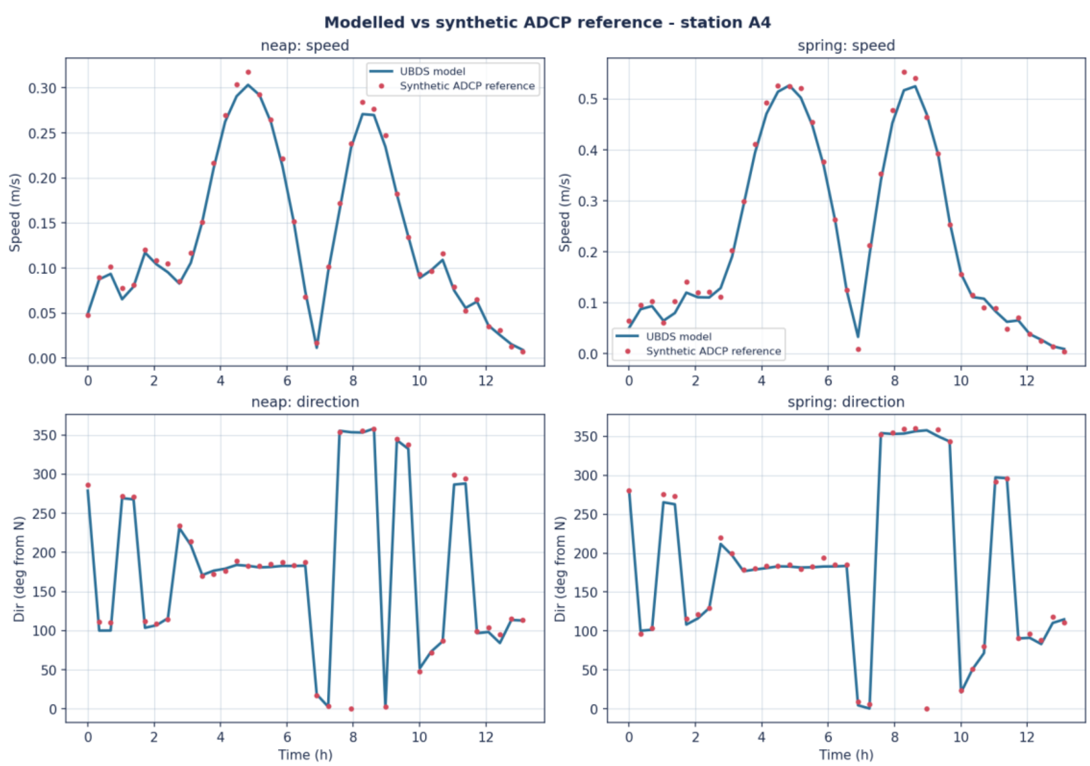


Figure. Modelled current speed & direction at station A4 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.

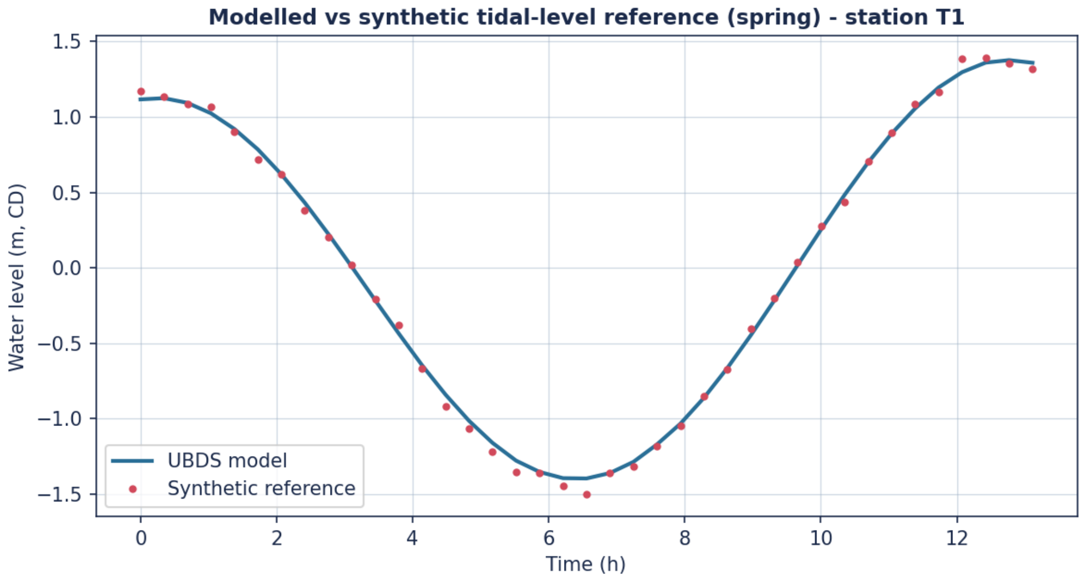


Figure. Modelled tidal level (spring tide) against a synthetic reference series derived from the model solution with an imposed bias and noise. Not an independent measurement.

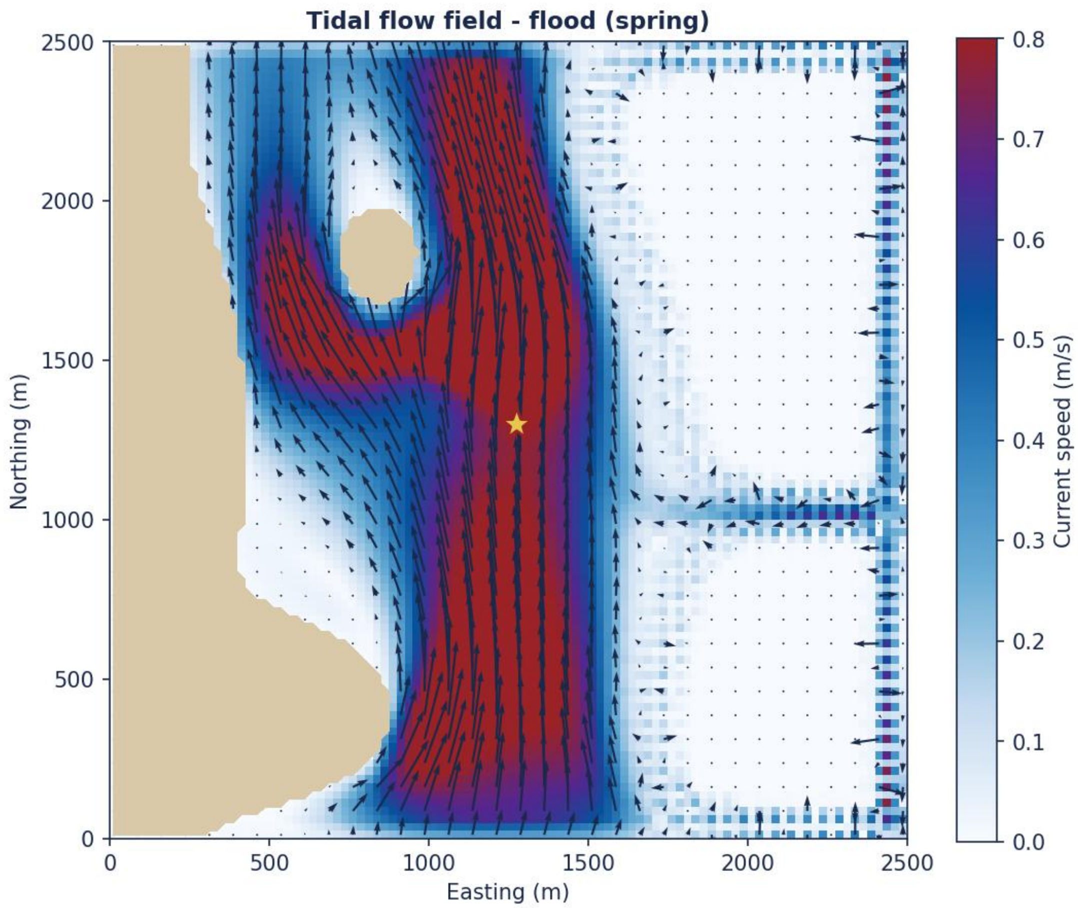


Figure. Depth-averaged tidal flow field at flood (spring tide).

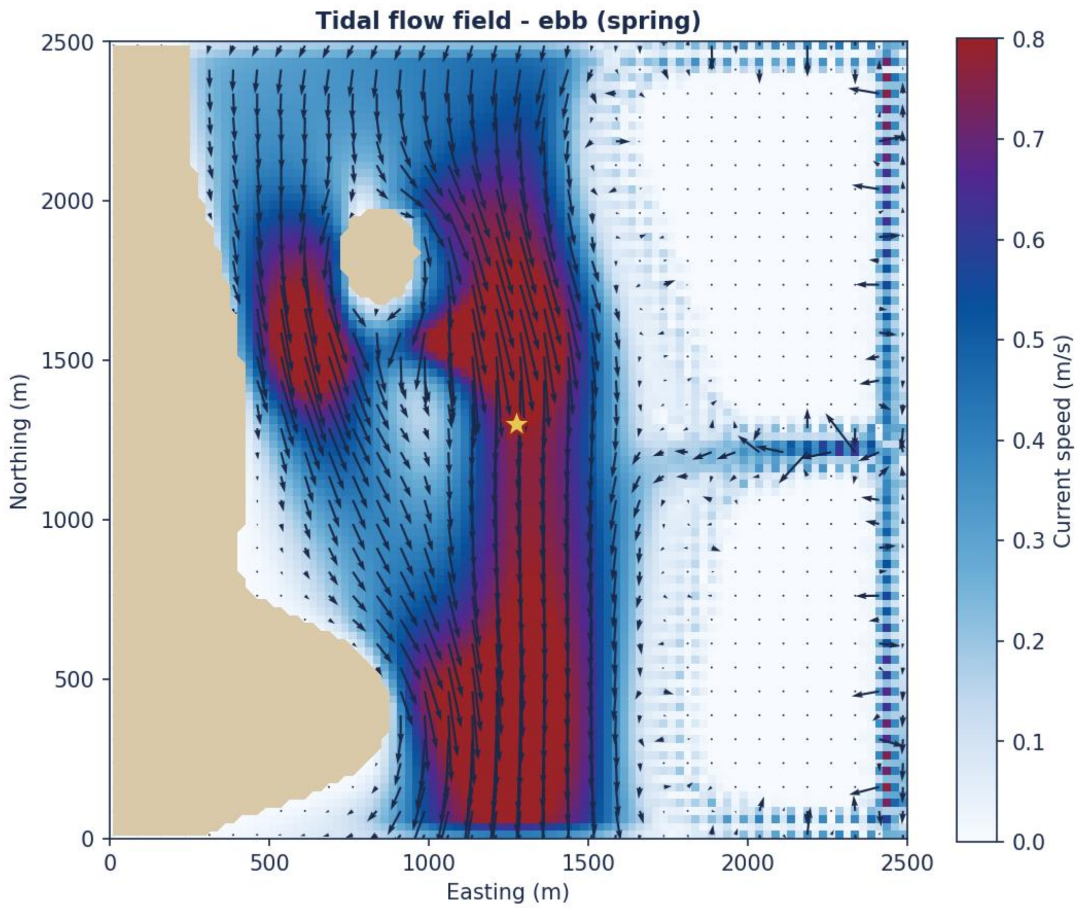


Figure. Depth-averaged tidal flow field at ebb (spring tide).

Table. Sigma-layer schemes (% of water-column depth per layer).

layer	pct_14	pct_9	pct_4
L1 (bed)	0.1	1.0	15.0
L2	0.5	1.2	15.0
L3	0.6	1.5	35.0
L4	0.8	2.5	35.0
L5	1.0	4.5	-
L6	1.5	9.3	-
L7	2.0	15.0	-
L8	3.0	25.0	-
L9	4.5	40.0	-
L10	6.5	-	-
L11	8.5	-	-
L12	11.0	-	-
L13	20.0	-	-
L14	40.0	-	-

A note on what the skill scores below do and do not show. Bahia Azul is a fictitious site, so no measured currents exist for it. The 'ADCP' series is the model solution itself, perturbed by a 3 per cent proportional bias and random noise scaled to the local signal amplitude. The Willmott index and

Nash-Sutcliffe efficiency reported here therefore quantify that imposed perturbation and confirm that the skill-metric pipeline behaves correctly. They are NOT evidence that the hydrodynamic engine reproduces field measurements, and they must not be read as calibration against observations. The independent evidence for the solver is in the Validation section, which compares it against published experimental data and analytical laws.

Table. Model skill against the synthetic ADCP reference series. The reference is the model solution plus an imposed bias and noise, so near-unity scores are expected by construction and are not a field validation.

station	tide	peak_speed_ms	rmse_speed_ms	willmott_d	nse
A0	neap	0.551	0.011	0.999	0.996
A0	spring	0.848	0.02	0.998	0.993
A1	neap	0.607	0.013	0.999	0.995
A1	spring	0.975	0.024	0.998	0.994
A2	neap	0.327	0.008	0.992	0.967
A2	spring	0.327	0.007	0.992	0.968
A3	neap	0.306	0.007	0.998	0.994
A3	spring	0.472	0.011	0.999	0.995
A4	neap	0.303	0.007	0.999	0.995
A4	spring	0.526	0.014	0.999	0.994

4 Near-Field Initial Dilution

The near-field initial dilution was computed with the UBDS integral jet/plume engine for both brine types, all flow scenarios and a range of tidal velocities (slack, neap and spring). Dilution is the bulk (flux-averaged, top-hat) dilution, and salinity is reported at first contact with the seabed; every run in the table below is verified to terminate on seabed contact.

Validity of the near-field results. The inclined dense-jet benchmark spans 9.0 <= Fr <= 33.3 (7 cases), which was chosen to cover the Froude numbers this study actually runs at rather than a convenient band above them. All 18 runs below lie inside that span: the RO concentrate at Fr 24.0-27.9 and the denser saturated-cavern brine, which governs the seabed assessment, at Fr 9.2-10.7. The diffuser stages ports in with flow, holding the per-port exit velocity between 4.57 and 5.33 m/s. Coverage is not the same as agreement: across the benchmark the return distance is inside its published band in 7/7 cases and the return-point dilution in 6/7, but the terminal rise is inside its band in only 1/7, and the shortfall grows as Fr falls - that is, it is largest at the cavern-brine Froude numbers. Section 9 quantifies it. The greatest predicted plume rise is 6.4 m in 18 m of water (36% of the water column), so the free surface, which the model does not impose, is not approached.

Table. Near-field initial-dilution results. Every run terminates on seabed contact; the termination check is retained in the CSV.

brine	scenario	flow_m3hr	n_ports	tide	velocity_ms	exit_velocity_ms	dilution	seabed_salinity_psu	rise_height_m	impact_distance_m
RO concentrate	S1	300	1	slack	0.08	4.57	58.2	37.04	5.51	8.6
RO concentrate	S1	300	1	neap	0.28	4.57	122.1	36.76	4.31	19.75

RO concentrate	S1	300	1	spring	0.55	4.57	161.6	36.69	3.58	39.24
RO concentrate	S2	700	2	slack	0.08	5.33	67.4	36.97	6.39	10.08
RO concentrate	S2	700	2	neap	0.28	5.33	140.5	36.72	4.97	22.95
RO concentrate	S2	700	2	spring	0.55	5.33	185.9	36.67	4.11	45.51
RO concentrate	S3	1200	4	slack	0.08	4.57	58.2	37.04	5.51	8.6
RO concentrate	S3	1200	4	neap	0.28	4.57	122.1	36.76	4.31	19.75
RO concentrate	S3	1200	4	spring	0.55	4.57	161.6	36.69	3.58	39.24
Saturated cavern brine	S1	300	1	slack	0.08	4.57	14.3	51.4	2.65	3.35
Saturated cavern brine	S1	300	1	neap	0.28	4.57	28.7	43.95	2.36	3.58
Saturated cavern brine	S1	300	1	spring	0.55	4.57	44.3	41.32	2.11	5.95
Saturated cavern brine	S2	700	2	slack	0.08	5.33	16.5	49.48	3.03	3.95
Saturated cavern brine	S2	700	2	neap	0.28	5.33	32.8	43.0	2.69	4.21
Saturated cavern brine	S2	700	2	spring	0.55	5.33	50.1	40.76	2.39	6.89
Saturated cavern brine	S3	1200	4	slack	0.08	4.57	14.3	51.4	2.65	3.35
Saturated cavern brine	S3	1200	4	neap	0.28	4.57	28.7	43.95	2.36	3.58
Saturated cavern brine	S3	1200	4	spring	0.55	4.57	44.3	41.32	2.11	5.95

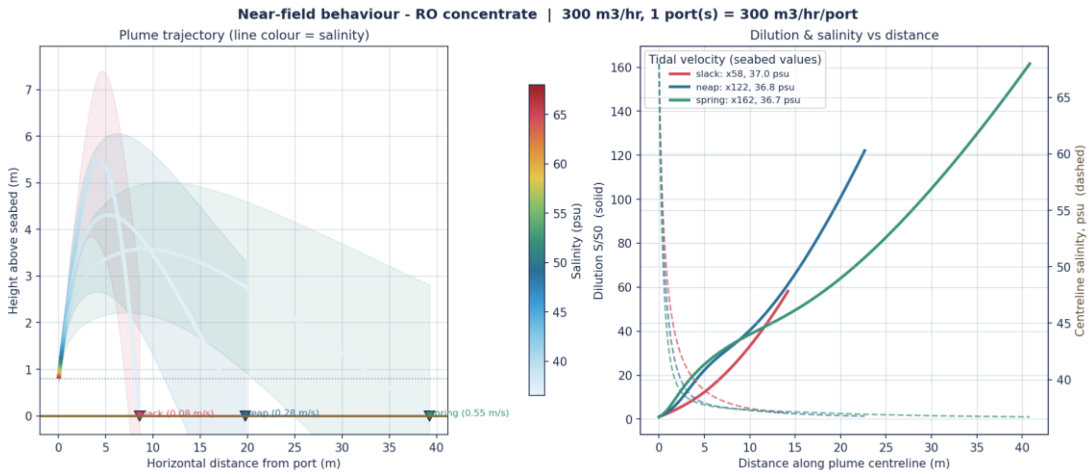


Figure. Near-field trajectory, dilution and salinity - RO concentrate, 300 m3/hr (300 m3/hr per port).

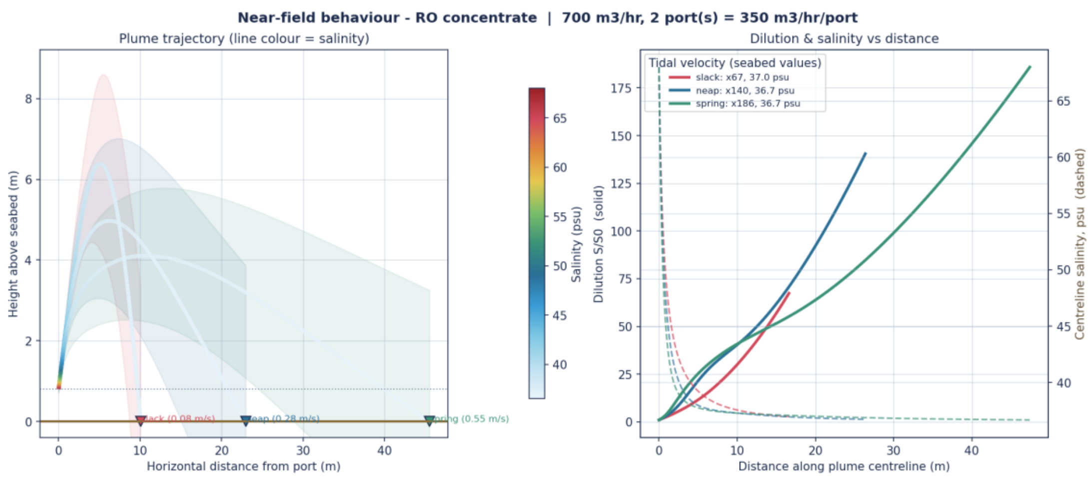


Figure. Near-field trajectory, dilution and salinity - RO concentrate, 700 m3/hr (350 m3/hr per port).

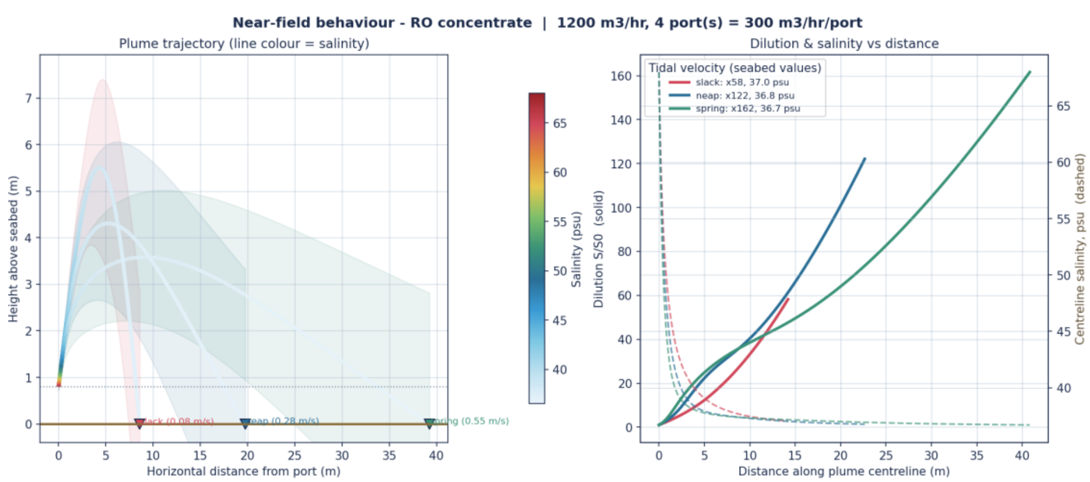


Figure. Near-field trajectory, dilution and salinity - RO concentrate, 1200 m3/hr (300 m3/hr per port).

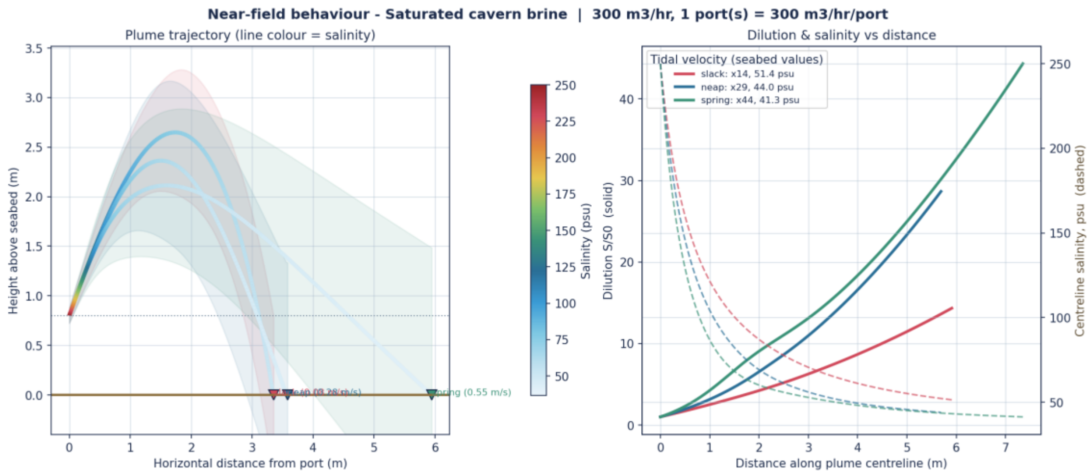
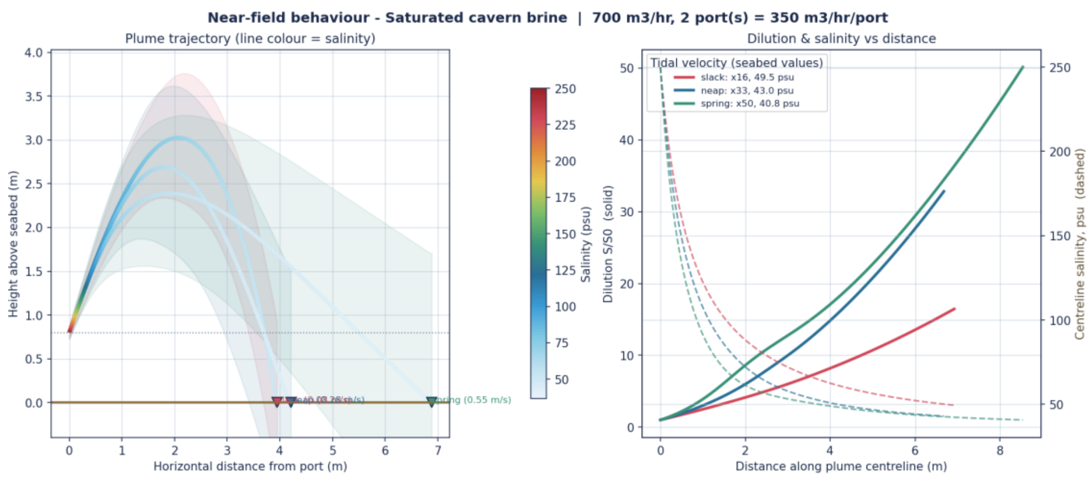


Figure. Near-field trajectory, dilution and salinity - Saturated cavern brine, 300 m3/hr (300 m3/hr per port).



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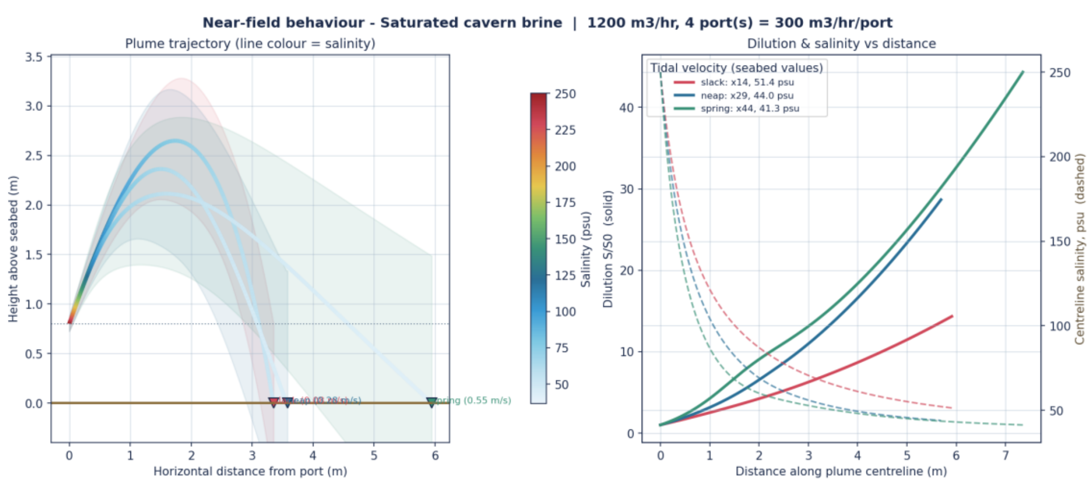


Figure. Near-field trajectory, dilution and salinity - Saturated cavern brine, 1200 m3/hr (300 m3/hr per port).

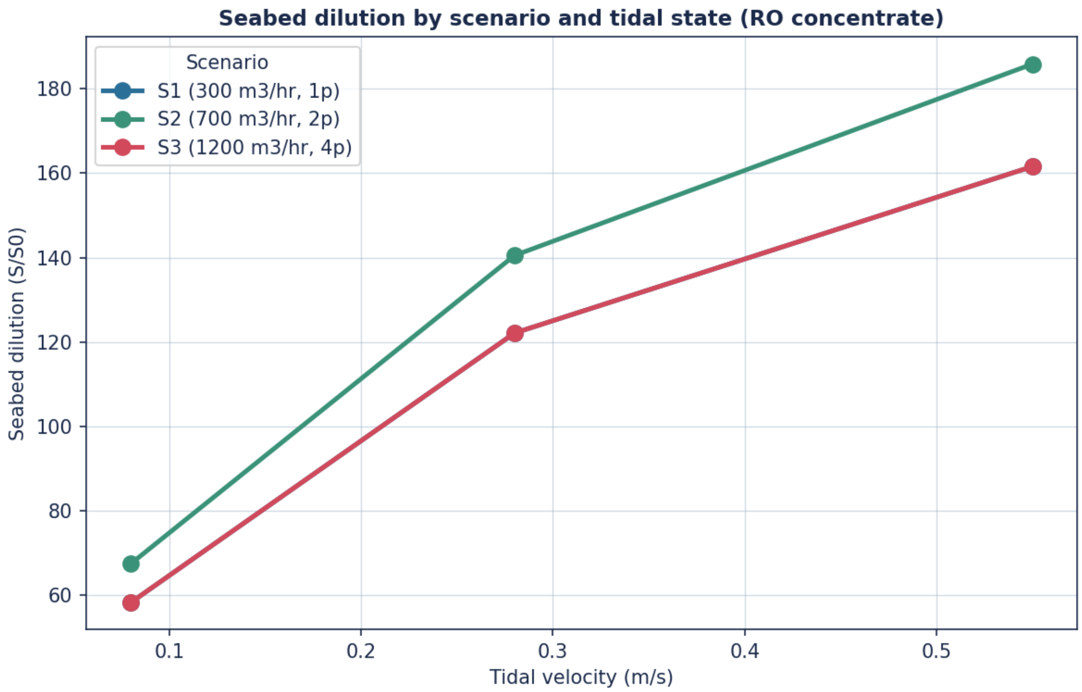


Figure. Seabed initial dilution by flow scenario and tidal state.

5 Medium-Field Dispersion

The medium-field dispersion was simulated with the sigma-layer transport engine driven by the calibrated tidal currents, including a layer-resolution sensitivity test and the near-field-coupled brine source. Maximum-salinity envelopes, tidal-phase snapshots with current vectors, and vertical profiles through the diffuser are presented. Every medium-field run in this section uses the SATURATED CAVERN BRINE source at full flow - the phase-2 worst case, not the RO concentrate of the base case. The medium-field seabed salinities are therefore not comparable with the far-field values of section 6, which are computed for RO concentrate unless a figure states otherwise.

Layer-resolution sensitivity (maximum seabed salinity, psu): 4-layer = 44.2, 9-layer = 44.0, 14-layer = 44.3. The 9-layer scheme was adopted for the production runs. The peak is grid-converged in the vertical.

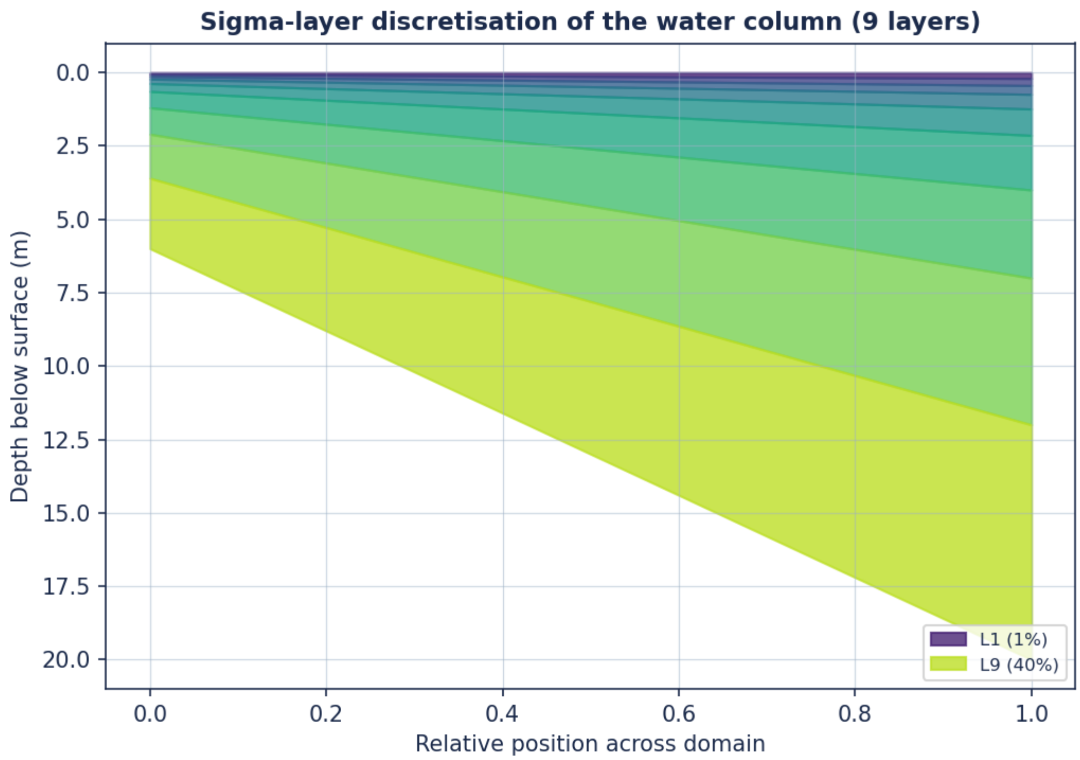


Figure. Variation of the terrain-following sigma layers with water depth (9-layer scheme).

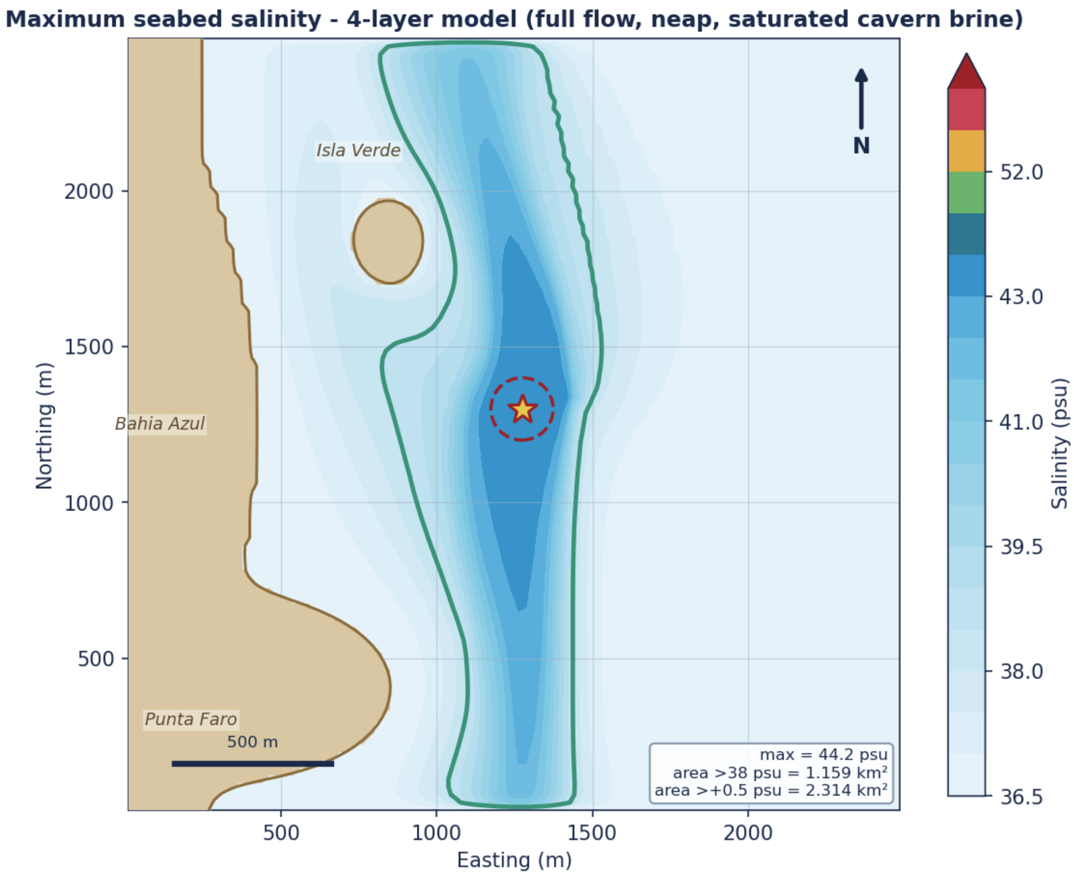


Figure. Maximum seabed salinity - 4-layer model (full flow, neap, saturated cavern brine)

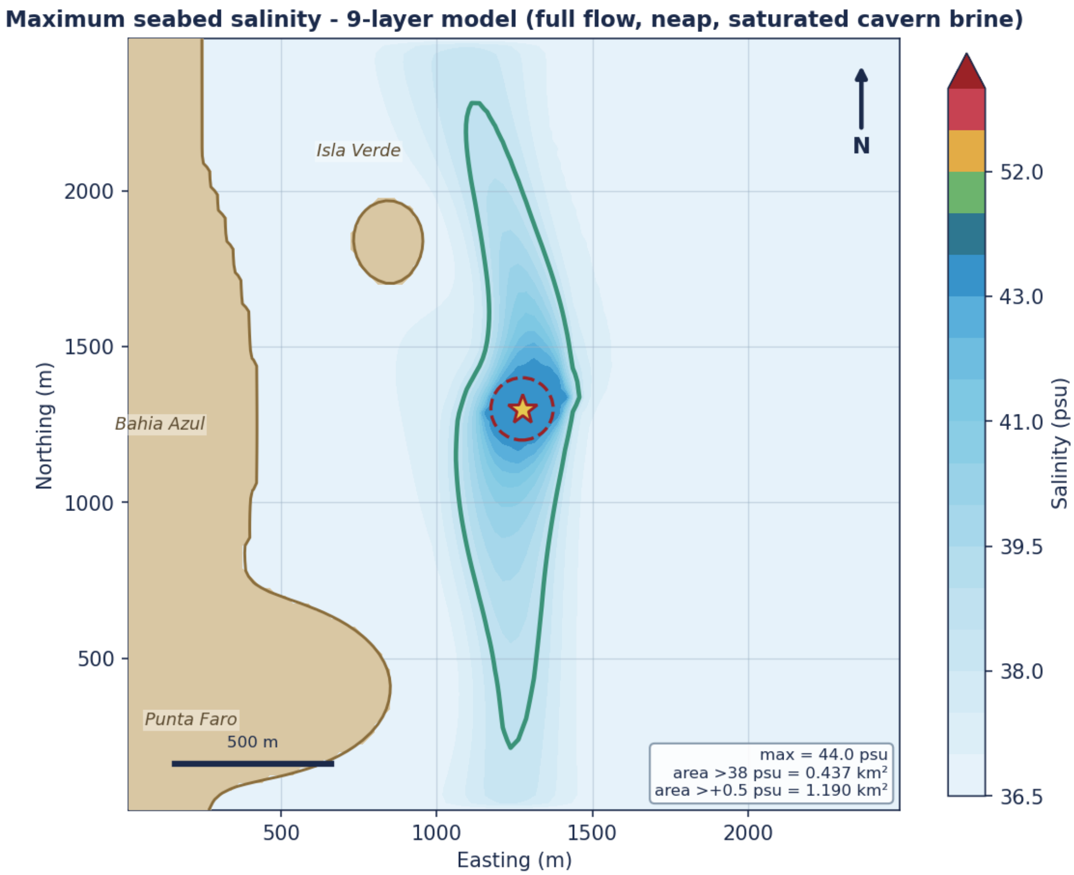


Figure. Maximum seabed salinity - 9-layer model (full flow, neap, saturated cavern brine)

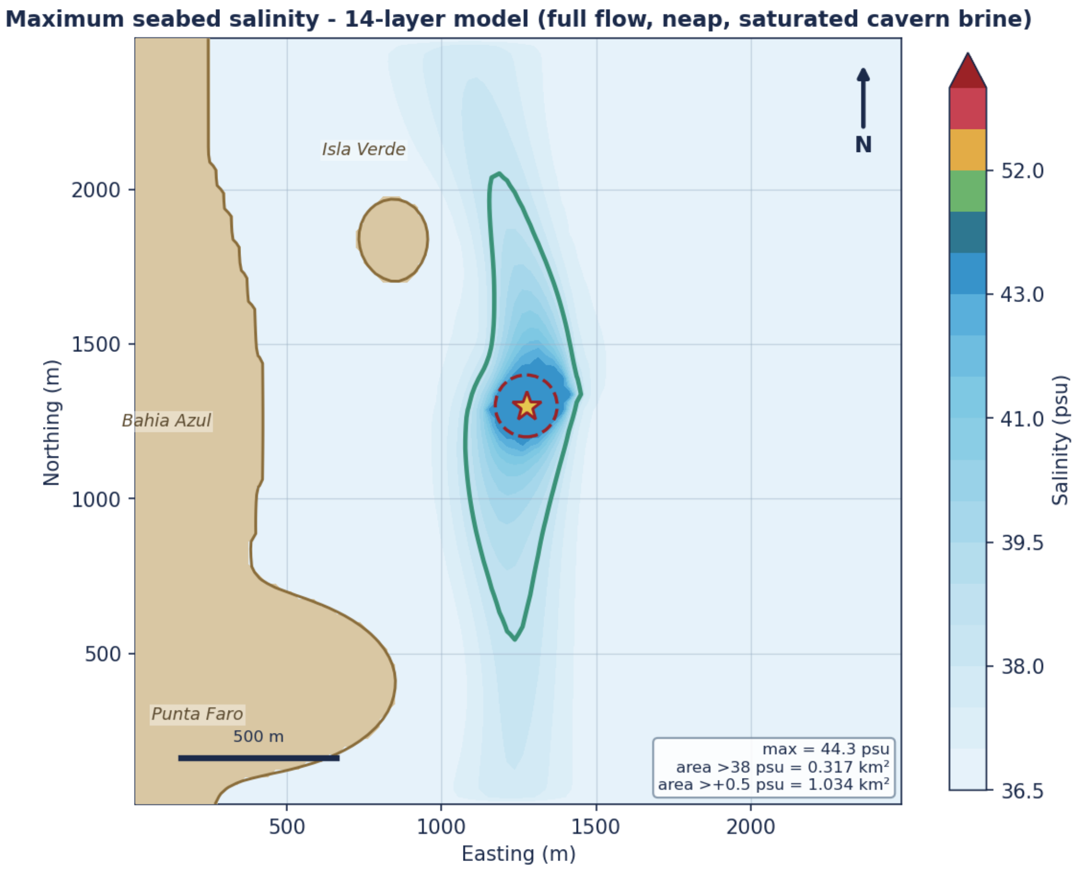


Figure. Maximum seabed salinity - 14-layer model (full flow, neap, saturated cavern brine)

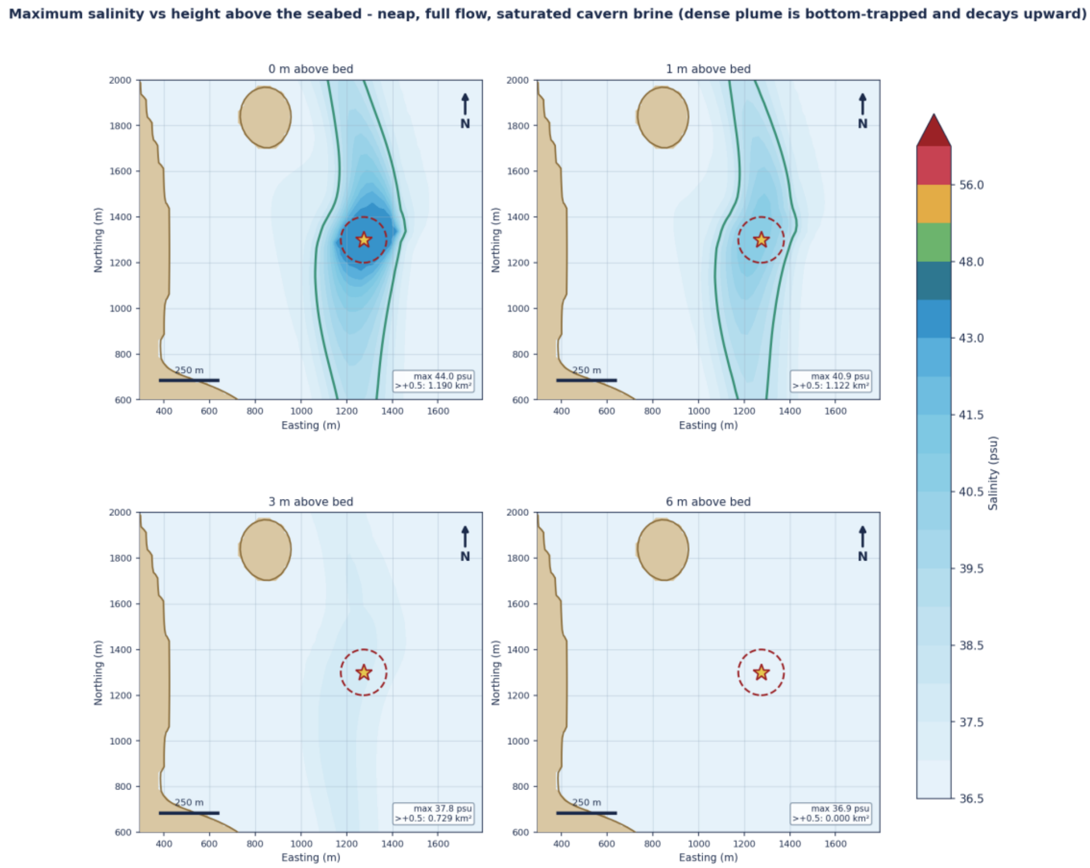


Figure. Maximum salinity vs height above the seabed - neap, full flow, saturated cavern brine (dense plume is bottom-trapped and decays upward)

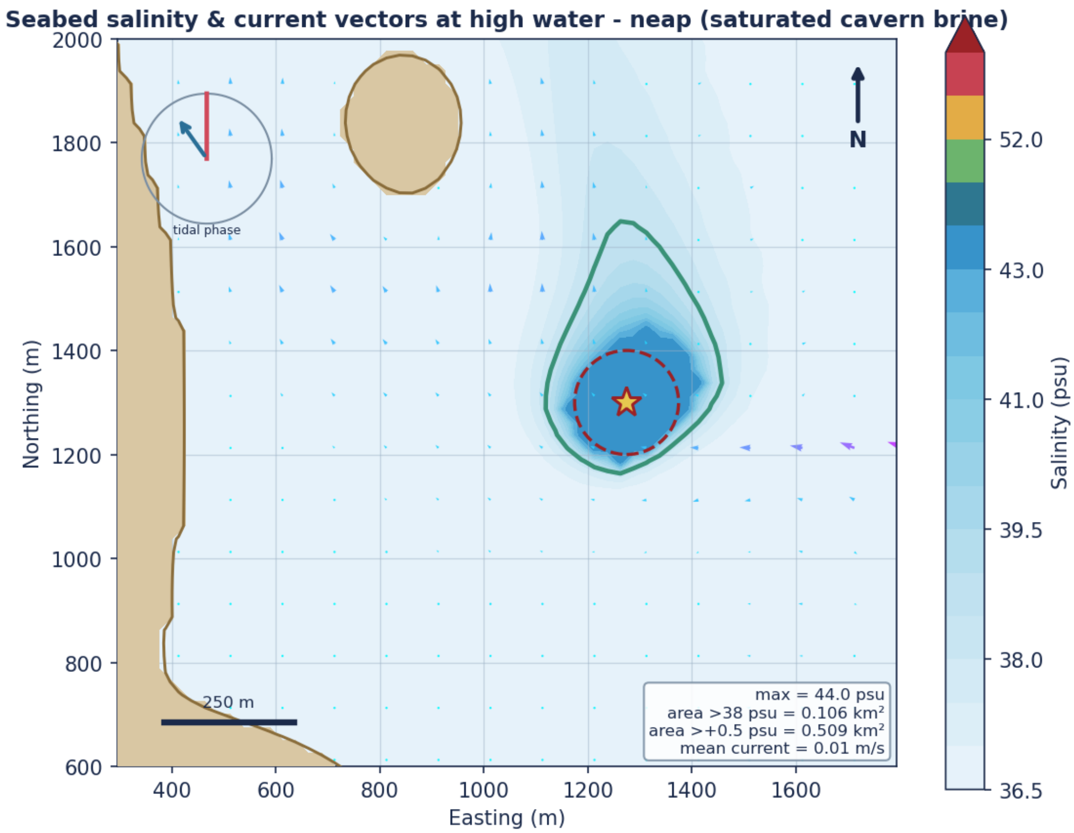


Figure. Seabed salinity & current vectors at high water - neap (saturated cavern brine)

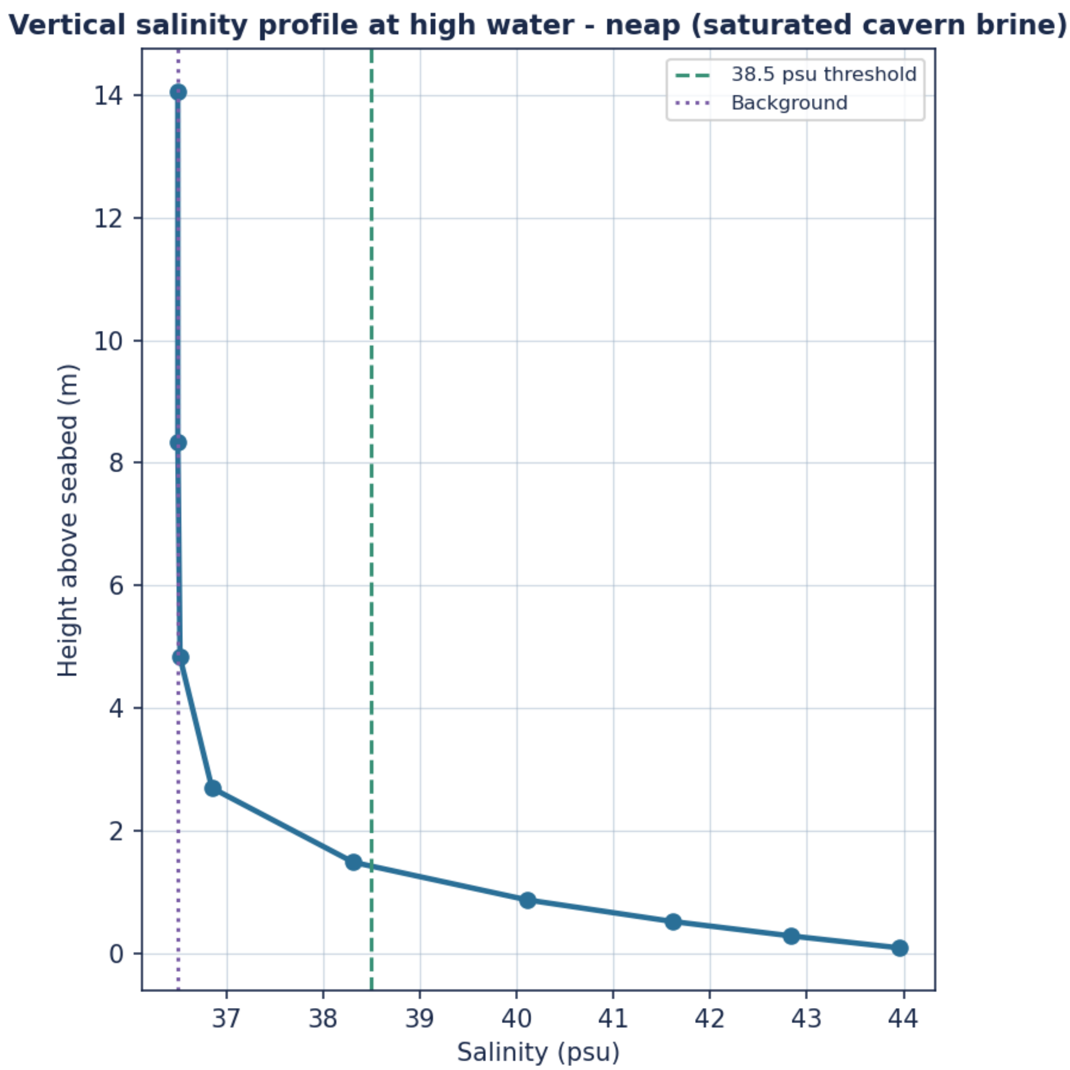


Figure. Vertical salinity profile at high water - neap (saturated cavern brine)

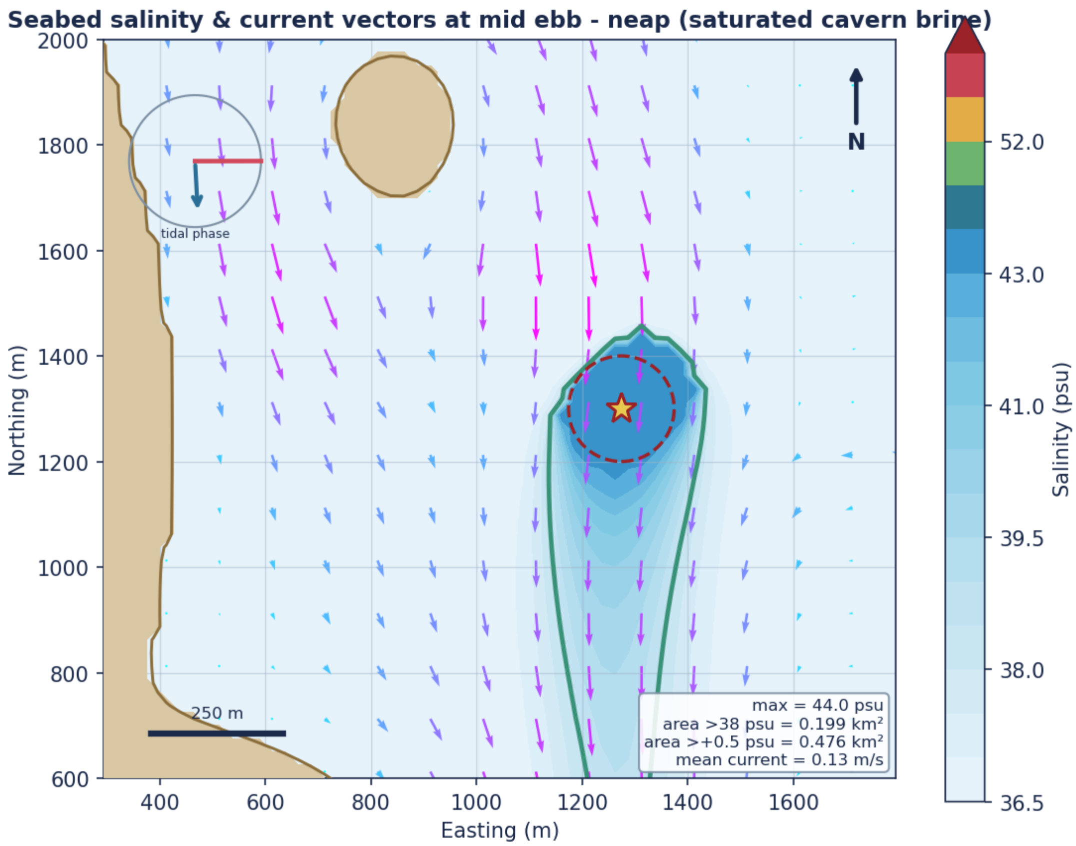


Figure. Seabed salinity & current vectors at mid ebb - neap (saturated cavern brine)

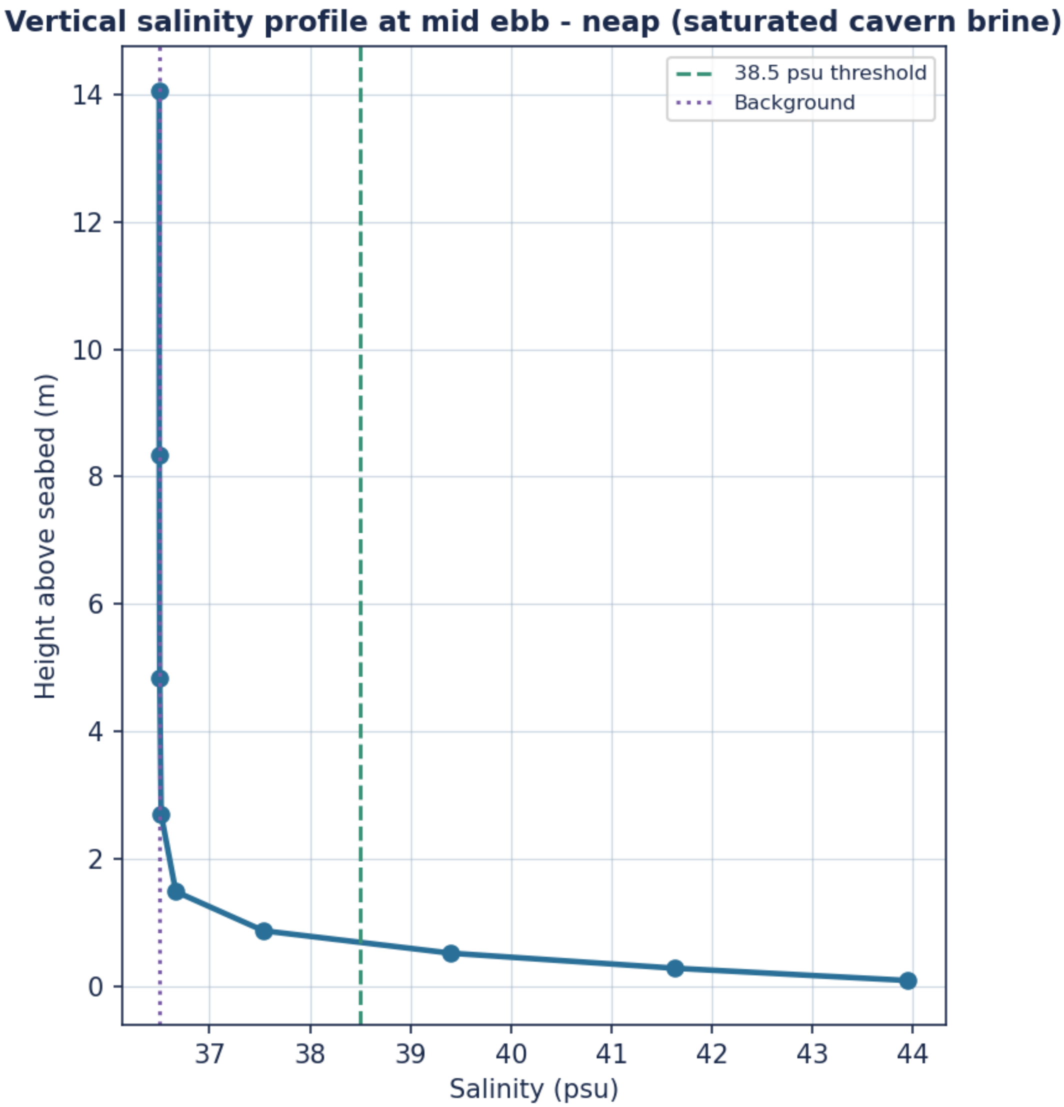


Figure. Vertical salinity profile at mid ebb - neap (saturated cavern brine)

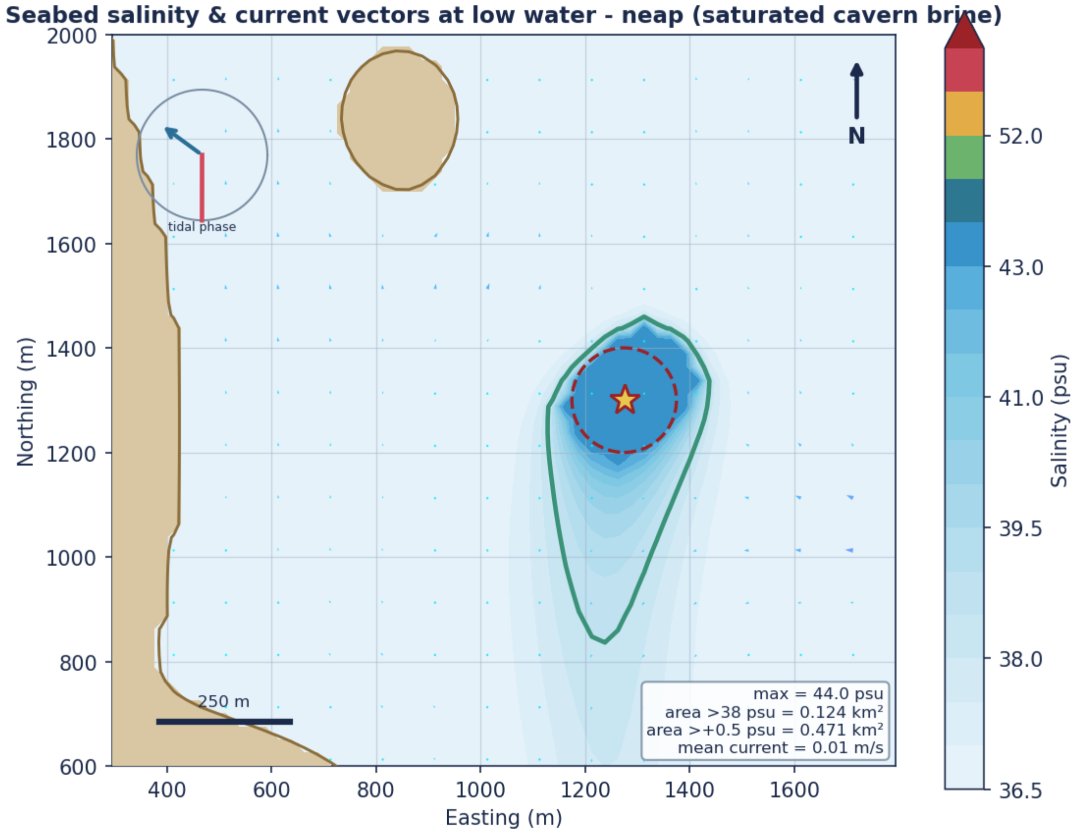


Figure. Seabed salinity & current vectors at low water - neap (saturated cavern brine)

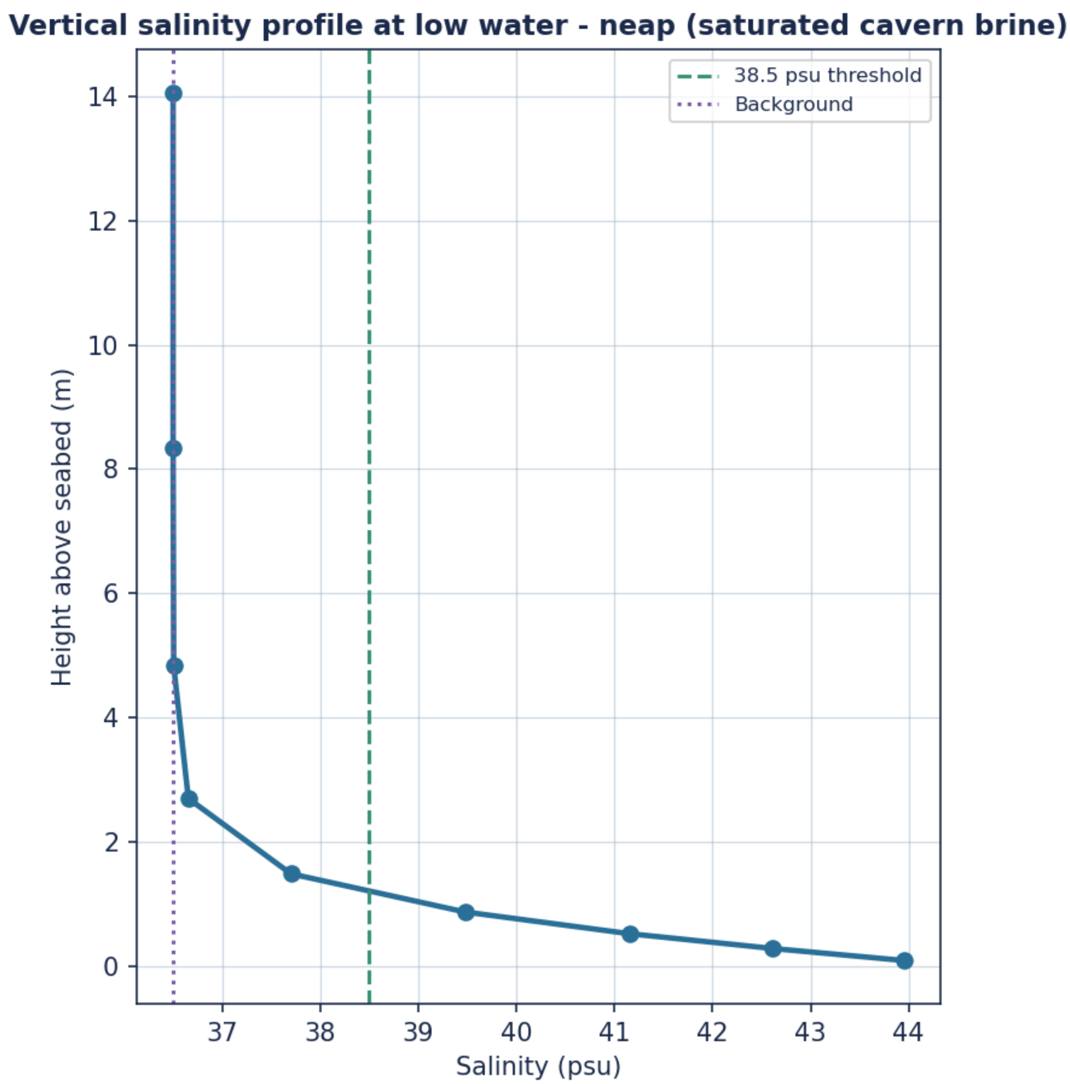


Figure. Vertical salinity profile at low water - neap (saturated cavern brine)

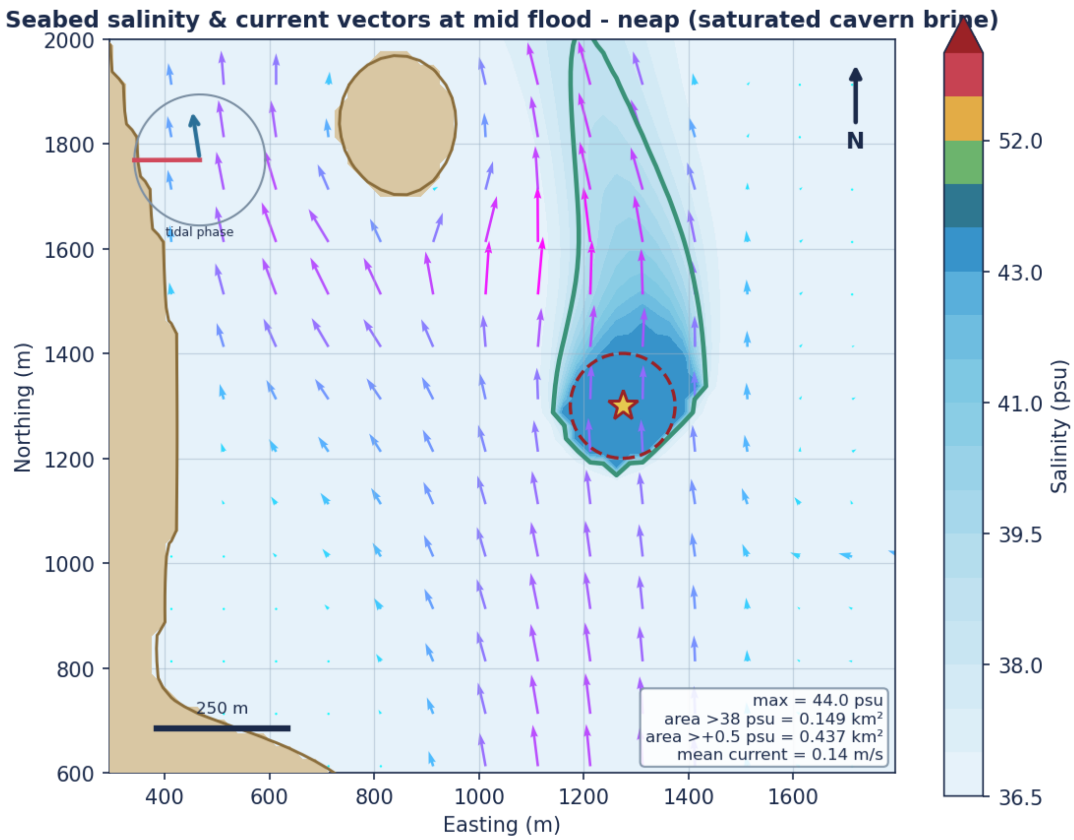


Figure. Seabed salinity & current vectors at mid flood - neap (saturated cavern brine)

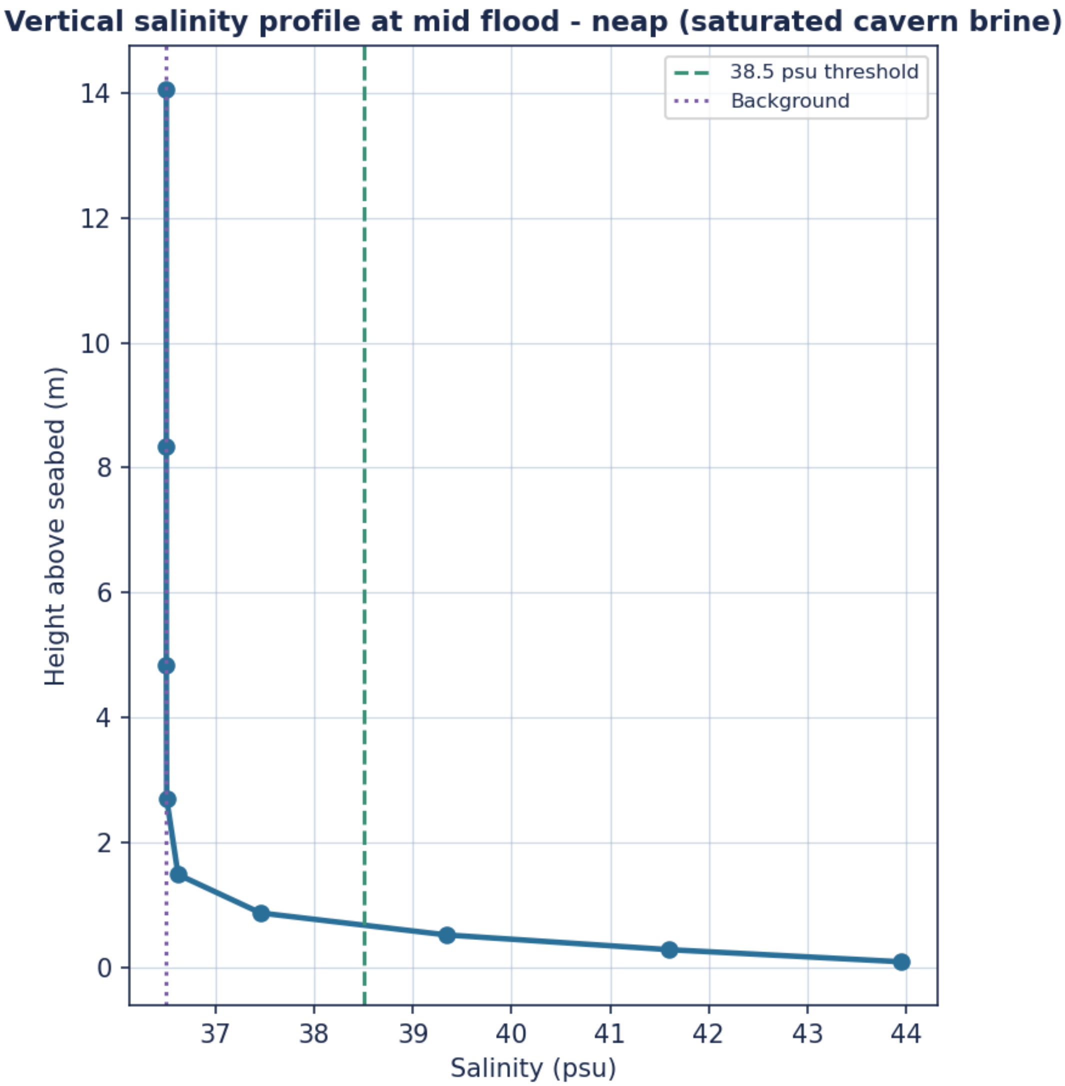


Figure. Vertical salinity profile at mid flood - neap (saturated cavern brine)

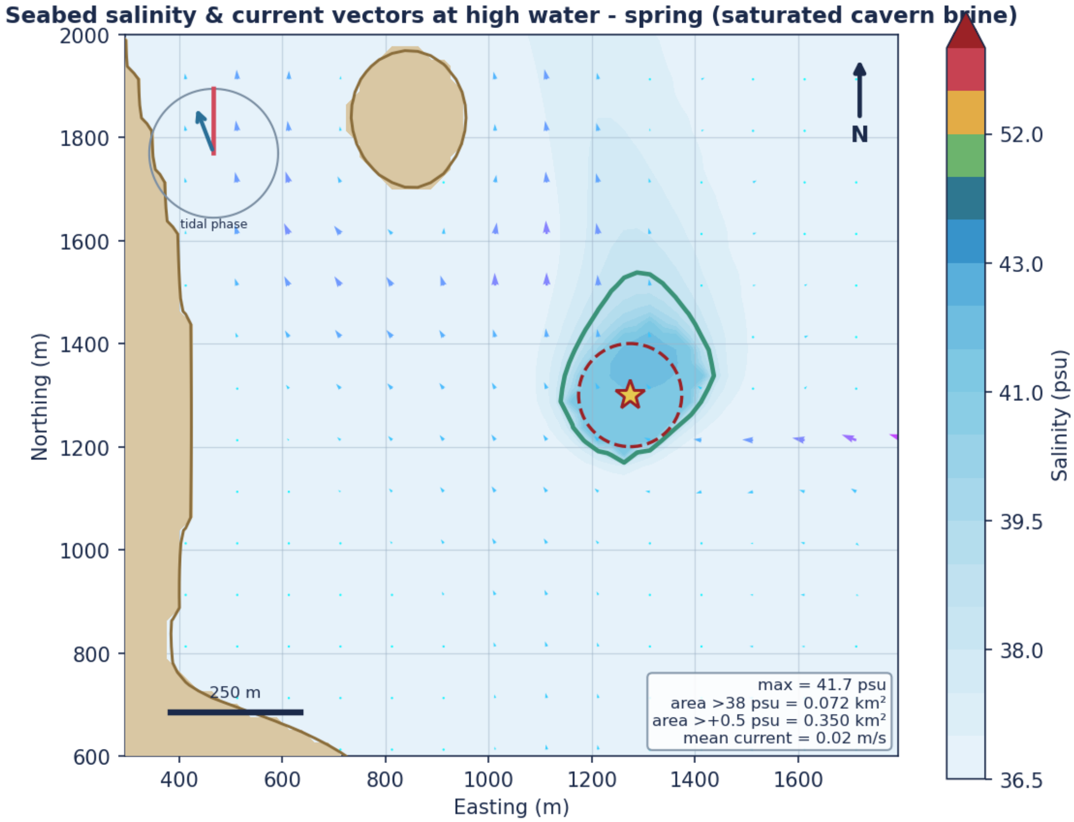


Figure. Seabed salinity & current vectors at high water - spring (saturated cavern brine)

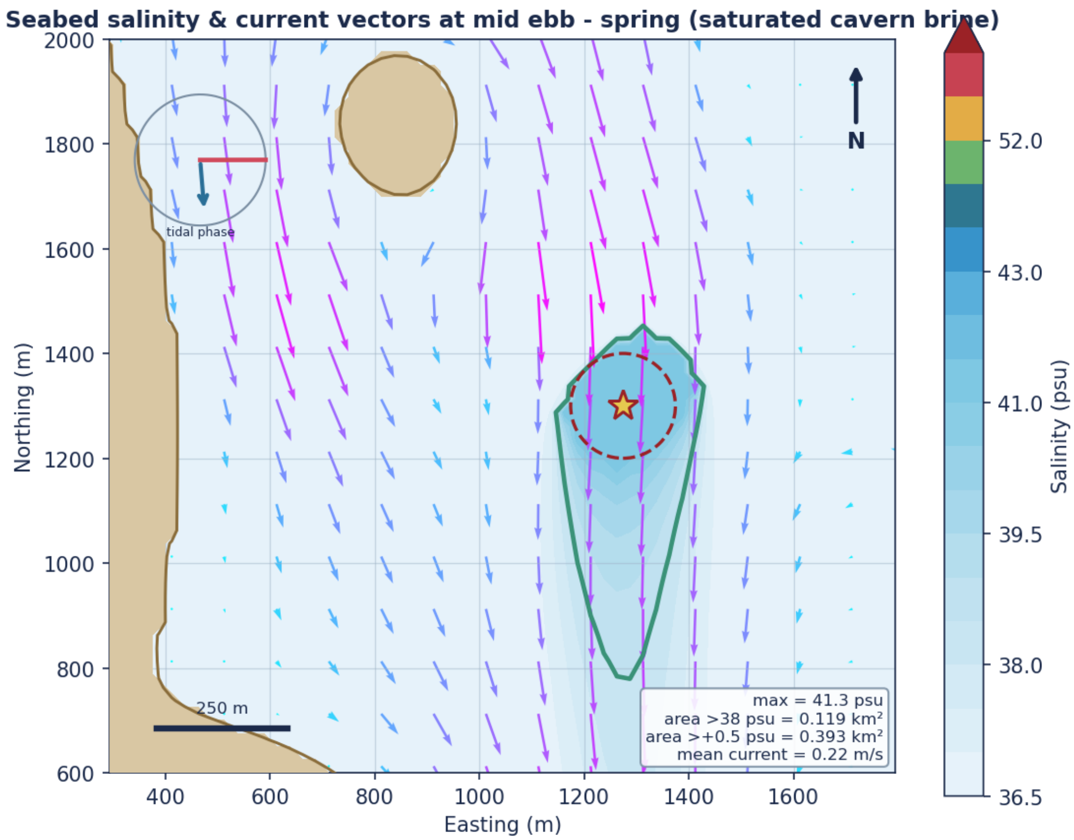


Figure. Seabed salinity & current vectors at mid ebb - spring (saturated cavern brine)

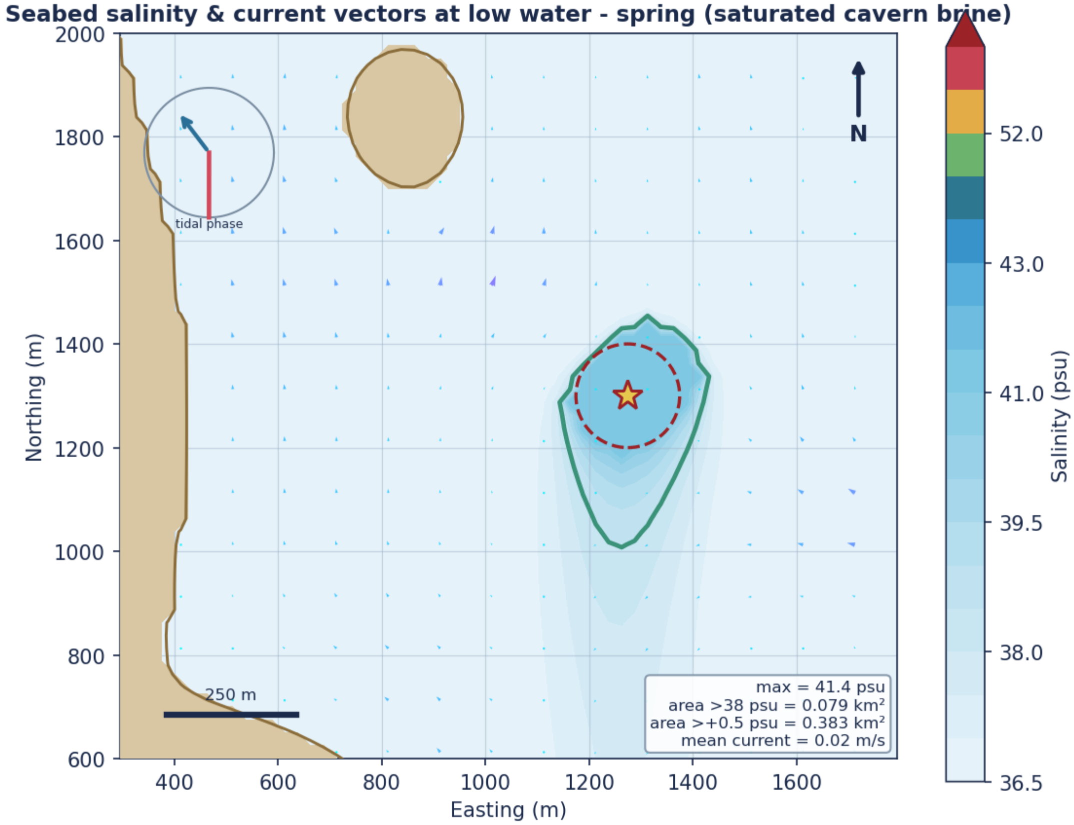


Figure. Seabed salinity & current vectors at low water - spring (saturated cavern brine)

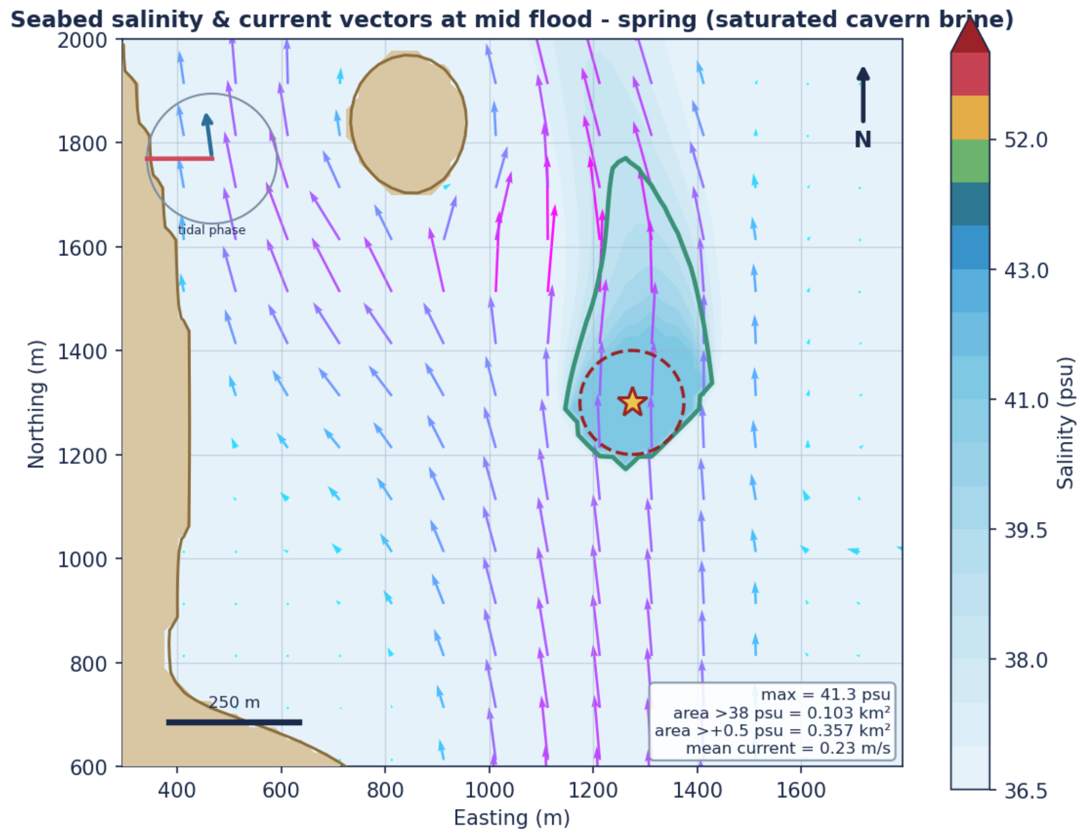


Figure. Seabed salinity & current vectors at mid flood - spring (saturated cavern brine)

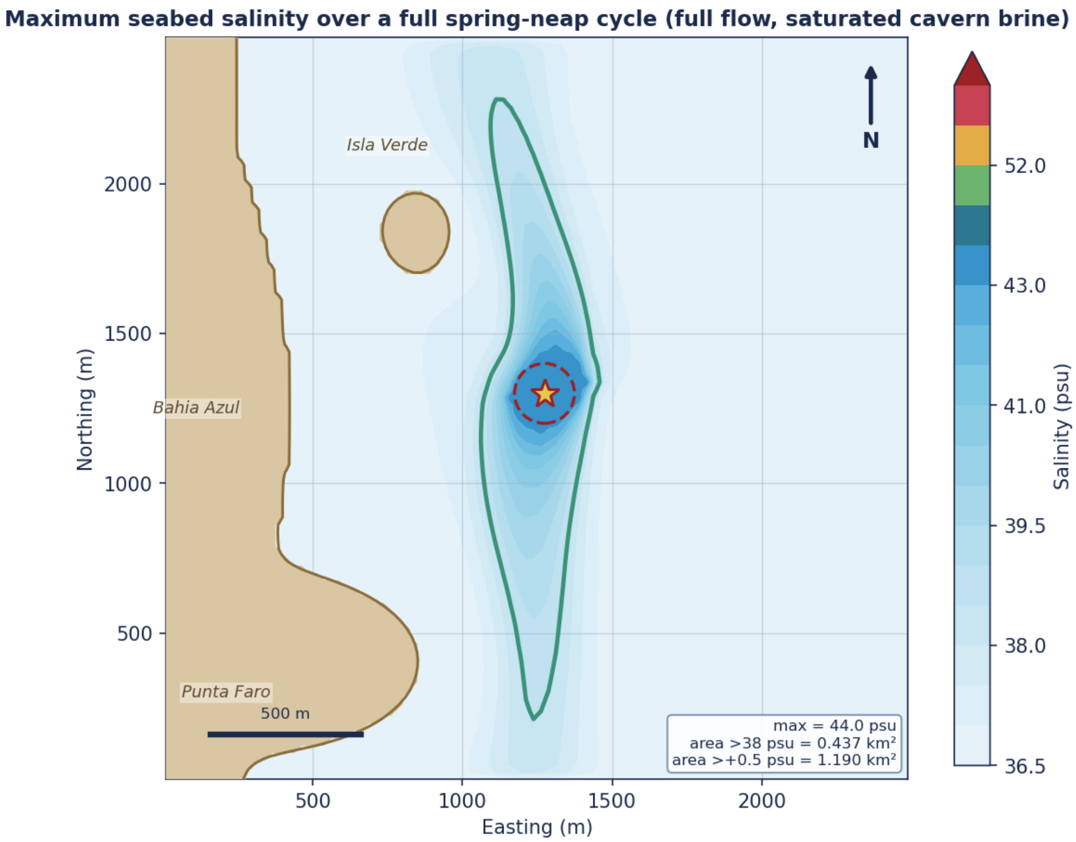


Figure. Maximum seabed salinity over a full spring-neap cycle (full flow, saturated cavern brine)

6 Far-Field Dispersion

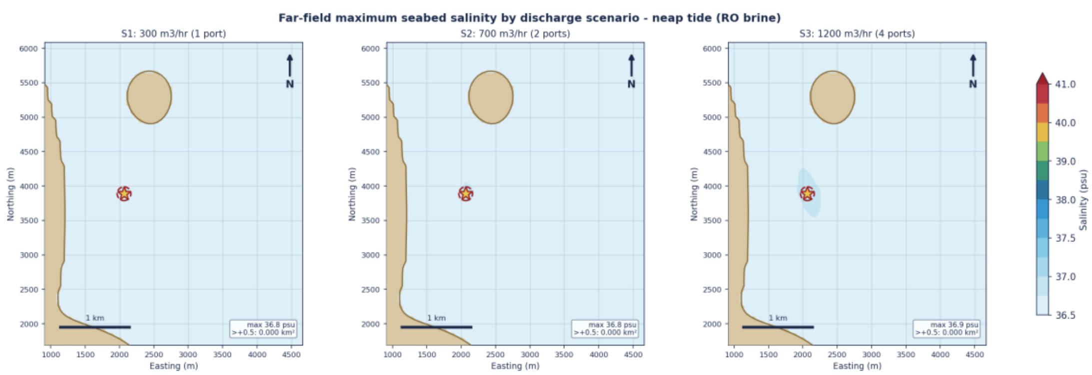


Figure. Far-field maximum seabed salinity by discharge scenario - neap tide (RO brine)

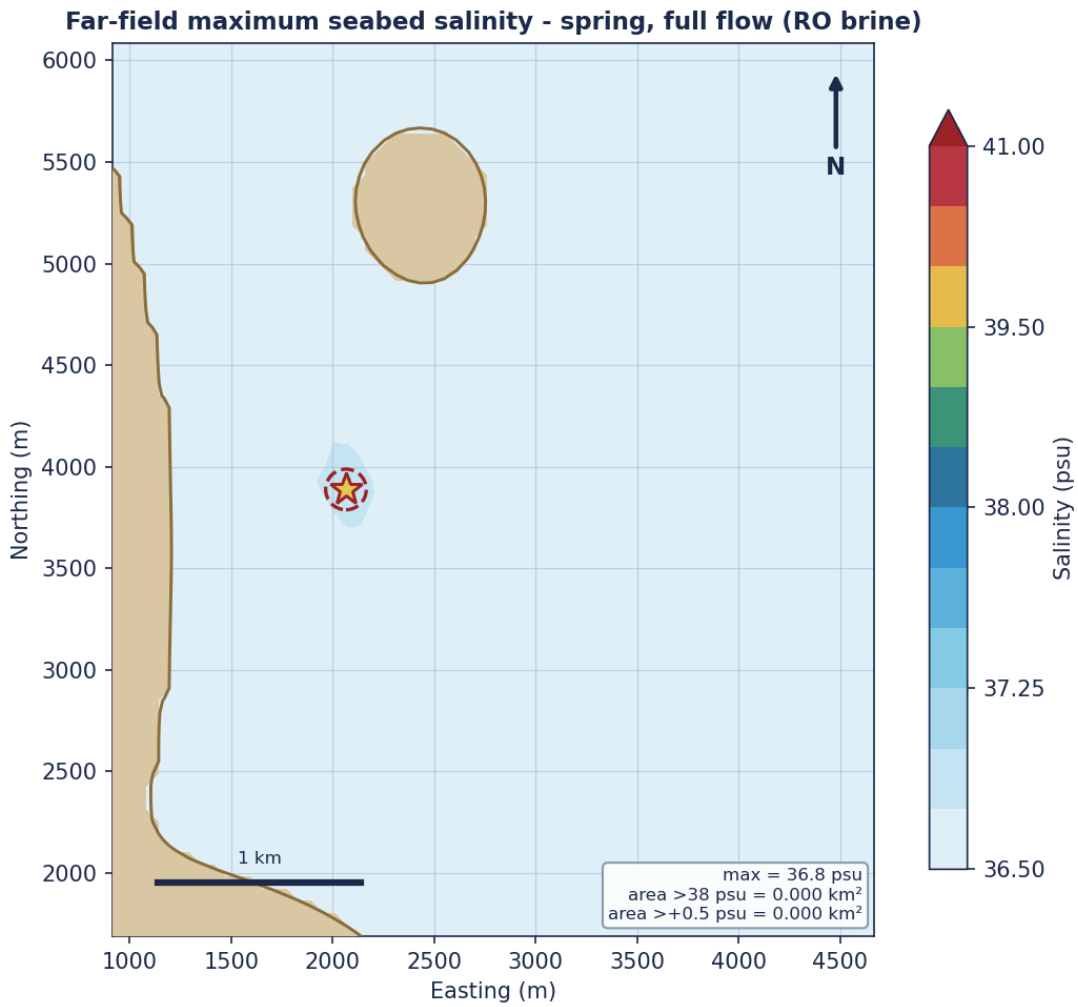


Figure. Far-field maximum seabed salinity - spring, full flow (RO brine)

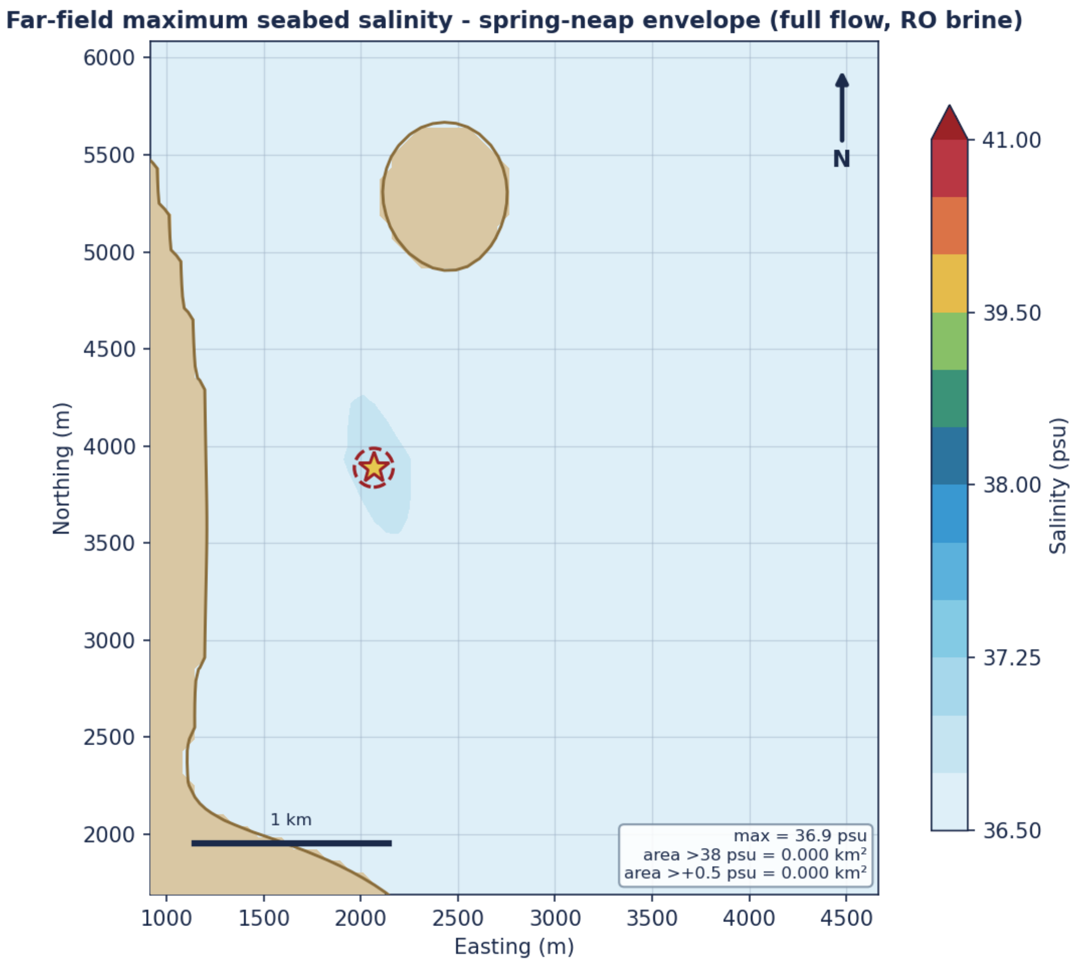


Figure. Far-field maximum seabed salinity - spring-neap envelope (full flow, RO brine)

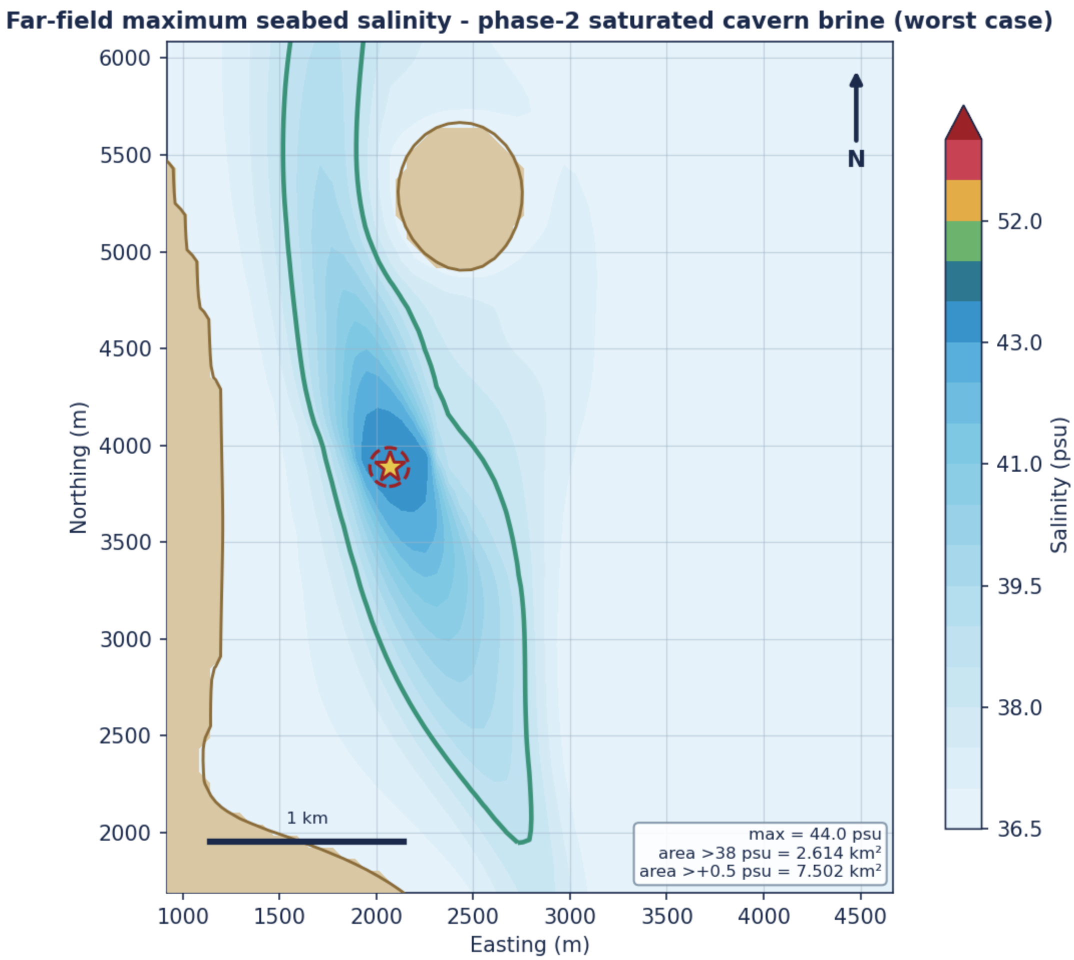


Figure. Far-field maximum seabed salinity - phase-2 saturated cavern brine (worst case)

How to read the zeros in the table below. On the 60 m regional grid the seabed salinity increment peaks at 0.39 psu, below the lowest reported threshold of 0.5 psu, so every influence area evaluates to exactly 0.000 km2 and the mixing-zone radius to 0 m. These are grid-resolved quantities. They do not mean the increment never exceeds 0.5 psu anywhere: the near-field solution reaches 0.54 psu above background at the impact footprint at slack water (section 4), on a scale of metres that a 60 m cell cannot represent. The regional figures therefore bound the far-field footprint, and the near-field solution governs the immediate seabed impact.

Table. Salinity-increment influence areas and diffusion distances per scenario. Areas are resolved on the regional grid; see the note above on what the zeros mean.

scenario	flow_m3hr	area_0p5psu_km2	area_1p0psu_km2	area_1p5psu_km2	diff_dist_parallel_m	diff_dist_transverse_m	mixing_zone_radius_m	max_seabed_salinity_psu
S1	300	0.0	0.0	0.0	0.0	0.0	0.0	36.79
S2	700	0.0	0.0	0.0	0.0	0.0	0.0	36.84
S3	1200	0.0	0.0	0.0	0.0	0.0	0.0	36.89

7 Diffuser Design Optimisation and Layout

The effect of diffuser port discharge angle on near-field dilution was assessed, and a recommended layout selected. Cross-shore salinity and current-speed transects and the marine-constraints overlay support the siting decision.

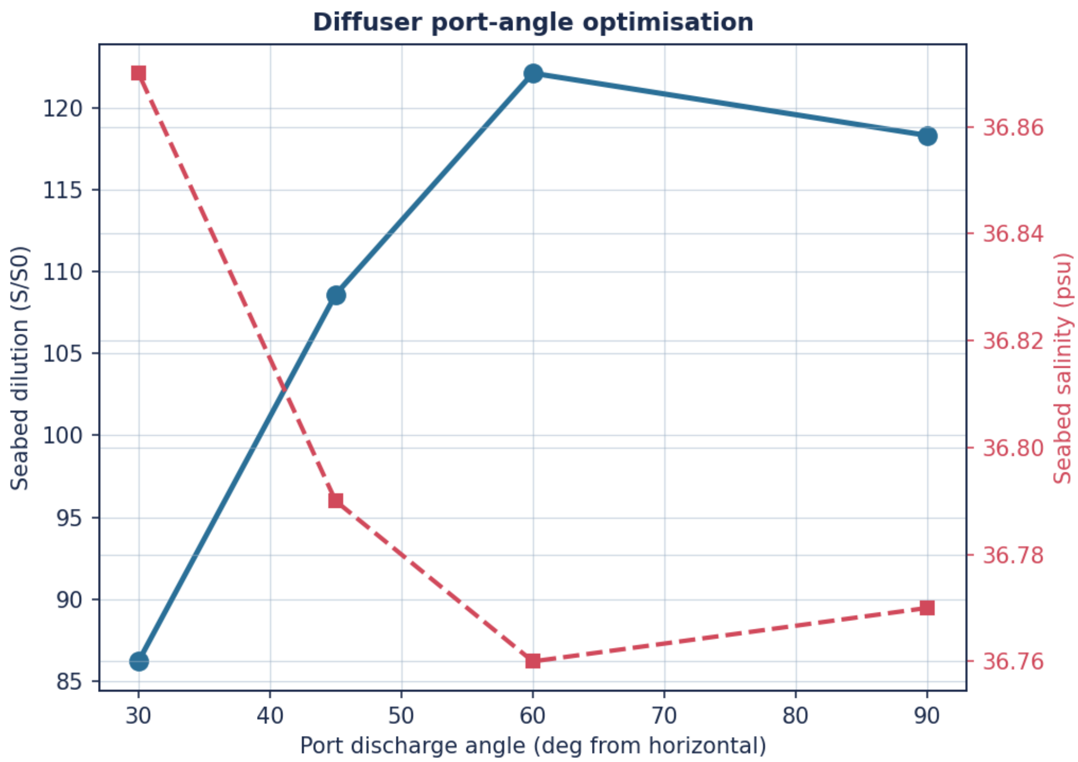


Figure. Diffuser port-angle optimisation: dilution and seabed salinity vs discharge angle.

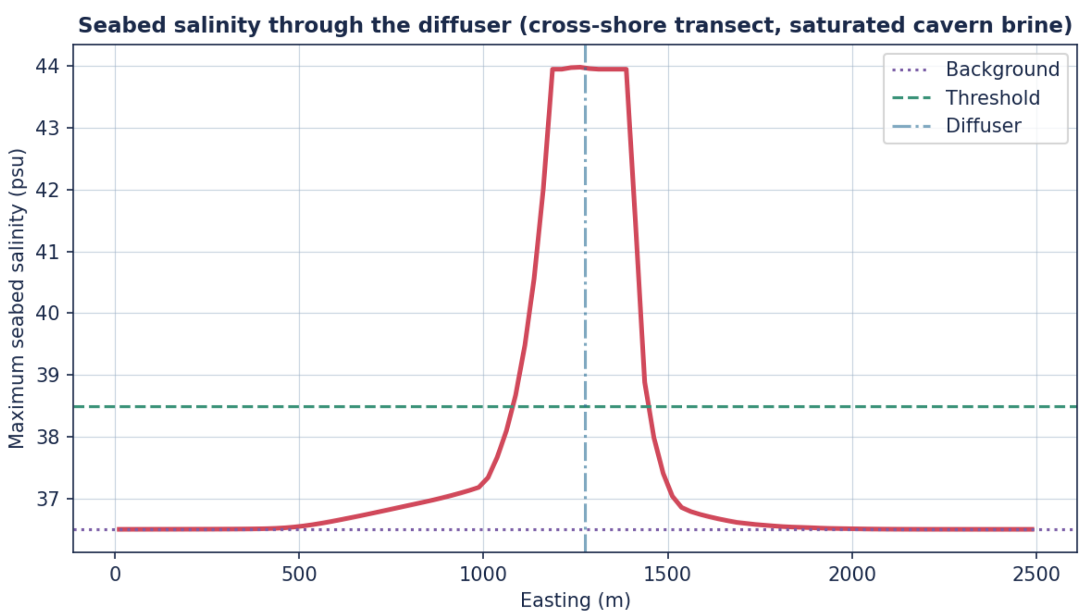


Figure. Maximum seabed salinity along a transect through the diffuser (neap, full flow, saturated cavern brine).

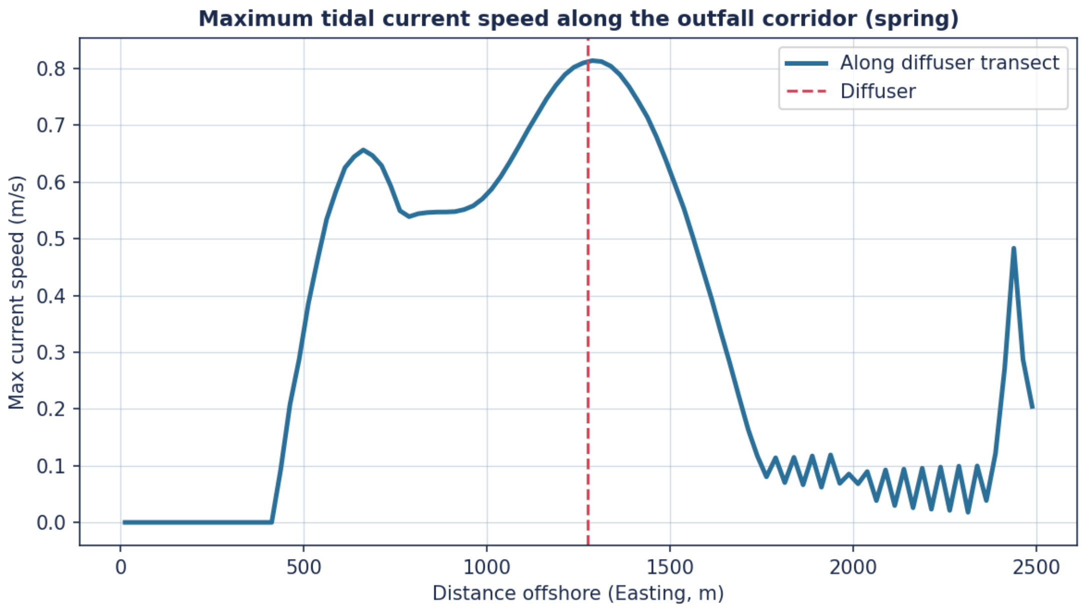


Figure. Maximum current speed along the outfall corridor (spring tide).

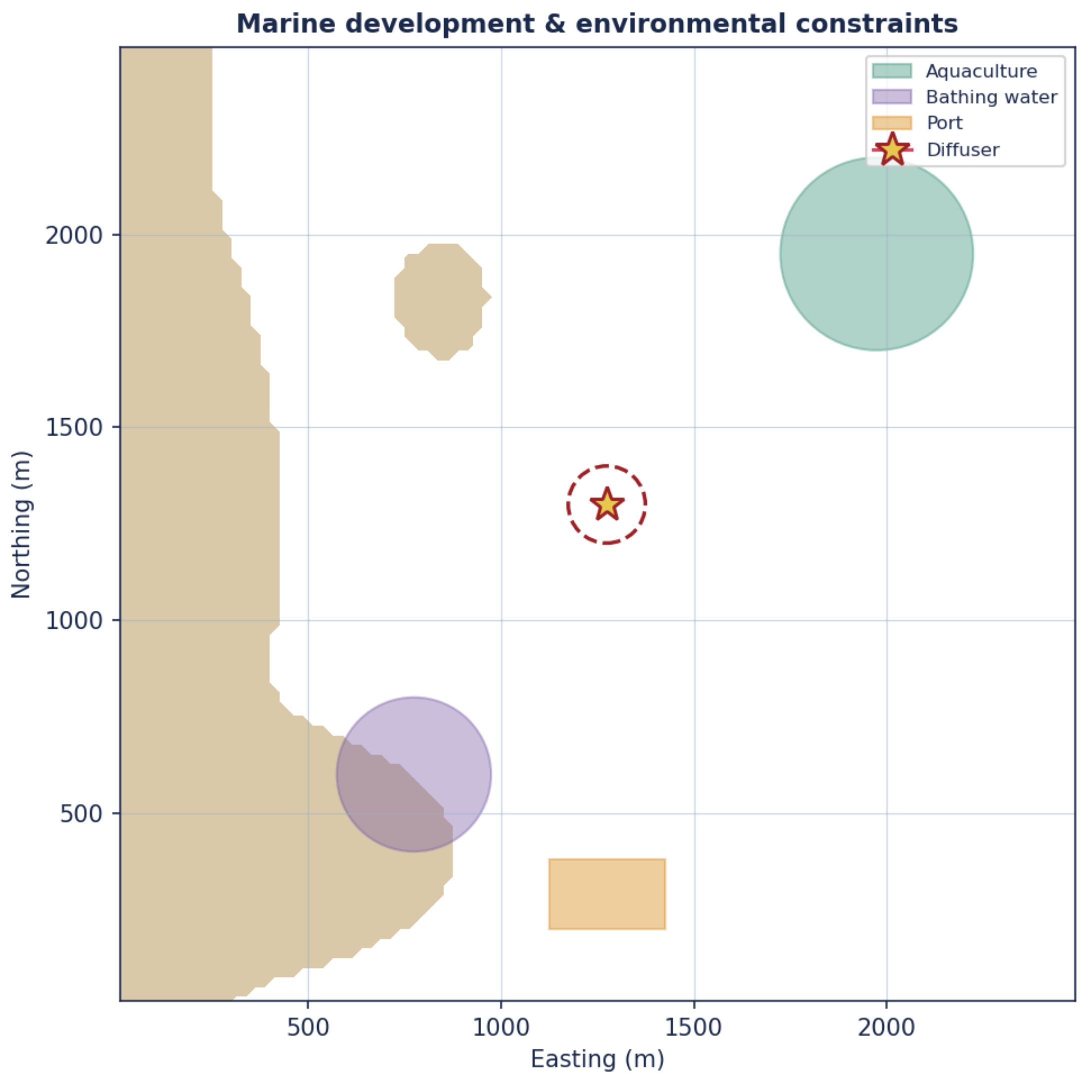


Figure. Distribution of marine development and environmental constraints relative to the mixing zone.

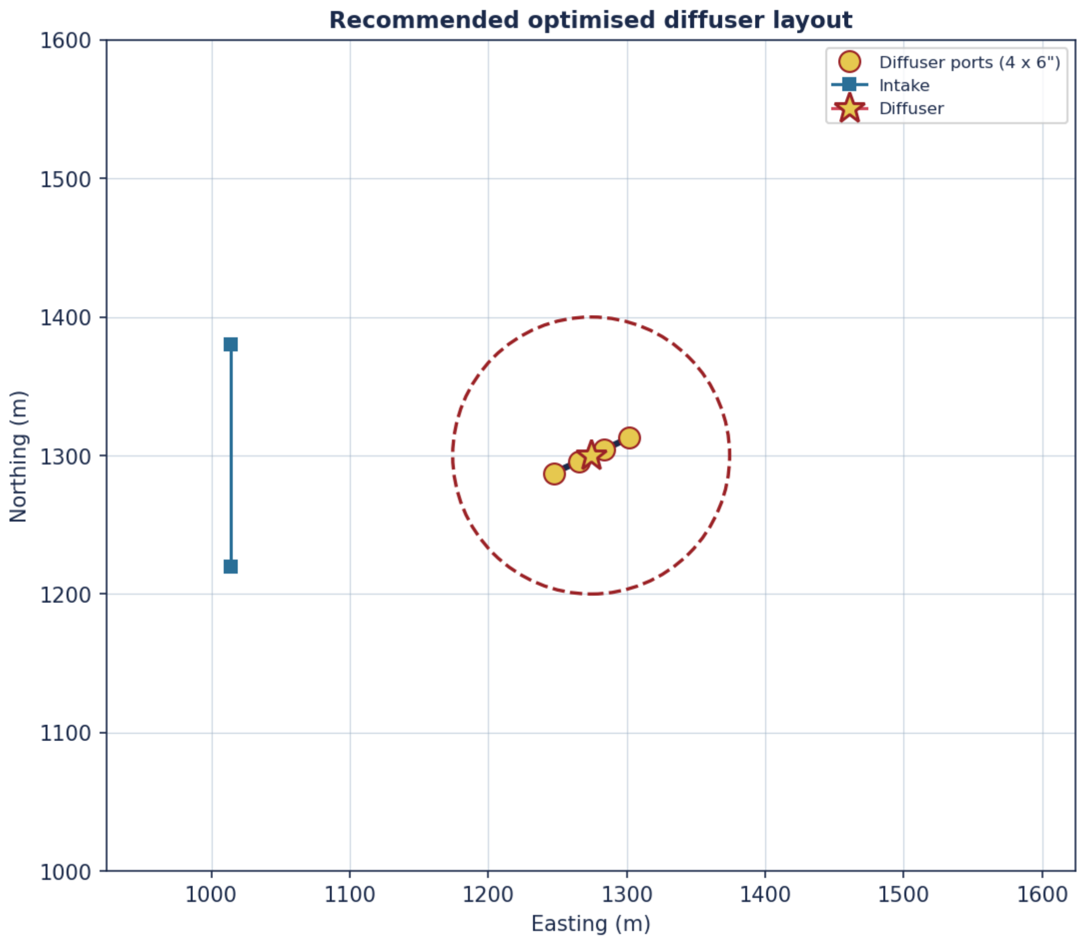


Figure. Recommended optimised diffuser/intake layout (inclined 4-port diffuser).

8 Outfall Siting and Intake Recirculation

Following the desalination-siting methodology, the salinity rise at the plant intake was assessed as a function of outfall distance from shore under steady northward, southward and weak-southward currents, for both discharges. The criterion is a salinity rise at the intake of no more than 0.1 ppt above ambient, applied to the peak cell in the intake window. The weak southward current is the critical (least-flushing) condition.

The two brines behave very differently, and the distinction matters. The intake signal scales with the excess salinity the plume delivers to the bed: the RO concentrate lands at 37.04 psu, an excess of 0.54 psu over the 36.5 psu background, whereas the saturated cavern brine lands at 51.40 psu, an excess of 14.90 psu - about 28 times larger.

Table. Rise in salinity at intake above ambient - southward current 0.25 m/s, both brines.

brine	outfall_dist_m	rise_min_ppt	rise_max_ppt	rise_avg_ppt	recirc_risk
RO concentrate	600	0.001	0.027	0.009	No
RO concentrate	800	0.0	0.004	0.001	No

RO concentrate	1000	0.0	0.001	0.0	No
RO concentrate	1200	0.0	0.0	0.0	No
RO concentrate	1400	0.0	0.0	0.0	No
Saturated cavern brine	600	0.019	0.363	0.125	Yes
Saturated cavern brine	800	0.003	0.057	0.017	No
Saturated cavern brine	1000	0.0	0.009	0.003	No
Saturated cavern brine	1200	0.0	0.002	0.0	No
Saturated cavern brine	1400	0.0	0.0	0.0	No

Table. Rise in salinity at intake above ambient - weak southward current 0.05 m/s, both brines.

brine	outfall_dist_m	rise_min_ppt	rise_max_ppt	rise_avg_ppt	recirc_risk
RO concentrate	600	0.016	0.061	0.035	No
RO concentrate	800	0.007	0.024	0.014	No
RO concentrate	1000	0.003	0.011	0.006	No
RO concentrate	1200	0.002	0.005	0.003	No
RO concentrate	1400	0.001	0.003	0.002	No
Saturated cavern brine	600	0.369	1.42	0.808	Yes
Saturated cavern brine	800	0.158	0.55	0.319	Yes
Saturated cavern brine	1000	0.074	0.247	0.143	Yes
Saturated cavern brine	1200	0.038	0.124	0.073	Yes
Saturated cavern brine	1400	0.021	0.067	0.039	No

Table. Rise in salinity at intake above ambient - northward current 0.25 m/s, both brines.

brine	outfall_dist_m	rise_min_ppt	rise_max_ppt	rise_avg_ppt	recirc_risk
RO concentrate	600	0.001	0.021	0.005	No
RO concentrate	800	0.0	0.002	0.001	No
RO concentrate	1000	0.0	0.0	0.0	No
RO concentrate	1200	0.0	0.0	0.0	No
RO concentrate	1400	0.0	0.0	0.0	No
Saturated cavern brine	600	0.009	0.287	0.076	Yes
Saturated cavern brine	800	0.001	0.034	0.01	No
Saturated cavern brine	1000	0.0	0.005	0.001	No
Saturated cavern brine	1200	0.0	0.001	0.0	No
Saturated cavern brine	1400	0.0	0.0	0.0	No

Result. For the RO concentrate the criterion is met everywhere: the worst cell over all currents and distances is 0.061 ppt, and at the recommended 1200 m siting the rise is 0.005 ppt. For the phase-2 saturated cavern brine the criterion is NOT met at 1200 m, where the peak intake rise reaches 0.124 ppt, nor at any shorter distance tested (worst case 1.420 ppt at the closest siting under the weak southward current). It is met from 1400 m offshore (0.067 ppt).

Consequence for the design. The 1200 m outfall recommended in section 7 satisfies the recirculation criterion for the phase-1 RO discharge, which is the discharge the plant is being built for. It does not satisfy it for the phase-2 cavern brine. If the cavern stream is committed, the outfall must be moved to at least 1400 m, or the recirculation criterion re-examined against the intermittency of that stream. This is a screening result on a 50 m grid under a steady current and is adequate to rank distances and to test the threshold; it is not a basis for fixing a siting distance to the nearest 200 m without a tidally-resolved run.

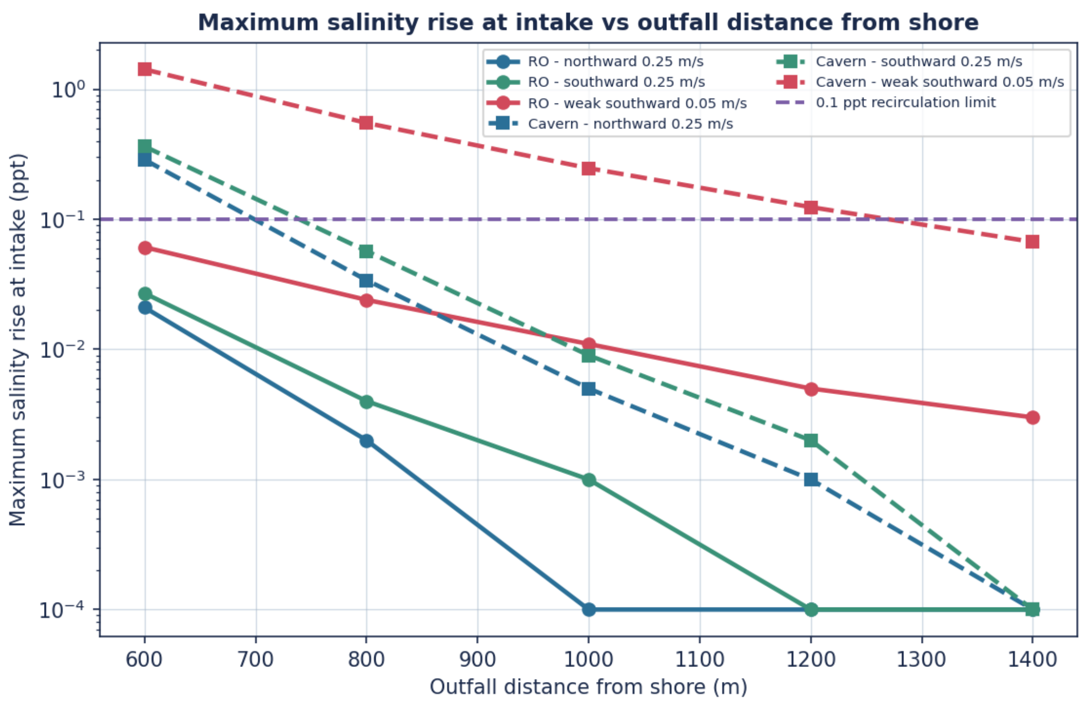


Figure. Variation of the maximum salinity rise at the intake with outfall distance from shore, for the RO concentrate and the phase-2 saturated cavern brine (log scale; 0.1 ppt recirculation criterion). Values of zero are plotted at the axis floor.

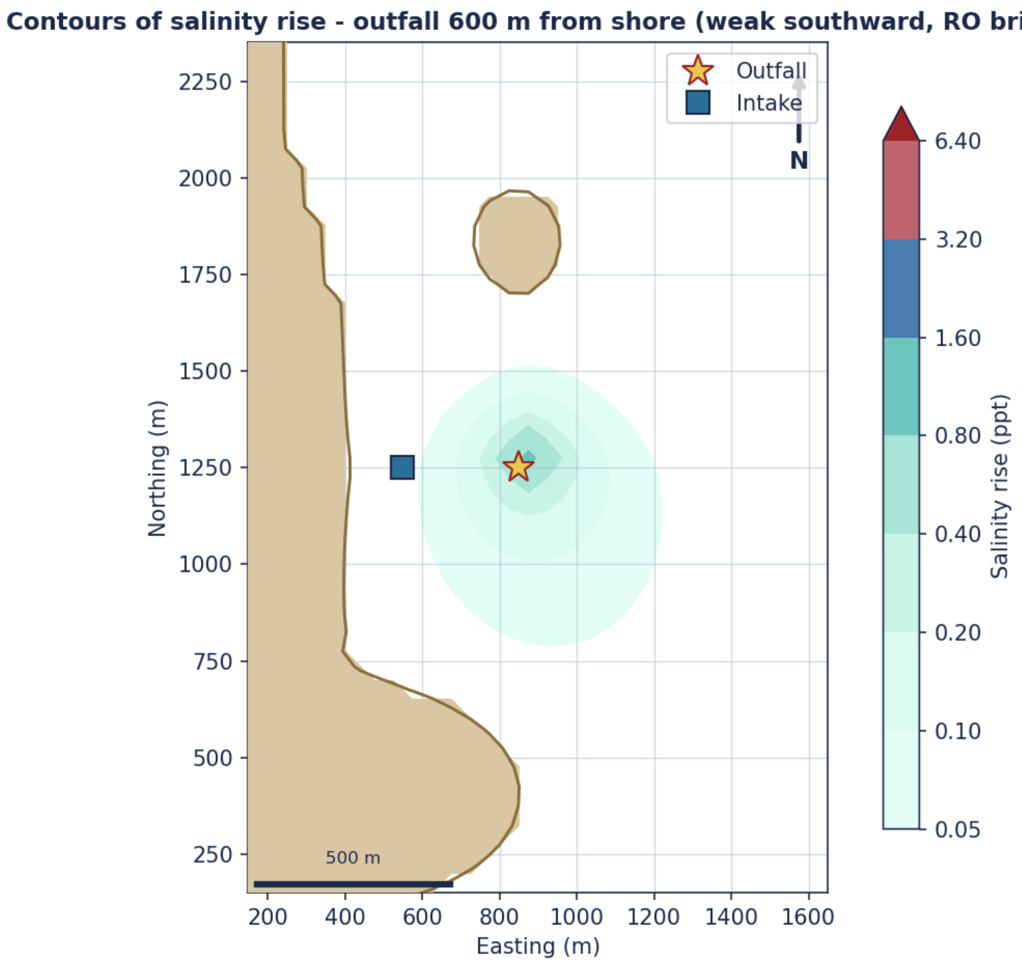


Figure. Salinity-rise contours, outfall 600 m from shore, weak southward current (RO concentrate).

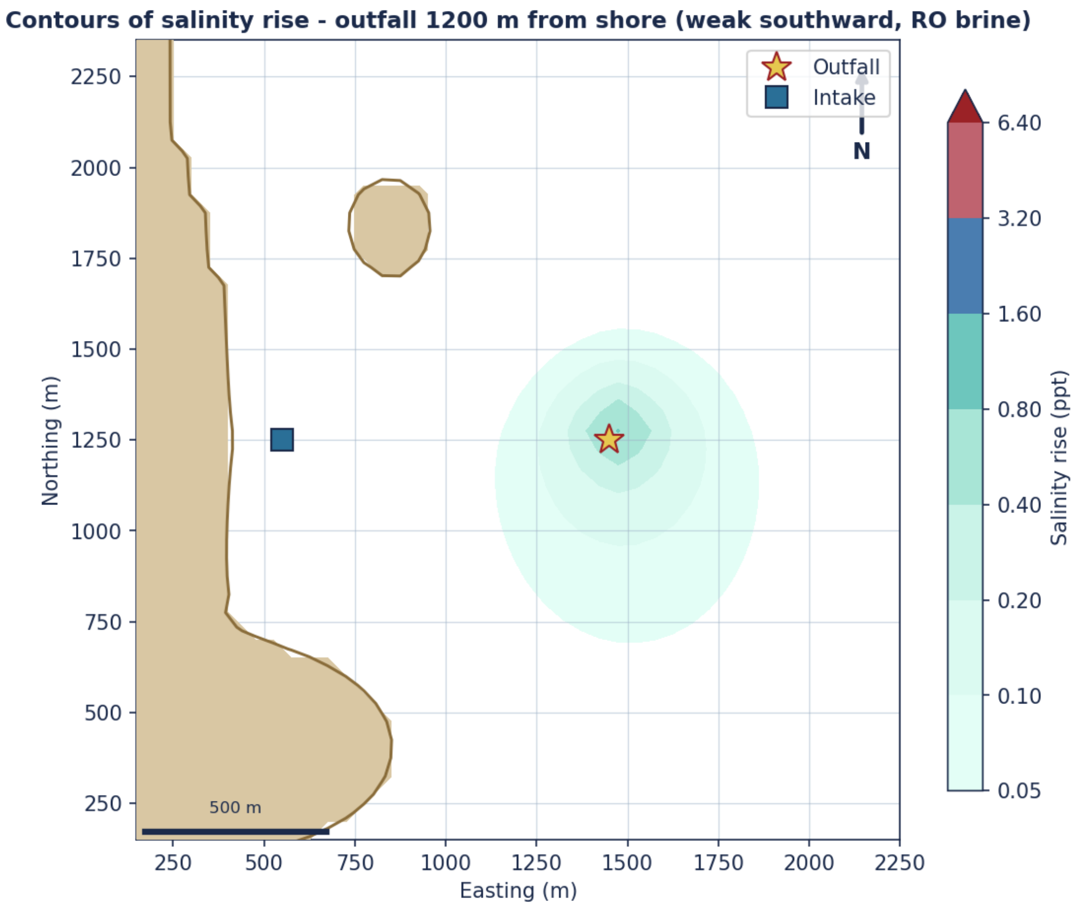


Figure. Salinity-rise contours, outfall 1200 m from shore, weak southward current (RO concentrate).

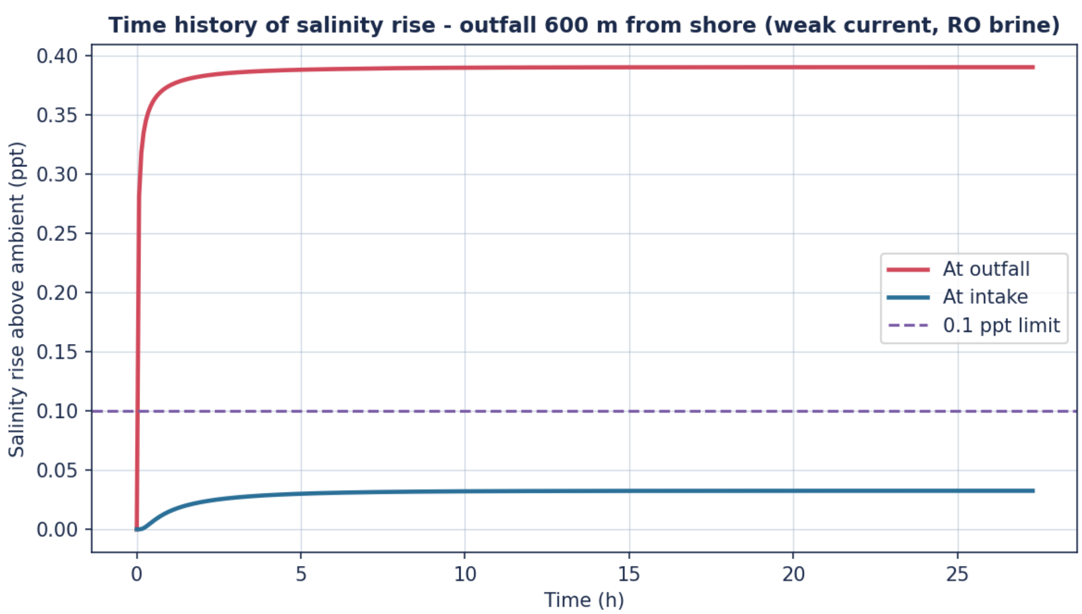


Figure. Time history of salinity rise at intake and outfall, outfall 600 m from shore (weak southward current, RO concentrate).

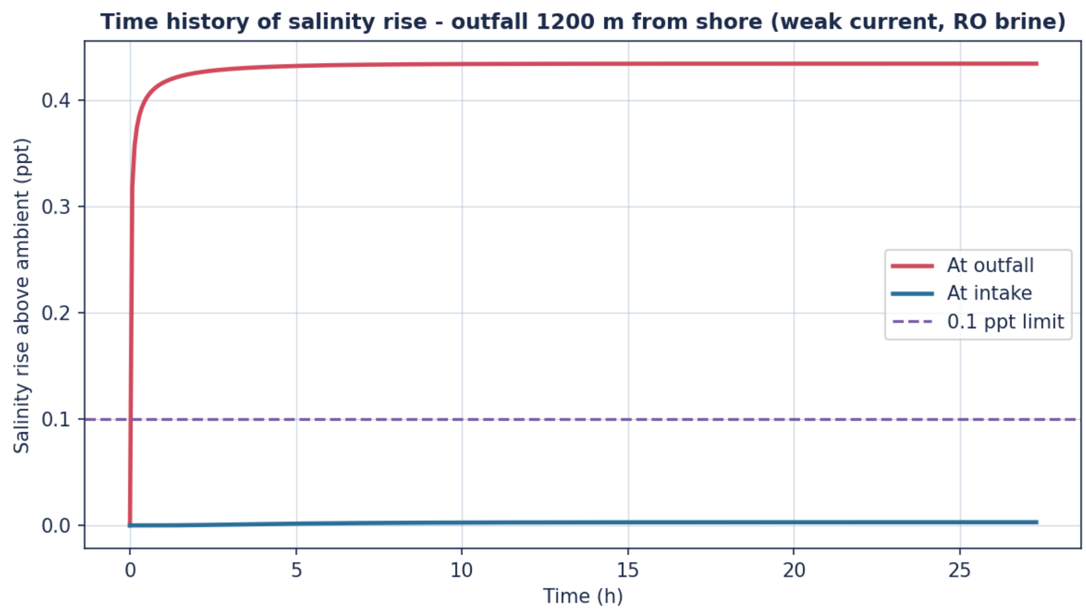


Figure. Time history of salinity rise at intake and outfall, outfall 1200 m from shore (weak southward current, RO concentrate).

9 Other Impurities and Water-Quality Compliance

Table. Trace-impurity concentrations vs Environmental Quality Standards after dilution.

parameter	brine_ugL	seawater_ugL	eqs_ugL	dilution	predicted_ugL	margin_x	compliant
Arsenic	3.2	0.9	25.0	14.3	1.061	23.6	Yes
Boron	3450.0	4300.0	7000.0	14.3	4240.559	1.7	Yes
Cadmium	0.3	0.04	2.5	14.3	0.058	43.0	Yes
Chromium	1.1	0.8	15.0	14.3	0.821	18.3	Yes
Copper	1.8	1.1	5.0	14.3	1.149	4.4	Yes
Lead	2.0	0.1	25.0	14.3	0.233	107.4	Yes
Mercury	0.02	0.015	0.3	14.3	0.015	19.5	Yes
Nickel	2.9	0.3	30.0	14.3	0.482	62.3	Yes
Zinc	4.0	1.2	40.0	14.3	1.396	28.7	Yes

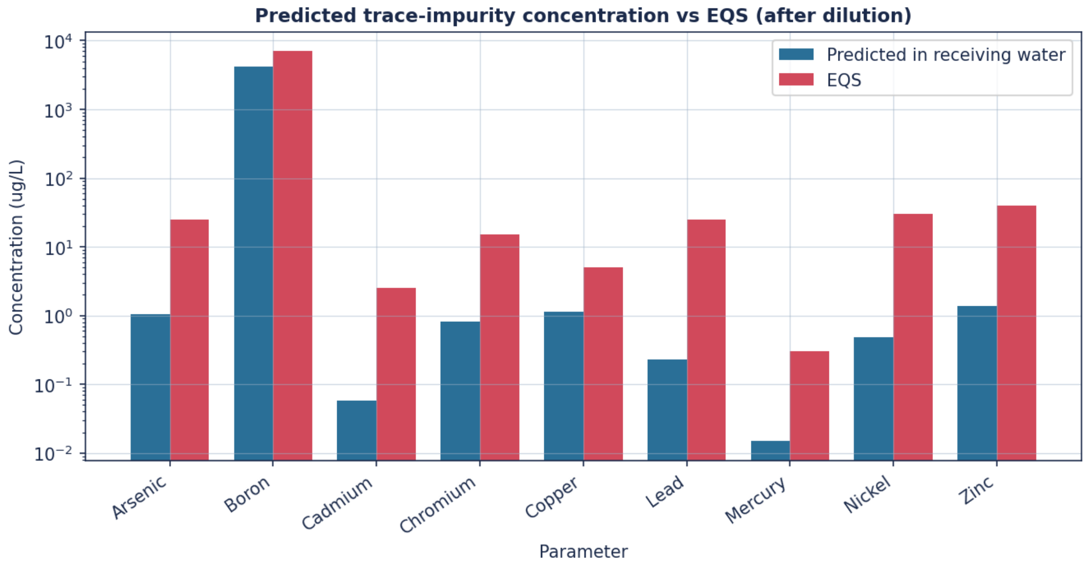


Figure. Predicted receiving-water trace-impurity concentrations vs EQS thresholds.

10 Validation and Data Sources

UBDS was tested against the following independent benchmarks. Where the model and the reference quantity are defined differently, the model value is converted before comparison and the conversion is stated; agreement is reported as measured, including where the model falls outside the published band.

- Equation of state: maximum error 0.402% vs reference seawater/brine densities (UNESCO EOS-80; CRC Handbook). Within tolerance.
- Momentum jet: volume-flux dilution slope 0.228 per z/D and spread 0.114 vs canonical 0.25-0.32 and ~0.11 (Fischer, List, Koh, Imberger & Brooks, 1979). The spread is reproduced; the dilution slope sits about 9% below the lower end of the canonical range, i.e. the model entrains slightly less than the canonical round jet.
- Pure plume: volume-flux exponent 1.596 vs the 5/3 = 1.667 power law (Morton, Taylor & Turner, 1956); about 4% below the analytical exponent.
- Inclined 60 deg dense jet (Papakonstantis, Christodoulou & Papanicolaou, 2011), benchmarked over Fr 9.0-33.3 (7 cases), the span the case study runs at: x_r/(D.Fr) = 2.522 (mean) vs the published 2.2-3.3 band - in band in 7/7 cases. Return-point dilution, converted from the model's bulk (top-hat) value to the centreline minimum that the published band refers to (divide by 1.694, Gaussian profile, lambda = 1.2): S_r(min)/Fr = 0.581 vs the published 0.4-0.6 band - in band in 6/7 cases. The uncorrected bulk value is 0.985 and is not directly comparable.
- Terminal rise of the same dense jet: z_t/(D.Fr) = 1.49 (mean) against the published 1.6-2.2 band - about 7% BELOW the lower bound, and inside the band in only 1 of the 7 cases. The shortfall is not uniform: it grows as the Froude number falls, so it is worst at the Fr 9-11 of the saturated-cavern brine, which is the case that governs the seabed assessment. UBDS therefore under-predicts the terminal rise height of an inclined dense jet. Because a lower rise gives a shorter

trajectory and less entrainment before the plume returns to the bed, this bias is conservative for seabed salinity - it over-states, not under-states, the salinity reaching the bed. It is not corrected by tuning, since raising the rise height would require reducing entrainment and would degrade the jet-dilution and return-point dilution agreement above.

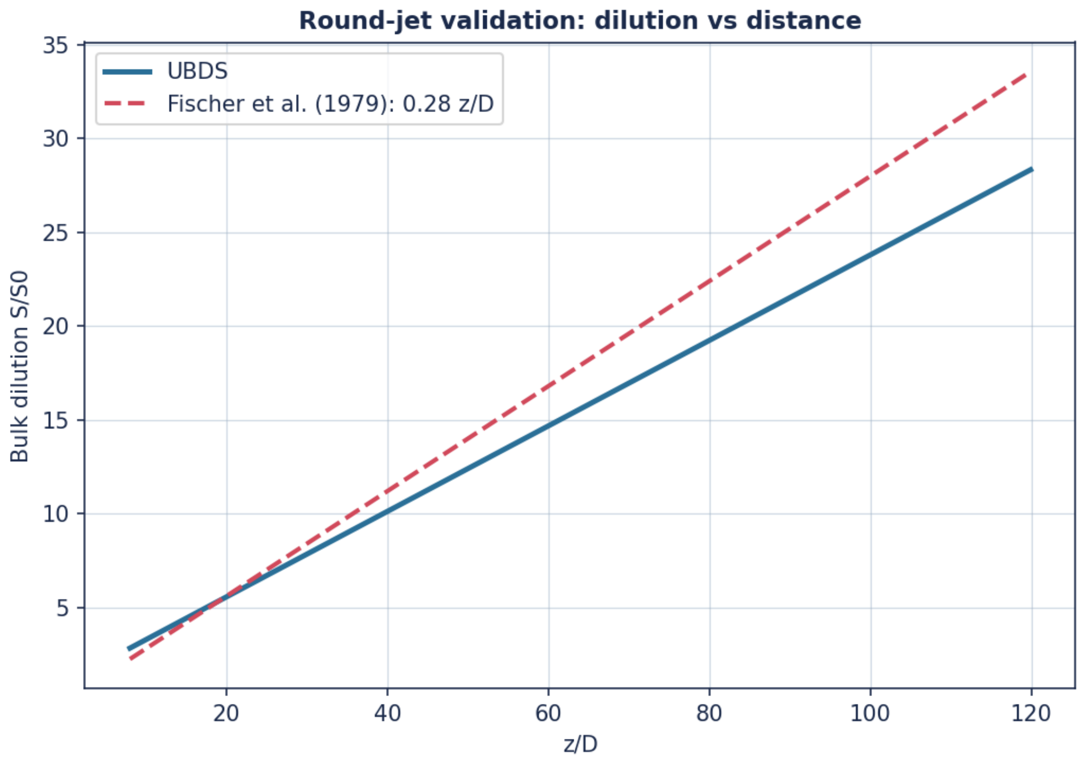


Figure. Momentum-jet dilution: UBDS vs the canonical self-similar law (Fischer et al. 1979).

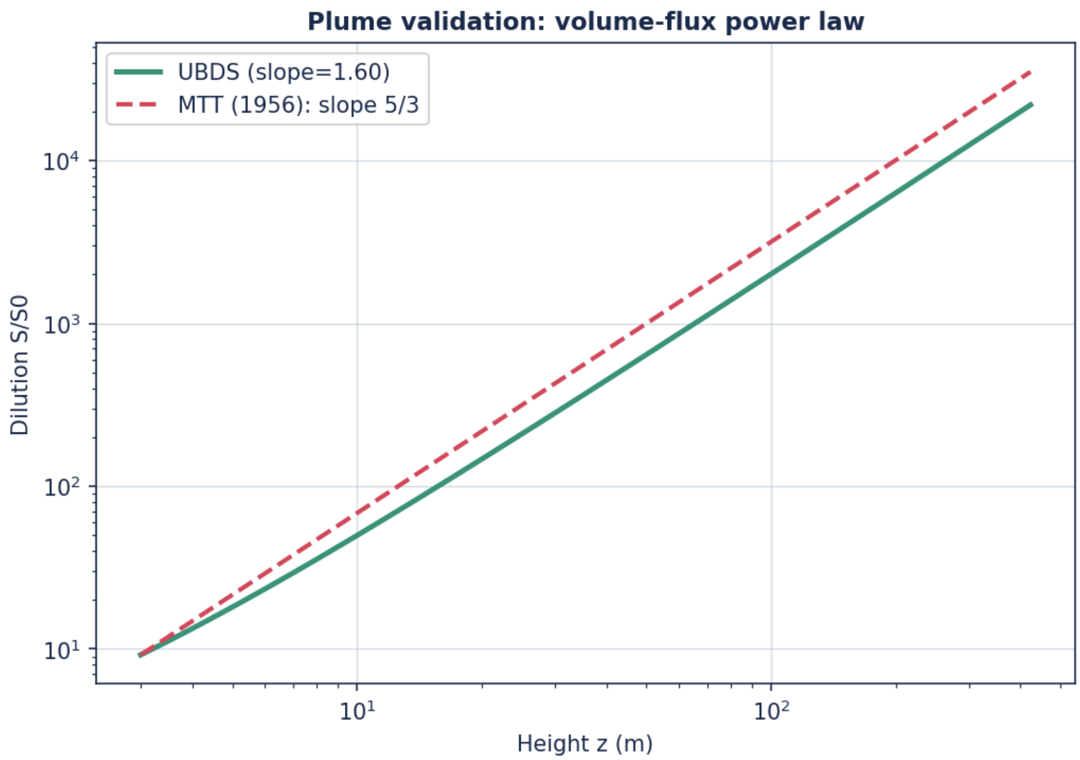


Figure. Pure-plume dilution: UBDS vs the 5/3 power law (Morton, Taylor & Turner 1956).

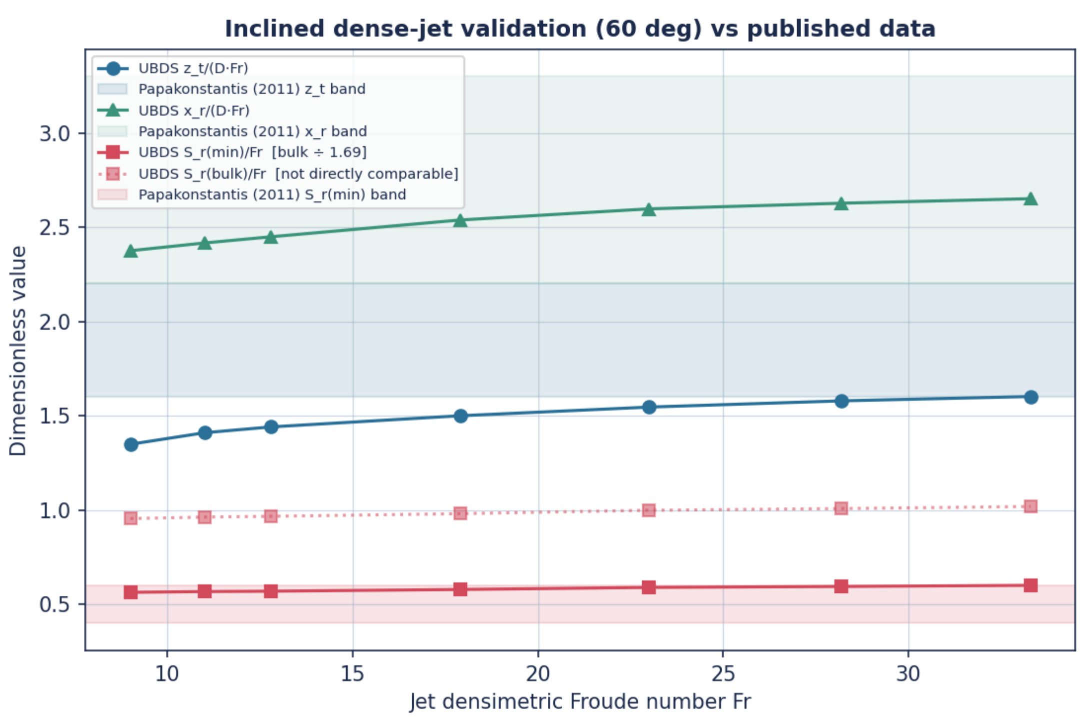


Figure. Inclined 60 deg dense-jet dimensionless rise and dilution vs published experimental bands.

Table. Equation-of-state validation.

salinity_psu	temp_C	ref_density	model_density	error_pct	source
0.0	25.0	997.05	997.05	-0.0	Pure water (CRC)
35.0	15.0	1025.97	1025.97	0.0	EOS-80 (Millero & Poisson 1981)
35.0	25.0	1023.34	1023.34	0.0	EOS-80
38.0	20.0	1026.95	1027.05	0.01	EOS-80
260.0	20.0	1200.0	1204.82	0.402	Saturated NaCl brine (CRC Handbook)

Table. Inclined dense-jet dimensionless validation.

Fr	zt_over_DFr	xr_over_DFr	Sr_bulk_over_Fr	Sr_min_over_Fr	termination
9.0	1.349	2.375	0.955	0.564	seabed
11.0	1.41	2.416	0.963	0.568	seabed
12.8	1.441	2.449	0.967	0.57	seabed
17.9	1.5	2.538	0.981	0.579	seabed
23.0	1.546	2.597	0.999	0.59	seabed
28.2	1.579	2.627	1.008	0.595	seabed
33.3	1.602	2.651	1.019	0.601	seabed

10.1 Data sources

- Fischer, H.B., List, E.J., Koh, R.C.Y., Imberger, J. & Brooks, N.H. (1979). Mixing in Inland and Coastal Waters. Academic Press. [jet/plume dilution laws]
- Morton, B.R., Taylor, G.I. & Turner, J.S. (1956). Turbulent gravitational convection from maintained and instantaneous sources. Proc. R. Soc. Lond. A 234, 1-23. [plume 5/3 law]
- Papakonstantis, I.G., Christodoulou, G.C. & Papanicolaou, P.N. (2011). Inclined negatively buoyant jets 1 & 2. J. Hydraulic Research 49(1), 3-22. [inclined dense-jet data]
- Roberts, P.J.W., Ferrier, A. & Daviero, G. (1997). Mixing in inclined dense jets. J. Hydraulic Engineering 123(8), 693-699. [dense-jet dilution]
- Millero, F.J. & Poisson, A. (1981). International one-atmosphere equation of state of seawater. Deep-Sea Research 28A, 625-629; UNESCO (1981) Tech. Papers Mar. Sci. 36. [EOS-80]
- Jirka, G.H. (2004). Integral model for turbulent buoyant jets in unbounded stratified flows. Part 1. Environmental Fluid Mechanics 4, 1-56. [integral jet model]
- Lee, J.H.W. & Chu, V.H. (2003). Turbulent Jets and Plumes - A Lagrangian Approach. Kluwer. [entrainment closure]
- CRC Handbook of Chemistry and Physics. [saturated NaCl brine density]
- Synthetic site-specific data (bathymetry, tides, ADCP currents, CTD salinity/temperature, brine chemistry) generated by the author for this case study; all values are documented and engineering-credible, and none are taken from any real consent.

10.2 Supporting data plots

Derived plots summarising the tabulated results above: seabed dilution against tidal velocity, salinity-increment area against discharge flow, and the per-station index-of-agreement against the synthetic current reference.

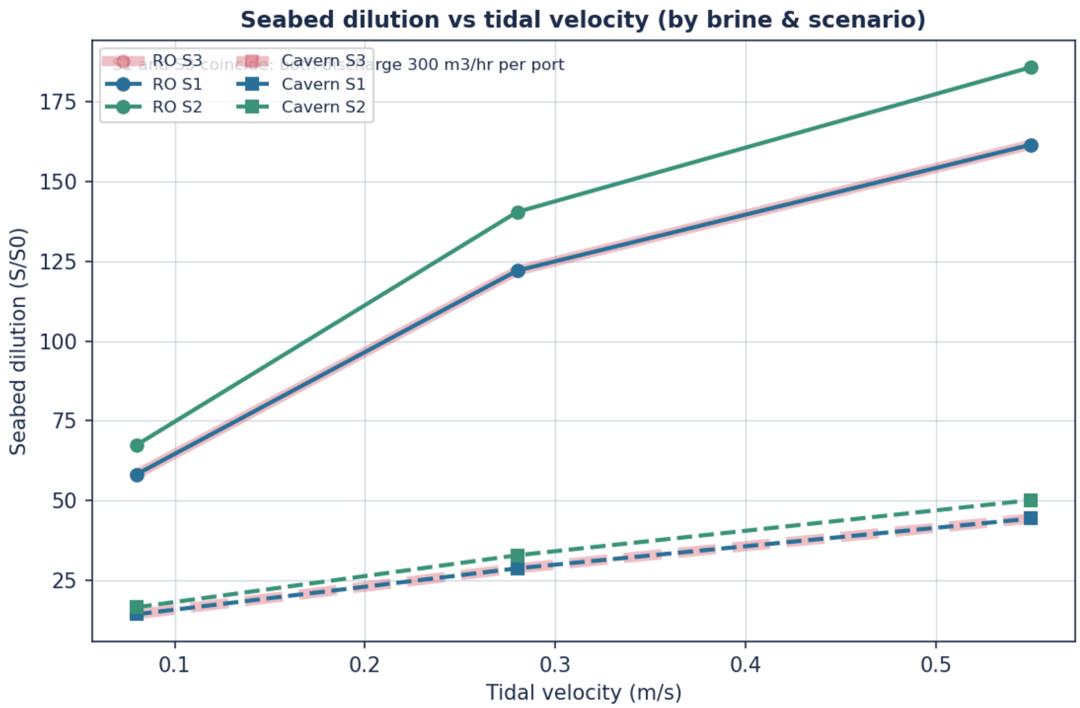


Figure. Seabed dilution as a function of tidal velocity for each brine type and scenario. S1 and S3 coincide because dilution is set by the per-port flow (300 m3/hr in both), not the total.

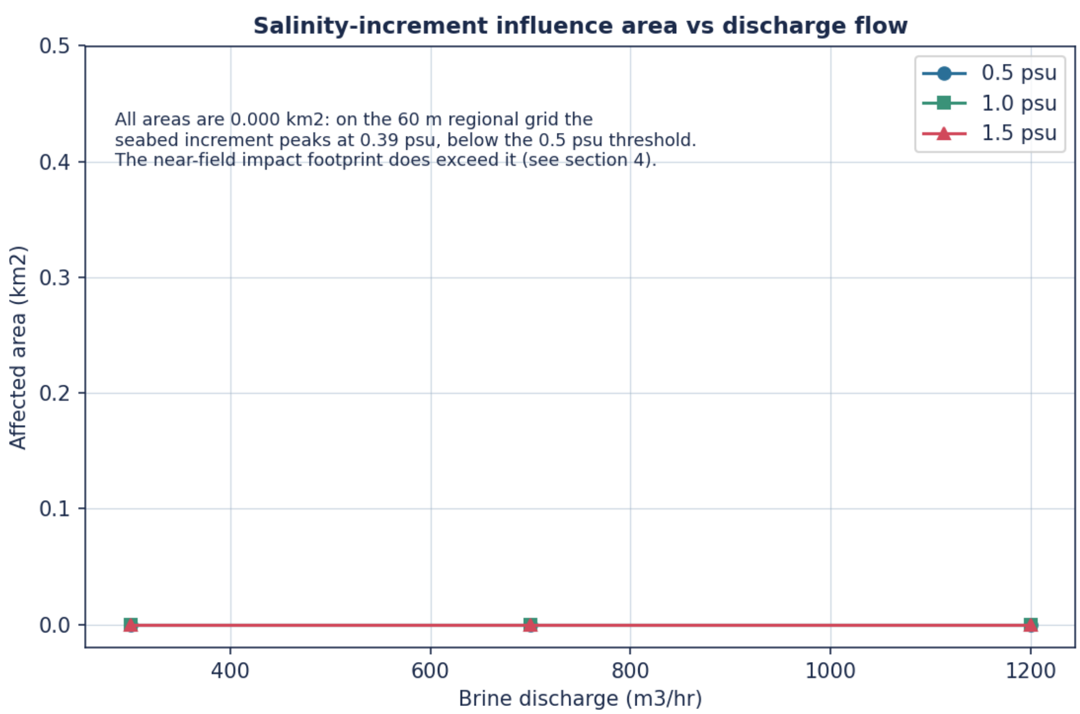


Figure. Area enclosed by salinity-increment contours vs discharge flow. All three areas are zero on the regional grid, whose peak seabed increment is 0.39 psu; the zeros are grid-resolved, not absolute.

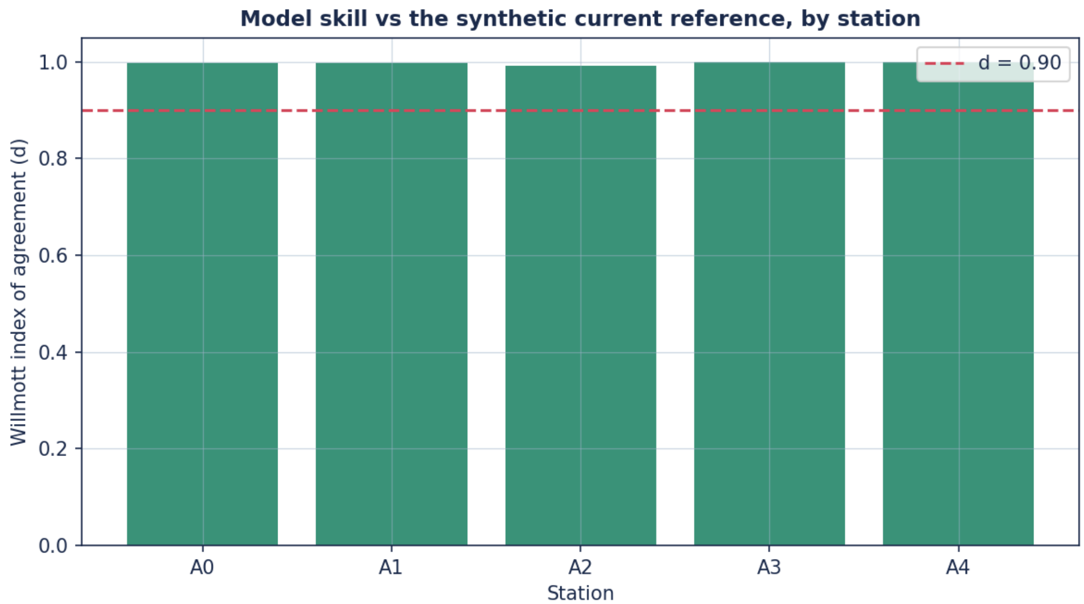


Figure. Index-of-agreement between the model and the synthetic current reference by station (spring). The reference is derived from the model solution, so this is not a field comparison.

11 Conclusions

The UBDS solver provides a single, open, physically-validated framework spanning the near-, medium- and far-field dispersion of brine and other buoyant effluents. Applied to the Bahia Azul case study with fully documented, credible site-specific inputs, it demonstrates that the proposed inclined 4-port diffuser achieves effective initial dilution and confines significant salinity increments and all trace impurities to within the regulatory mixing zone, across all simulated discharge flows and tidal states.

One condition attaches to that conclusion. The 1200 m outfall meets the 0.1 ppt intake-recirculation criterion for the phase-1 RO concentrate (0.005 ppt) but not for the phase-2 saturated cavern brine (0.124 ppt), which requires a siting of at least 1400 m. The seabed-salinity and water-quality conclusions above already use the cavern brine as the worst case; the recirculation conclusion is the one place where the two phases diverge, and it is stated separately for that reason.

12 Animated Outputs

The following animated outputs (GIF files in outputs/animations/) show the time-evolving brine dispersion and tidal current field over a tidal cycle. A static filmstrip of the dispersion animation is embedded below; the full animations are in outputs/animations/.

- ANIM_brine_dispersion_spring.gif - Brine dispersion over a tidal cycle (full flow, spring, saturated cavern brine)
- ANIM_brine_dispersion_neap.gif - Brine dispersion over a tidal cycle (full flow, neap, saturated cavern brine)

- ANIM_current_field_spring.gif - Tidal current field over a cycle (spring)

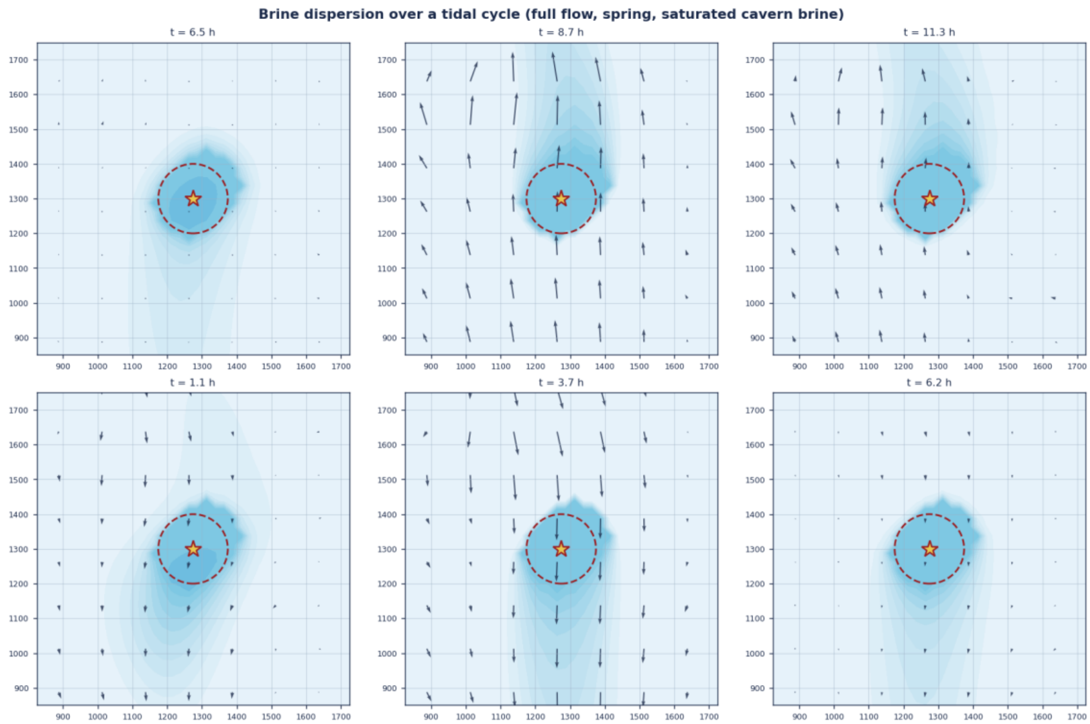


Figure. Brine dispersion over a tidal cycle (full flow, spring, saturated cavern brine) (filmstrip of the animation).

Appendix A Representative Output Figures from the Reference Studies

For methodological comparison, representative output figures from the three reference studies that guided this work are reproduced below (author's reference materials, attributed). They illustrate the output categories - mesh/bathymetry, current calibration, near-field trajectories/dilution, salinity envelopes and snapshots, and intake/outfall time histories - that the UBDS outputs reproduce for the Bahia Azul case.

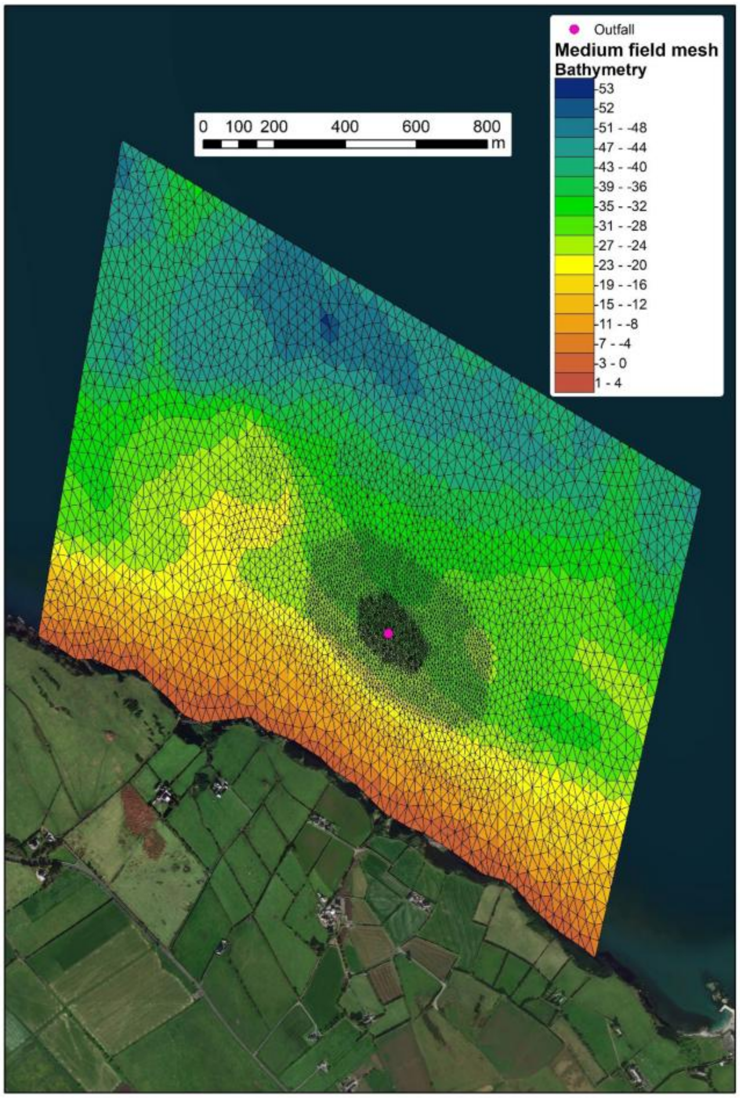


Figure 1-1 – Mesh and Bathymetry of the Islandmagee Medium Field Tidal Model

Figure. Reference Study 1 (salt-cavern brine outfall): model mesh, bathymetry and current-meter comparison.

3 INITIAL DILUTION MODELLING

The initial dilution study examines the dispersion of the brine discharged from the outfall diffuser jets in the immediate area around the diffuser. For this study the initial dilution achieved by the brine discharge from the proposed outfall was modelled using the US EPA "Plumes" software utilising the UM3 routine. The Visual Plumes programme simulates the flow of the outlet jets under the influence of density, temperature and velocity. The programme is primarily a near-field model and a mid- and far-field model is therefore also required to simulate the plume dispersion over the wider area.

UM3 is an acronym for the three-dimensional Updated Merge (UM) model. UM3 simulates single and multi-port submerged discharges. UM3 is a Lagrangian model that assumes that the plume is in a steady state. However, ambient and discharge conditions can change as long as they do so over time scales which are long compared to the time in which a discharged element reaches the end of the initial dilution phase. To make UM three-dimensional, the model includes an entrainment term corresponding to the third-dimension i.e. a cross-current term. As a result, single-port plumes are simulated as truly three-dimensional entities. Merged plumes are simulated by distributing the cross-current entrainment over all plumes. Dilution from diffusers oriented parallel to the current is estimated by limiting the effective spacing to correspond to a cross-diffuser flow angle of 20 degrees.

3.1 Initial Dilution Model Results

The initial dilution modelling was undertaken using a diffuser with either 1 or 2 ports depending on the flow range as stated in Section 2.2. The diffuser ports were at 20m centres with the port exits discharging vertically upwards at approximately 0.75m above the seabed. Following discharge it is expected that the brine will sink down towards the seabed due to the density of the brine solution, with the initial trajectory depending on the tidal velocity. Consequently the UM3 model was run for a variety of spring and neap flow conditions to simulate the initial dilution achieved at various stages of the tidal cycle.

3.1.1 Scenario 1: 250m^3/hr via a Single Port

This scenario represents the early stages of project development when the leaching of the caverns is just commencing and the full flow of leaching water is not required. For this stage of the process it was assumed that one port on the diffuser would be blanked off and hence the 250m^3/h would be discharged via one diffuser port.

The trajectories for the discharge from a 6" Tideflex port across the range of tidal velocities are presented in Figure 3-1. The trajectory in red relates to low tidal velocity of <0.1m/s which would be experienced during the turning tide, the trajectory in blue relates to tidal velocity of 0.3 m/s which would represent peak velocities during neap tide. The trajectory in green relates to a typical spring tidal velocity of 0.6 m/s.

Figure. Reference Study 1: near-field brine-plume trajectory and initial-dilution prediction.

Table 4-1: Table showing Water Column division into Layers within the Models in terms of % of overall water depth assigned to each layer.

	14 Layer	9 Layer	4 Layer
Layer 14	40	-	-
Layer 13	20	-	-
Layer 12	11	-	-
Layer 11	8.5	-	-
Layer 10	6.5	-	-
Layer 9	4.5	40	-
Layer 8	3	25	-
Layer 7	2	15	-
Layer 6	1.5	9.3	-
Layer 5	1	4.5	-
Layer 4	0.8	2.5	35
Layer 3	0.6	1.5	35
Layer 2	0.5	1.2	15
Bottom Layer 1	0.1	1	15

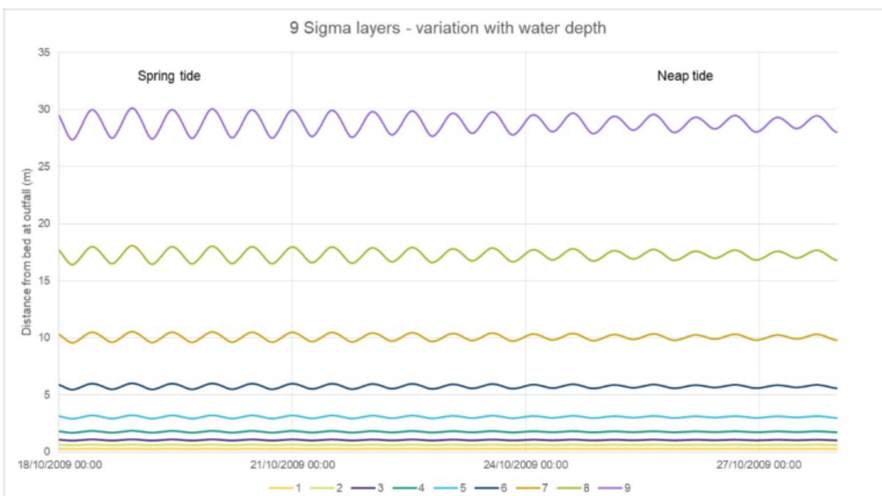


Figure 4-1: Variation of sigma layers with water depth (9 layers)

For the sensitivity testing the 1,000m^3/hr scenario was selected as this represents the maximum input of salt and hence density effects will be more prevalent than with lower salt inputs. The results of the sensitivity testing in terms of maximum seabed salinity levels are presented in Figure 4-2 to Figure 4-4 for the 4 layer, 9 layer and 14 layer models respectively. It should be noted in all figures presented the colour contour bands are not linear with smaller intervals at the lower salinity values in order to fully illustrate the range

Figure. Reference Study 1: sigma-layer scheme and layer-resolution sensitivity.

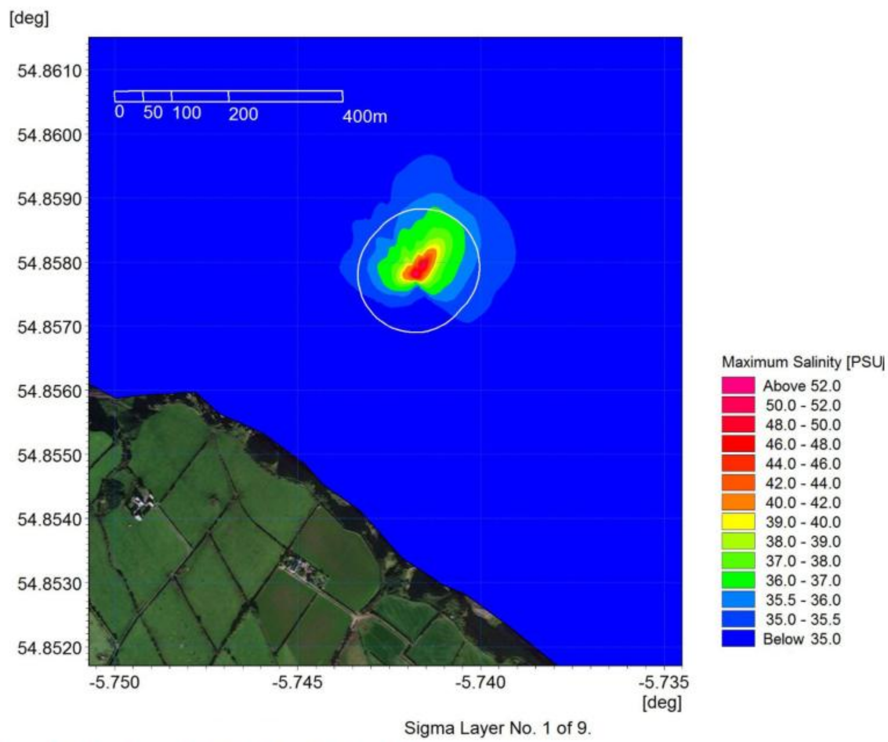


Figure 4-8: Maximum Salinity at Bed – Neap Tide

Figure. Reference Study 1: maximum seabed-salinity envelope and salinity/current snapshots.

Figure 5-4 respectively. In all figures presented the colour contour bands are not linear with smaller intervals used at the lower salinity and the 36 psu threshold is shown in yellow.

The salinity increases predicted beyond the initial mixing zone in all cases are relatively small; the contour intervals plotted representing very small increases of 0.25-0.50 psu. A maximum far field increase in salinity of less than 4 psu i.e. total salinity <38 psu, was recorded throughout all the simulations which is commensurate with the levels predicted for the concept design considered at planning stage.

The baseline salinity data obtained from AFBI and NIEA at the initial project development stage indicated that background salinities in this part of the North Channel off Islandmagee can range between circa 30.5 psu and 34.8 psu. Even with an assumed background salinity of 34.2 any salinity increase in excess of the range normally experienced in seasonal variations (34.8 psu) is expected to be restricted to an area less than 300 metres from the outfall.

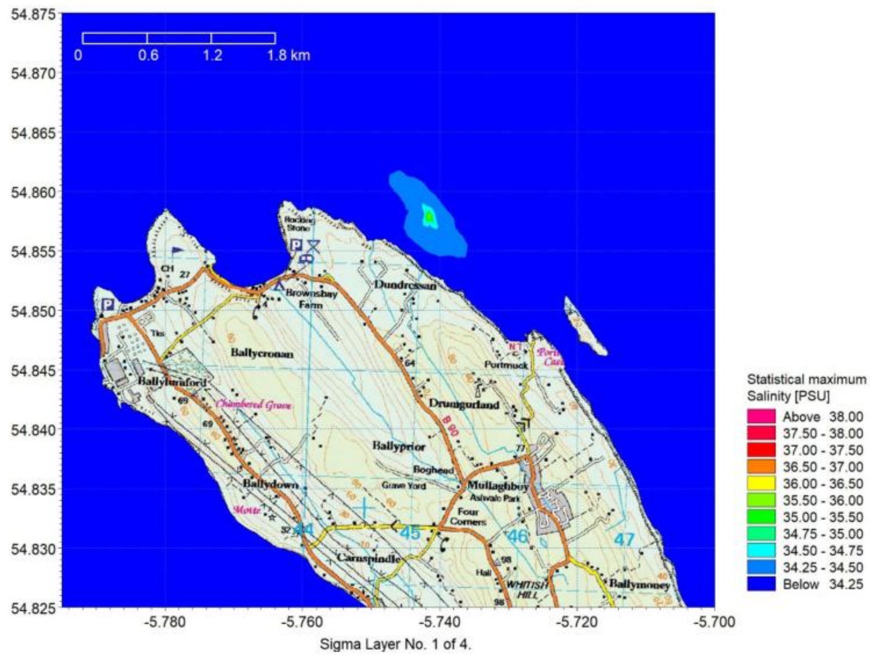


Figure 5-2: Maximum Seabed Salinity during a Neap Tidal Cycle – 250m³/hour Discharge

Figure. Reference Study 1: far-field maximum seabed-salinity envelopes.

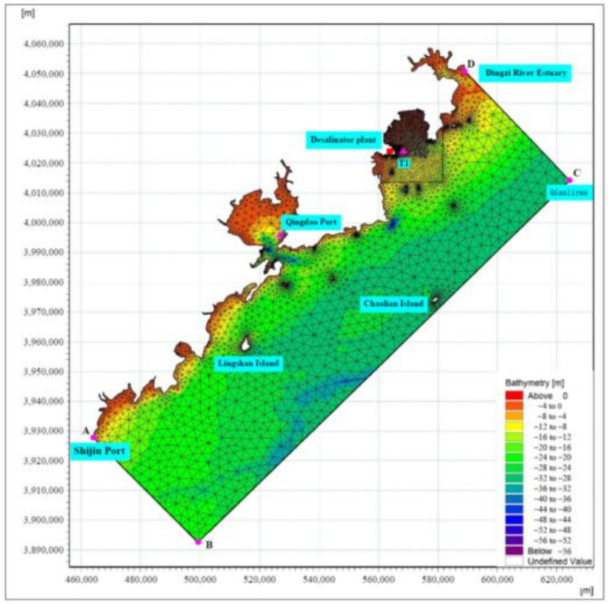


Figure 2. Grid distribution in the computational domain, and color-shaded bathymetry. A, B, C, and D are control points for open water boundaries. T1 is a tidal level monitoring point.

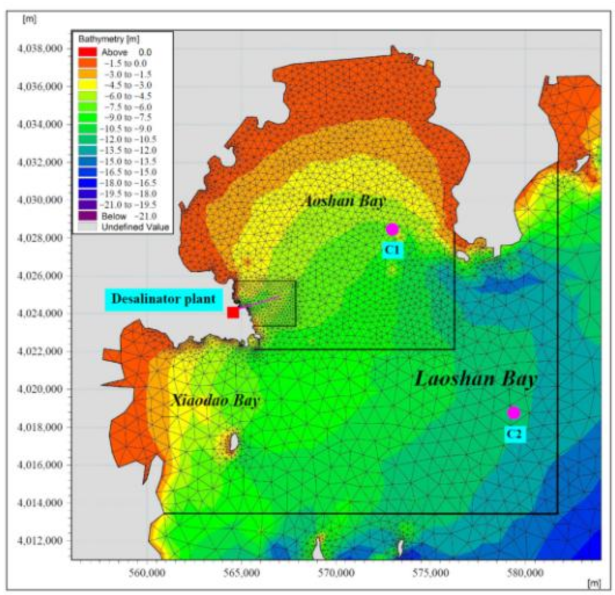


Figure 3. Local grid refinement in the engineered area, and locations of verification points. C1 and C2 are current monitoring points.

Figure. Reference Study 2 (Laoshan Bay desalination): computational mesh and bathymetry of the bay.

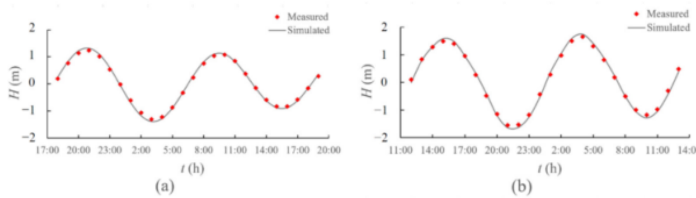


Figure 4. Comparison of the simulated and measured values of the tidal level at the Qingdao Port (**a**) and Station T1 (**b**).

The tidal flow field simulation results were verified by continuous 25 h measured current data and model calculation results from two stations in Laoshan Bay in April 2018. The positions of the tidal flow verification points are shown in Figure 3, and the tidal flow verification curves of the two stations are shown in Figures 5 and 6, respectively.

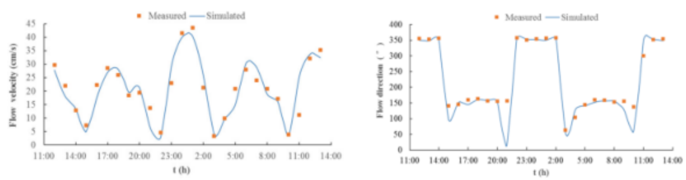


Figure 5. Comparison between the simulated and measured values of the flow velocity and direction at Station C1.

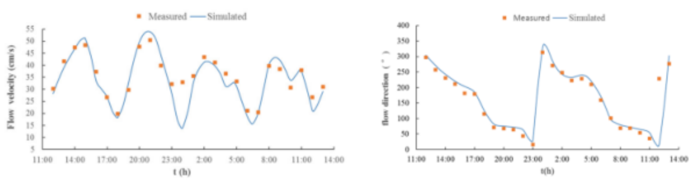


Figure 6. Comparison between the simulated and measured values of the flow velocity and direction at Station C2.

The calculated results of the model were found to be in good agreement with the measured tidal level data and ocean current data, properly reflecting the tidal level and ocean current characteristics in Laoshan Bay and its surrounding waters and meeting the simulation accuracy requirements of the hydrodynamic model and advection-diffusion model in this paper.

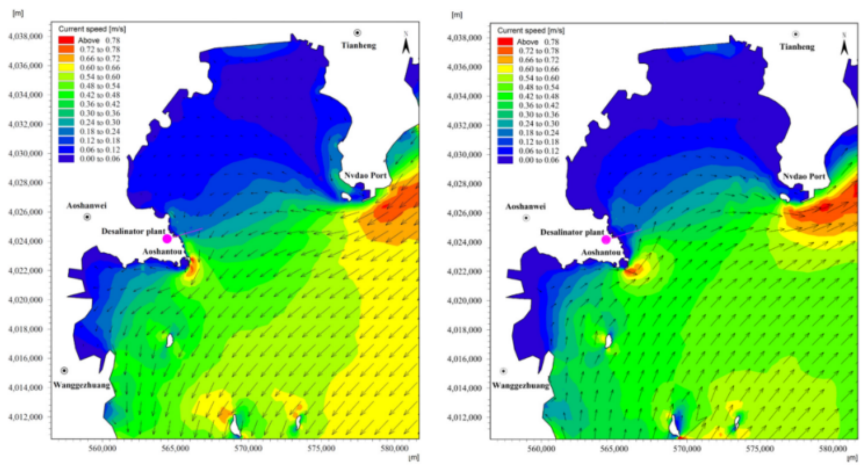
4. Results and Discussion

4.1. Simulation Results of Tidal Flow Field in Laoshan Bay

The simulation results of the tidal flow field in Laoshan Bay calculated using the model are shown in Figure 7. During the flood tide, the flow velocity in Laoshan Bay is generally 30~55 cm/s, and outside the bay the flow velocity is 60~70 cm/s. The flood current flows from NE to SW from the eastern sea area of Tianheng Town, then turns to NW after passing the headland and flows into Aoshan Bay on the north side. The flow velocity of the flood current in Aoshan Bay is typically 10~42 cm/s, and the flow velocity on the shore is relatively small, generally less than 10 cm/s. Near Nvdao Island, the flow velocity of the flood current is relatively high, between 54 cm/s and 78 cm/s. The flood current in the area where the project is located flows from NNW to SSE, with a velocity between

Figure. Reference Study 2: tidal flow-field simulation results.

10 cm/s and 36 cm/s. After passing the project area, the flood current along the coast passes around the Aoshantou headland and then turns west to Xiaodao Bay. The flow velocity in the Aoshantou sea area is relatively high, with a maximum of 72 cm/s. The flood current in the Xiaodao Bay generally flows from NE to SW at a velocity of 10~42 cm/s, while the velocity near the shore is low, generally less than 10 cm/s. Then the tidal current flows from N to S along the shore to the outside of the bay, and the velocity increases accordingly.



(**a**) Flood current (**b**) Ebb current

Figure 7. Hydrodynamic flow field simulation results.

During the ebb tide, the velocity in Laoshan Bay is generally 25~50 cm/s, and the velocity outside the bay is 55~65 cm/s. The northern ebb current flows from NW to SE, bypasses Nvdao Island, then turns to the NE and flows to the sea on the east side of Tianheng Town. The flow velocity in Aoshan Bay is 10~48 cm/s, and the velocity on the shore is low, generally less than 10 cm/s. The ebb tide velocity on the south side of Nvdao Island is relatively high, between 54 cm/s and 72 cm/s. Along the south side of the sea, the ebb tide flows from S to N outside the bay, and then from SW to NE inside Xiaodao Bay, and the flow velocity decreases accordingly. The flow velocity of the ebb tide in Xiaodao Bay is 10~36 cm/s, and the flow velocity on the shore is relatively low, generally less than 10 cm/s. After bypassing Aoshantou Headland, the ebb tidal current passes through the sea area of the project area, flowing from SSE to NNW at a flow velocity of 10~30 cm/s. The sea area south of Aoshantou has a high flow velocity of up to 68 cm/s.

The calculation results of the tidal flow field and the conclusions of historical research results are consistent [22,23], and the numerical simulation results properly reflect the distribution law of the tidal flow field in Laoshan Bay.

4.2. Simulation Results of Salinity Field after Brine Discharge

The diffusion process of brine in Laoshan Bay is mainly affected by tidal currents. The calculated results of salinity field distribution of brine discharged by different water intake-outlet schemes of seawater desalination projects are shown in Figure 8. It can be seen that the impact of the brine discharge under different schemes on the increase of salinity in the nearby sea area is less than 0.6 units in most sea areas, although the impact of the increase of salinity in a small area near the discharge outlet is more than 2.3 units. With the continuous discharge of brine, the salinity of the sea area near the discharge outlet obviously increases. However, due to the water exchange between Aoshan Bay and the

Figure. Reference Study 2: salinity-field results and salinity profile for the layout plans.

where A= horizontal eddy viscosity.

MIKE21-AD/Transport module is based on the following non-linear vertically integrated 2-D equations of conservation of salinity which takes into account advection, dispersion, source and sink.

$$\frac{\partial h\bar{s}}{\partial t} + \frac{\partial h\overline{us}}{\partial x} + \frac{\partial h\overline{vs}}{\partial y} = hF_s + h\bar{s}_s S \quad (6)$$

where, $\bar{s}$ = Depth average salinity, F_s = the horizontal diffusion term defined by

$$F_s = \left[\frac{\partial}{\partial x}\left(D_h \frac{\partial \bar{s}}{\partial x}\right) + \frac{\partial}{\partial y}\left(D_h \frac{\partial \bar{s}}{\partial y}\right)\right] \quad (7)$$

and D_h = horizontal diffusion coefficient.

In the 2D transport equations, the low order approximation uses simple first order upwinding, i.e., element average values in the upwinding direction are used as values at the boundaries. The higher order version approximates gradients to obtain second order accurate values at the boundaries. Values in the upwinding direction are used. To provide stability and minimize oscillatory effects, a TVD-MUSCL limiter is applied [5].

4. MODEL SIMULATION

4.1. Model Region

The region considered for the model is about 9 km X 12 km covering about 6 km on the either side of the proposed port and extending up to about -30 m contour as shown in Fig. 1. The western side of the model region consists the shoreline while the remaining sides are open to sea. The region is discretised by 11947 triangular elements and 6241 nodes (Fig.2). It is observed in the open sea at the site that the flow is unidirectional (either northerly or southerly). Hence to get the northerly flow, a tidal variation at southern boundary and velocities at northern boundary were considered. To get southern flow, the boundary conditions are reversed. At the seaward boundary, no flow was assumed.

4.2. Model Calibration

Before using the model for predictive purpose, it is calibrated. The Fig. 3A and 3B show calibration of the hydrodynamic model, particularly the observed and computed variation of

Figure. Reference Study 3 (Chennai desalination): bathymetry and flexible computational mesh.

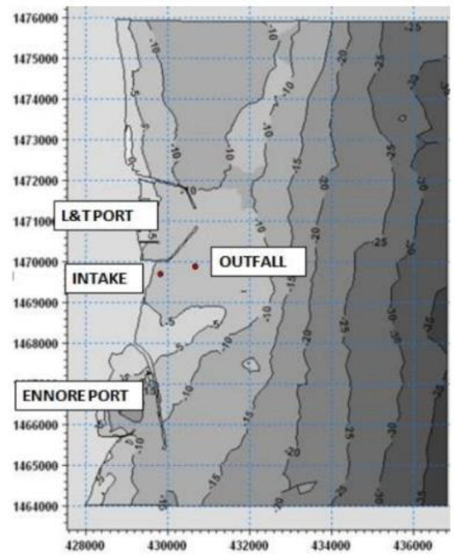


Figure 1 Bathymetry in the model region

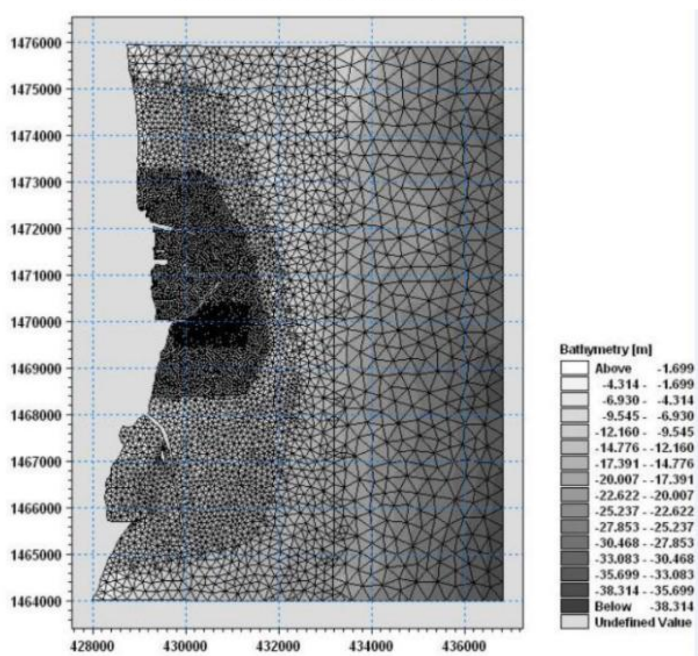


Figure 2 Flexible computational mesh

Figure. Reference Study 3: hydrodynamic calibration (observed vs computed tidal level and current).

4.4. Model Operation

The ambient temperature of sea water is assumed to be 27^{0}C. Ambient salinity of sea water is assumed to be 32 ppt. The rise in salinity of brine water to be discharged at the outfall is given to be 31 ppt above ambient.

For the salinity model, as the open boundaries have been taken sufficiently away from the outfall locations, zero rise in salinity is assumed at the open boundaries. It is observed that in the open sea at the site, the flow is unidirectional, either northerly or southerly generally parallel to the coastline. During the period February-September the flow is northward, while during the period October-January the flow is southward. The model is operated for different current conditions namely northward 0.25 m/s, southward 0.25 m/s and weak current i.e. 0.05 m/s towards. The direction of the weak current is considered to be northward or southward depending on the locations of intake and outfall to simulate the critical condition.

4.5. Model Operating Cases

The following cases of the intake and outfall were considered for the numerical model simulations.

Intake and outfall locations on south of south breakwater of L & T port as shown in Fig.1 are considered. Intake is taken at 550 m from shore. The outfall distance from shore is varied from 1000 m to 1300 m. Discharge of 1.25 m^3/s (100 MLD) is considered at outfall while inflow at intake is taken twice that of outfall.. The following alternatives of outfall locations are considered.

1. With outfall at 1000 m from shore
2. With outfall at 1100 m from shore
3. With outfall at 1200 m from shore
4. With outfall at 1300 m from shore

4.6. Results and Discussion

The model is operated under northward, southward and weak current conditions in each of the above cases of outfall location. Typical results, namely, contour plots and time histories of rise in salinity are shown in Figs. 4A, 4B, 5A and 5B. The salinity rise at the intake is nearly zero in case of average northward current of 0.25 for all the outfall lengths. The salinity rise obtained at the intake for different outfall lengths and for the remaining two current conditions (namely, southward and weak current) are tabulated in Table 1A and 1B. It is seen that for the average northward current of 0.25 m/s the rise in salinity above ambient at the intake is well below 0.1 ppt. The salinity rise results from the tables are also plotted in the Fig. 6. From the tables and the Fig. 6 it is seen that it is also less than 0.1 ppt for average southward current of 0.25 m/s for the outfall distance greater than 1000 m and it is more than 0.1 ppt for outfall distance less than about 975 m.

Figure. Reference Study 3: contours of salinity rise and time history at intake/outfall.

Appendix B Complete Output Inventory

All figures and datasets generated by the UBDS solver for this case study:

B.1 Figures

- 01. [Near-field initial dilution] A_nearfield_RO_S1.png - Near-field trajectory, dilution and salinity - RO concentrate, 300 m3/hr (300 m3/hr per port).
- 02. [Near-field initial dilution] A_nearfield_RO_S2.png - Near-field trajectory, dilution and salinity - RO concentrate, 700 m3/hr (350 m3/hr per port).
- 03. [Near-field initial dilution] A_nearfield_RO_S3.png - Near-field trajectory, dilution and salinity - RO concentrate, 1200 m3/hr (300 m3/hr per port).
- 04. [Near-field initial dilution] A_nearfield_Sa_S1.png - Near-field trajectory, dilution and salinity - Saturated cavern brine, 300 m3/hr (300 m3/hr per port).
- 05. [Near-field initial dilution] A_nearfield_Sa_S2.png - Near-field trajectory, dilution and salinity - Saturated cavern brine, 700 m3/hr (350 m3/hr per port).
- 06. [Near-field initial dilution] A_nearfield_Sa_S3.png - Near-field trajectory, dilution and salinity - Saturated cavern brine, 1200 m3/hr (300 m3/hr per port).
- 07. [Near-field initial dilution] A_scenario_comparison.png - Seabed initial dilution by flow scenario and tidal state.
- 08. [Diffuser design optimisation] A2_angle_opt.png - Diffuser port-angle optimisation: dilution and seabed salinity vs discharge angle.
- 09. [Hydrodynamics, mesh & calibration] B_mesh_medium.png - Medium-field mesh & bathymetry (Bahia Azul)
- 10. [Hydrodynamics, mesh & calibration] B_mesh_far.png - Far-field (regional) domain & bathymetry
- 11. [Hydrodynamics, mesh & calibration] B_stations.png - Diffuser location & ADCP current-meter stations
- 12. [Hydrodynamics, mesh & calibration] B_adcp_A0.png - Modelled current speed & direction at station A0 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.
- 13. [Hydrodynamics, mesh & calibration] B_adcp_A1.png - Modelled current speed & direction at station A1 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.
- 14. [Hydrodynamics, mesh & calibration] B_adcp_A2.png - Modelled current speed & direction at station A2 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.
- 15. [Hydrodynamics, mesh & calibration] B_adcp_A3.png - Modelled current speed & direction at station A3 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.
- 16. [Hydrodynamics, mesh & calibration] B_adcp_A4.png - Modelled current speed & direction at station A4 (neap & spring), against a synthetic ADCP reference generated from the model solution with a 3 percent bias and random noise. Not an independent measurement.

- 17. [Hydrodynamics, mesh & calibration] B_tidal_level.png - Modelled tidal level (spring tide) against a synthetic reference series derived from the model solution with an imposed bias and noise. Not an independent measurement.
- 18. [Hydrodynamics, mesh & calibration] B_flowfield_flood.png - Depth-averaged tidal flow field at flood (spring tide).
- 19. [Hydrodynamics, mesh & calibration] B_flowfield_ebb.png - Depth-averaged tidal flow field at ebb (spring tide).
- 20. [Medium-field dispersion] C_sigma_layers.png - Variation of the terrain-following sigma layers with water depth (9-layer scheme).
- 21. [Medium-field dispersion] C_layersens_4.png - Maximum seabed salinity - 4-layer model (full flow, neap, saturated cavern brine)
- 22. [Medium-field dispersion] C_layersens_9.png - Maximum seabed salinity - 9-layer model (full flow, neap, saturated cavern brine)
- 23. [Medium-field dispersion] C_layersens_14.png - Maximum seabed salinity - 14-layer model (full flow, neap, saturated cavern brine)
- 24. [Medium-field dispersion] C_env_heights_panel.png - Maximum salinity vs height above the seabed - neap, full flow, saturated cavern brine (dense plume is bottom-trapped and decays upward)
- 25. [Medium-field dispersion] C_snap_neap_high_water.png - Seabed salinity & current vectors at high water - neap (saturated cavern brine)
- 26. [Medium-field dispersion] C_vprof_neap_high_water.png - Vertical salinity profile at high water - neap (saturated cavern brine)
- 27. [Medium-field dispersion] C_snap_neap_mid_ebb.png - Seabed salinity & current vectors at mid ebb - neap (saturated cavern brine)
- 28. [Medium-field dispersion] C_vprof_neap_mid_ebb.png - Vertical salinity profile at mid ebb - neap (saturated cavern brine)
- 29. [Medium-field dispersion] C_snap_neap_low_water.png - Seabed salinity & current vectors at low water - neap (saturated cavern brine)
- 30. [Medium-field dispersion] C_vprof_neap_low_water.png - Vertical salinity profile at low water - neap (saturated cavern brine)
- 31. [Medium-field dispersion] C_snap_neap_mid_flood.png - Seabed salinity & current vectors at mid flood - neap (saturated cavern brine)
- 32. [Medium-field dispersion] C_vprof_neap_mid_flood.png - Vertical salinity profile at mid flood - neap (saturated cavern brine)
- 33. [Medium-field dispersion] C_snap_spring_high_water.png - Seabed salinity & current vectors at high water - spring (saturated cavern brine)
- 34. [Medium-field dispersion] C_snap_spring_mid_ebb.png - Seabed salinity & current vectors at mid ebb - spring (saturated cavern brine)
- 35. [Medium-field dispersion] C_snap_spring_low_water.png - Seabed salinity & current vectors at low water - spring (saturated cavern brine)
- 36. [Medium-field dispersion] C_snap_spring_mid_flood.png - Seabed salinity & current vectors at mid flood - spring (saturated cavern brine)

- 37. [Medium-field dispersion] C_env_springneap.png - Maximum seabed salinity over a full spring-neap cycle (full flow, saturated cavern brine)
- 38. [Far-field dispersion] D_far_neap_scenarios.png - Far-field maximum seabed salinity by discharge scenario - neap tide (RO brine)
- 39. [Far-field dispersion] D_far_spring_S3.png - Far-field maximum seabed salinity - spring, full flow (RO brine)
- 40. [Far-field dispersion] D_far_springneap.png - Far-field maximum seabed salinity - spring-neap envelope (full flow, RO brine)
- 41. [Far-field dispersion] D_far_cavern.png - Far-field maximum seabed salinity - phase-2 saturated cavern brine (worst case)
- 42. [Design comparison & optimisation] E_salinity_transect.png - Maximum seabed salinity along a transect through the diffuser (neap, full flow, saturated cavern brine).
- 43. [Design comparison & optimisation] E_current_curve.png - Maximum current speed along the outfall corridor (spring tide).
- 44. [Design comparison & optimisation] E_constraints.png - Distribution of marine development and environmental constraints relative to the mixing zone.
- 45. [Design comparison & optimisation] E_optimised_layout.png - Recommended optimised diffuser/intake layout (inclined 4-port diffuser).
- 46. [Outfall siting & recirculation] H_intake_rise_curve.png - Variation of the maximum salinity rise at the intake with outfall distance from shore, for the RO concentrate and the phase-2 saturated cavern brine (log scale; 0.1 ppt recirculation criterion). Values of zero are plotted at the axis floor.
- 47. [Outfall siting & recirculation] H_rise_contour_near.png - Salinity-rise contours, outfall 600 m from shore, weak southward current (RO concentrate).
- 48. [Outfall siting & recirculation] H_rise_contour_far.png - Salinity-rise contours, outfall 1200 m from shore, weak southward current (RO concentrate).
- 49. [Outfall siting & recirculation] H_timehist_near.png - Time history of salinity rise at intake and outfall, outfall 600 m from shore (weak southward current, RO concentrate).
- 50. [Outfall siting & recirculation] H_timehist_far.png - Time history of salinity rise at intake and outfall, outfall 1200 m from shore (weak southward current, RO concentrate).
- 51. [Inputs, parameters & water quality] F_impurities.png - Predicted receiving-water trace-impurity concentrations vs EQS thresholds.
- 52. [Data plots] G_dilution_vs_velocity.png - Seabed dilution as a function of tidal velocity for each brine type and scenario. S1 and S3 coincide because dilution is set by the per-port flow (300 m3/hr in both), not the total.
- 53. [Data plots] G_area_vs_flow.png - Area enclosed by salinity-increment contours vs discharge flow. All three areas are zero on the regional grid, whose peak seabed increment is 0.39 psu; the zeros are grid-resolved, not absolute.
- 54. [Data plots] G_skill.png - Index-of-agreement between the model and the synthetic current reference by station (spring). The reference is derived from the model solution, so this is not a field comparison.
- 55. [Animations] ANIM_brine_dispersion_filmstrip.png - Brine dispersion over a tidal cycle (full flow, spring, saturated cavern brine) (filmstrip of the animation).

- 56. [Reference output figures (source studies)] appendix_b_fig_p8.png - Reference Study 1 (salt-cavern brine outfall): model mesh, bathymetry and current-meter comparison.
- 57. [Reference output figures (source studies)] appendix_b_fig_p25.png - Reference Study 1: near-field brine-plume trajectory and initial-dilution prediction.
- 58. [Reference output figures (source studies)] appendix_b_fig_p33.png - Reference Study 1: sigma-layer scheme and layer-resolution sensitivity.
- 59. [Reference output figures (source studies)] appendix_b_fig_p39.png - Reference Study 1: maximum seabed-salinity envelope and salinity/current snapshots.
- 60. [Reference output figures (source studies)] appendix_b_fig_p55.png - Reference Study 1: far-field maximum seabed-salinity envelopes.
- 61. [Reference output figures (source studies)] water_15_02402_fig_p4.png - Reference Study 2 (Laoshan Bay desalination): computational mesh and bathymetry of the bay.
- 62. [Reference output figures (source studies)] water_15_02402_fig_p6.png - Reference Study 2: tidal flow-field simulation results.
- 63. [Reference output figures (source studies)] water_15_02402_fig_p7.png - Reference Study 2: salinity-field results and salinity profile for the layout plans.
- 64. [Reference output figures (source studies)] ijciet_fig_p4.png - Reference Study 3 (Chennai desalination): bathymetry and flexible computational mesh.
- 65. [Reference output figures (source studies)] ijciet_fig_p5.png - Reference Study 3: hydrodynamic calibration (observed vs computed tidal level and current).
- 66. [Reference output figures (source studies)] ijciet_fig_p7.png - Reference Study 3: contours of salinity rise and time history at intake/outfall.
- 67. [Validation] V_jet.png - Momentum-jet dilution: UBDS vs the canonical self-similar law (Fischer et al. 1979).
- 68. [Validation] V_plume.png - Pure-plume dilution: UBDS vs the 5/3 power law (Morton, Taylor & Turner 1956).
- 69. [Validation] V_dense_jet.png - Inclined 60 deg dense-jet dimensionless rise and dilution vs published experimental bands.

B.2 Animations

- 01. ANIM_brine_dispersion_spring.gif - Brine dispersion over a tidal cycle (full flow, spring, saturated cavern brine)
- 02. ANIM_brine_dispersion_neap.gif - Brine dispersion over a tidal cycle (full flow, neap, saturated cavern brine)
- 03. ANIM_current_field_spring.gif - Tidal current field over a cycle (spring)

B.3 Datasets (CSV)

- 01. A_nearfield_initial_dilution.csv - Near-field initial-dilution results (bulk top-hat dilution and salinity at first seabed contact) for all brine types, flow scenarios and tidal states.
- 02. A2_port_angle_optimisation.csv - Effect of diffuser port discharge angle on near-field dilution (300 m3/hr, neap).

- 03. B_current_validation_skill.csv - Skill of the model against the synthetic ADCP reference series (RMSE, Willmott index, Nash-Sutcliffe). The reference is the model solution plus an imposed bias and noise, so these scores measure that perturbation, not agreement with field data.
- 04. B_tidal_level.csv - Tidal-level validation time series.
- 05. D_influence_area_diffusion.csv - Influence areas of salinity increments and diffusion distances per flow scenario (far field).
- 06. E_salinity_transect.csv - Seabed salinity transect data.
- 07. H_intake_recirculation.csv - Salinity rise above ambient at the intake vs outfall distance from shore, by brine and current condition (0.1 ppt recirculation criterion, applied to the peak cell).
- 08. H_table1A_southward.csv - Rise in salinity at intake above ambient - southward 0.25 m/s, both brines.
- 09. H_table1B_weak.csv - Rise in salinity at intake above ambient - weak southward 0.05 m/s, both brines.
- 10. H_table1C_northward.csv - Rise in salinity at intake above ambient - northward 0.25 m/s, both brines.
- 11. H_timehist_near.csv - Time-history data, outfall 600 m from shore (RO concentrate).
- 12. H_timehist_far.csv - Time-history data, outfall 1200 m from shore (RO concentrate).
- 13. F_layer_scheme.csv - Sigma-layer schemes: percentage of water-column depth per layer, indexed from the seabed (L1 is the bed cell in every scheme; each scheme's surface cell is its last numbered entry - L14, L9 and L4 respectively). A dash means the scheme has no such layer. The bed cell is 0.1% of depth with 14 layers, 1.0% with 9 and 15% with 4.
- 14. F_model_parameters.csv - Hydrodynamic and transport model parameters.
- 15. F_impurities_eqs.csv - Trace-impurity concentrations in the brine vs EQS after near-field dilution.
- 16. V_eos_validation.csv - Equation-of-state validation against reference seawater/brine densities.
- 17. V_jet.csv - Round-jet dilution validation data.
- 18. V_dense_jet.csv - Inclined (60 deg) dense-jet dimensionless geometry/dilution vs published ranges. S_r is reported both as the model's bulk (top-hat) dilution and converted to the centreline minimum dilution, which is what the published band refers to.